



Chips: The biggest AI bottleneck

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- This week's upcoming Xi-Trump summit throws the geopolitical issues surrounding semiconductors back into the spotlight.
- In this paper, we argue that the key bottlenecks are in advanced chip and, to a lesser extent, specialized equipment (e.g., EUV lithography tools) and manufacturing, located predominantly in Taiwan and the Netherlands.
- Technology, particularly advanced chips, is now seen as the heart of an intense competition to retool economies, reform militaries, and serve as a source of national strength.

Our new series, "The Great Rebalancing," focuses on five "critical chokepoints" that we see as redefining global power: **Currencies, Critical Minerals, Chips, Cables, and Cloud.**

Each chapter of our series will explore **how and why these chokepoints define a new matrix of global power.** Currencies allocate risk, critical minerals anchor the physical economy, chips generate compute power, cables move data and finance, and cloud platforms organize markets. Influence over this layered system shapes geopolitical standing – determining which nations can scale their economies, exert control, and apply strategic pressure.

This series argues that a "great rebalancing" is underway, shifting the global order from liberal institutionalism, where interdependence fosters cooperation, toward what we term "infrastructure realism". In this new paradigm, control over key systems and inputs shapes outcomes. Under "infrastructure realism", material and digital infrastructures become decisive instruments of state power. The US-China trade tensions of 2025 starkly illustrated this shift. The financial system — from bond markets to payment rails to digital money — acts as the meta-infrastructure that coordinates and prices all others.

Thus, this series addresses the wider question: which major powers are best positioned for influence and what are their strategic advantages? The 2025 US National Intelligence Annual Threat Assessment identified China as the primary actor challenging US global interests, citing its use of "asymmetric and conventional hard power tactics" and its drive to establish "alternative systems" to compete with the US¹. While the competitive balance of power is longstanding, 2025 highlighted the asymmetric economic dependencies and capabilities shaping the relationship between the two countries.

As these five chapters will demonstrate, we believe power now rests on the control of critical infrastructure. The US and China are in a race to reduce strategic dependencies on each other, and 'we believe the outcome of this dynamic' will define the next era.

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¹ Review of ANNUAL THREAT ASSESSMENT of the U.S. INTELLIGENCE COMMUNITY. 2025. Office of the Director of National Intelligence.



Introduction: From compute to chokepoints, the competition for industrial control

In Chapter 1: Currency, we argued that the US is successfully adapting to maintain leadership in digital finance (or, as we call it, “[meta-infrastructure](#)”), whilst in Chapter 2: Critical Minerals, we [analysed China’s dominance](#) across different segments of the critical minerals supply chains and how the central government has been able to leverage this advantage in recent national security disputes. These case studies illustrate each layer of the infrastructure-realist stack, which we describe as digital, invisible, and embedded within complex, globally distributed supply chains.

In this paper, we now move on to the next layer: semiconductors, otherwise colloquially referred to as chips. As the contours of modern conflict shape themselves around the AI race, chips have become central to great power competition given their crucial bottleneck in the AI stack.

We examine the cutting-edge chip technologies that underpin cloud computing and modern defence systems, with chipmakers emerging as some of the most visible beneficiaries of the AI-driven technological cycle. We also examine the US-China conflict around legacy chips, the mature nodes (>22nm) that power everyday appliances.

Because both the US and China have used their respective industrial strengths in chips and critical minerals for retaliatory actions, and because chips themselves are a direct end-product of critical minerals, it is tempting to draw neat parallels between the two countries’ economic toolkits. However, there are key differences to consider. First, the scale of investment and engineering complexity in semiconductors is far greater than that of rare earths. Second, unlike critical mineral supply chains – where China unquestionably exerts dominance across multiple stages – the semiconductor ecosystem is more fragmented, with different countries controlling distinct parts of the value chain. Apple’s complex production process illustrates this asymmetry: chips are designed by Apple in the US, manufactured in Taiwan, and assembled in China².

At the centre of this competition lies Taiwan. US Treasury Secretary Scott Bessent summed up the Taiwan risk at Davos earlier this year: “The single biggest threat to the world economy” he said, “is that 97% of the world’s most advanced chips are manufactured on the island. If that island were blockaded, if that capacity were destroyed, it would be an economic apocalypse³.” An equally critical though less immediate bottleneck is the Dutch company ASML; without its unique lithography equipment, Taiwan cannot build the fabs that produce these integrated circuits. However, ASML has indicated a potential to scale up production by 3x by 2030, if necessary.

Ultimately, building a vertically integrated domestic semiconductor industry has so far proven to be an unrealistic goal for any single nation. As we illustrate, each stage of the chip supply chain is a potential chokepoint for another. This creates a competitive landscape with no clear overarching winners, resulting in structural interdependence, supply fragility, and a stalemate that keeps both the US and China in an asymmetric power struggle.

² For a thorough breakdown of Apple’s supply chain, read Chris Miller and Vishnu Venugopalan’s work here: Chris Miller and Vishnu Venugopalan, “Apple’s Supply Chain: Economic and Geopolitical Implications,” American Enterprise Institute, June 3, 2025, <https://www.aei.org/research-products/report/apples-supply-chain-economic-and-geopolitical-implications/>.

³ Tripp Mickle, “The Looming Taiwan Chip Disaster That Silicon Valley Has Long Ignored,” *The New York Times*, February 24, 2026, <https://www.nytimes.com/2026/02/24/technology/taiwan-china-chips-silicon-valley-tsmc.html>.



1. Chips: Infrastructure as statecraft

Although China dominates in mineral processing and magnets manufacturing, the US maintains advantage in their downstream outputs: chip software, intellectual property (IP), and manufacturing equipment. As a result, the US has been able to counter China's critical minerals leverage by deploying its own toolkit of semiconductor dominance. This dynamic was starkly illustrated in April 2025, when the US imposed export controls on Nvidia's H20 chips to China in response to Beijing's own rare earth export restrictions on April 4⁴.

The ongoing chip competition illustrates how crucial semiconductors have become in contemporary economic statecraft. Here, we trace their historical emergence as strategic assets, analyze their diverse economic functions and market structures, and look at how the race for supply chain sufficiency has narrowed to a contest over cutting-edge chip technology.

1.1 Semiconductors: The physical layer of the digital economy

Semiconductors – miniature electronic devices primarily made from silicon or germanium – are indispensable components of nearly every modern electronic device. They power everything from smartphones, computers and home appliances to critical infrastructure such as telecommunications networks, data centers, autonomous vehicles, and industrial automation systems. At the highest level, semiconductors are designed to sense, compute, store, transmit, and control information and energy.

The overall semiconductor market is expected to generate ~\$1trn in revenue in 2026, and over \$1.5trn in 2030, compared to ~\$791bn in 2025⁵. Today's semiconductor industry is highly concentrated, with a few dozen major players producing a wide range of specialized chips with distinct economic functions:

- **Logic chips (“Compute”)**: These act as the “brains” of digital systems, processing information and executing tasks. This category includes CPUs (central processing units), GPUs (graphics processing units), and AI accelerators used in data centers, personal computers, smartphones, and cloud infrastructure.
- **Memory chips (“Storage”)**: These chips store and retrieve data, enabling systems to retain information and operate efficiently. Key examples include DRAM and NAND flash, which are ubiquitous across servers, PCs, and mobile devices⁶.
- **Analog and power chips (“Control”)**: These manage electrical signals and power flows, allowing digital systems to interact with the physical world. They are essential in automotive systems, industrial machinery, energy infrastructure, and defense applications.
- **Specialized and integrated chips (“Optimized Systems”)**: These are designed for specific tasks or highly integrated functions. This includes application-specific integrated circuits (ASICs) tailored for particular workloads (e.g. crypto mining or networking) and systems-on-a-chip (SoCs) that combine multiple components into a single, power-efficient unit.

⁴ Marion Laboure and Camilla Siazon, “Thematic Research: US vs. China: The Biggest Bargaining Points”, Deutsche Bank Research, April 17, 2025, https://research.db.com/research/Article?rid=e0c6c6ad-47e1-41cb-8162-edf715f0355a-604&kid=RP0001&documentType=R&wt_cc1=IND-3146993-0000.

⁵ Dylan Peterson, “Global Annual Semiconductor Sales Increase 25.6% to \$791.7 Billion in 2025 - Semiconductor Industry Association,” Semiconductor Industry Association, February 6, 2026. <https://www.semiconductors.org/global-annual-semiconductor-sales-increase-25-6-to-791-7-billion-in-2025/>. For the 2030 projections, see Company, Manufacturing. “2026 Technology Symposium Highlights.” Tsmc.com. Taiwan Semiconductor Manufacturing Company. https://www.tsmc.com/english/symposium_highlights/2026.

⁶ DRAM and NAND are types of memory semi-components, but DRAM memory manages data and requires power, therefore making it volatile, whereas NAND stores data and does not require power.



1.2 A brief history of semiconductors: From Cold War origins to global bedrock

Since their invention in the 1940s, semiconductors have been intrinsically linked to national security and strategic competition. Their role, however, has evolved from state-led innovation to market-driven globalization, and more recently back to geopolitical rivalry. Semiconductors were also once in the wheelhouse of US strength, but over the 1990s-2010s, the country's economic model was guided by the belief that integrating global trade and supply chains promoted peace and economic prosperity. As we note later, the symbiotic relationship that flourished between Taiwan's foundries and US design firms exemplified this era.

State-led origins: Semiconductors as military technology (1940s–1980s)

Semiconductors emerged from Cold War US defence programs, most notably with the invention of the transistor (a single, fundamental chip component used to switch or amplify electrical signals) at Bell Labs (1947-58). Early innovation was heavily subsidized and driven by military and space exploration demand.⁷

In the late 1950s, the invention of the integrated circuit (a chip that contains millions or billions of interconnected transistors) helped semiconductors move from niche military applications into the rapidly expanding commercial computer industry, fuelling the rise of Silicon Valley. By the 1980s, semiconductors were embedded across everyday technologies like cars and personal computers, as exponential improvements in computing power and cost efficiency led to a rigorous and sustained cycle of innovation.

During the initial years of chip innovation, the US remained the relatively unchallenged global leader in semiconductor production, accounting for 60% of supply in the 1970s.⁸ However, as global interest in chip production grew and other countries began to develop their own capabilities, Japan's rapid ascent in the global DRAM market stirred concern among US policymakers, who began to view the country as a primary strategic competitor. By the mid-1980s, Japan had overtaken the US in semiconductor production, and towards the end of the decade, Japanese firms held leading positions in key segments of the semiconductor equipment market.⁹

In response to what it viewed as a national security threat, the US forced the issue. The resulting 1986 US-Japan Semiconductor Agreement sought to halt the dumping of Japanese DRAMs in third markets at below-cost prices and increased foreign access to Japan's market, with Japan agreeing to help secure 20% of its domestic semiconductor market for foreign producers within five years. Concurrently, the US moved to strengthen its own industry by establishing SEMATECH in 1987, a government-backed research consortium comprising of 14 different domestic chip companies.¹⁰

This intense rivalry proved short-lived. Japan's chip ascent stalled following its financial crisis in the early 1990s, and by the end of the 20th century, other US technology competitors like the Soviet Union and South Korea had also faded.

Market-led globalization: Fragmentation and the rise of Taiwan (1990s–2010)

With the dissolution of the USSR in 1991, the US was at its apogee as the standard-bearer for a Western model of liberal democracy and capitalism. This unipolar moment ushered in an era defined by liberal institutionalism, where successive US administrations promoted open markets, free trade, and neoliberal policies through multilateral efforts.

⁷ For more background on the history of semiconductors, see Chris Miller, *Chip War: The Fight for the World's Most Critical Technology* (New York: Simon and Schuster, 2022).

⁸ Emily G. Blevins, Michaela D. Platzer, Karen M. Sutter, "Semiconductors: U.S. Industry, Global Competition, and Federal Policy," CRS Report R46581 (Washington, DC: Congressional Research Service, October 26, 2025), <https://www.congress.gov/crs-product/R46581>.

⁹ Japan was supplying 70% of the world's lithography equipment. See Chris Miller's *Chip War*.

¹⁰ Government funding for SEMATECH stopped in the 1990s, after which it relied on industry funding. In 2015, the research group was absorbed by the State University of New York Polytechnic Institute. See Blevins, Platzer, and Sutter, "Semiconductors."



Under this new world order, the semiconductor industry became increasingly globalized and specialized. Its value chain fragmented into distinct segments—design, manufacturing, and equipment—each optimized for efficiency and scale. Landmark trade agreements, including the World Trade Organization’s Information Technology Agreement in 1996 and its 2015 expansion, helped bring semiconductor tariffs close to zero in many countries.¹¹

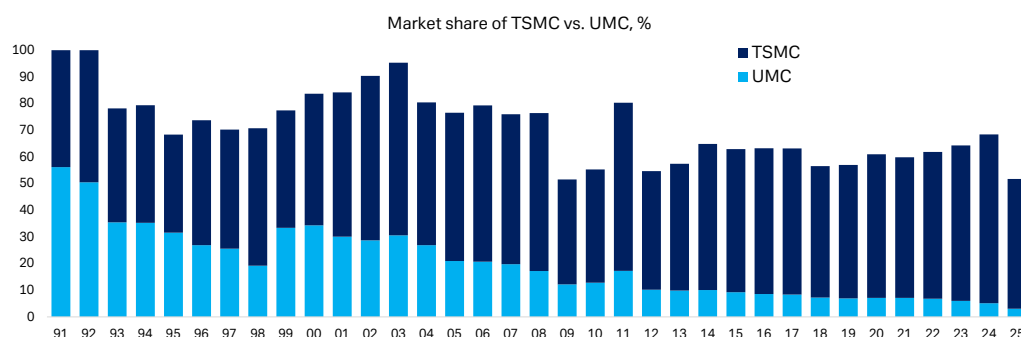
Within this liberalized framework, the industry’s centre of gravity shifted toward Taiwan. Its rise was spearheaded by the Taiwan Semiconductor Manufacturing Company (TSMC), founded in 1987 by Morris Chang, a veteran of US company Texas Instruments. TSMC pioneered the foundry model – manufacturing chips designed by external customers – which in turn enabled the rise of “fabless” design firms, particularly in the US. Prior to this innovation, most chip companies had operated as integrated device manufacturers, a capital-intensive model that required them to design and produce their circuits in-house.

Rather than a strategic rivalry, this division of labour created a powerful symbiotic relationship: the US led in design, while Taiwan specialized in manufacturing. Supported by talent flows from Silicon Valley, this model underpinned two decades of rapid innovation and global integration. TSMC’s rise as a leading foundry enabled the growth of US companies in turn; and this symbiotic relationship sustained both sides for the next two decades. While the increasing reliance on a “fabless” business model would eventually be viewed as a strategic vulnerability for US manufacturing, this trade-off was accepted by US leaders during the post-Cold War era of globalization and liberal institutionalism.

Return of geopolitics: Semiconductors as strategic infrastructure (2010–present)

By 2010, the growing importance of semiconductors in cloud computing, and advanced military systems brought their strategic importance back to the forefront. Chips were no longer viewed merely as commercial inputs, but as foundational infrastructure underpinning economic and military power. This shift marked the beginning of a new phase of geopolitical competition, particularly between the US and China. Consequently, governments have re-engaged with the sector through industrial policy, export controls, and national security frameworks, transforming the industry from a globalized, efficiency-driven system to one increasingly defined by strategic competition and the race to control critical supply chain chokepoints.

Figure 1: TSMC quickly surpassed competitors in the foundry market after its founding, largely due to continued investment even during downturns and a strong drive to innovate beyond companies like UMC.



Source: Deutsche Bank Research, Bloomberg Finance LP.

¹¹ See Blevins, Platzer, and Sutter, “Semiconductors.”



1.3 China's semiconductors policy: From integration to strategic self-reliance

Today, Huawei is a central player in China's drive for both chip design and manufacturing, largely in response to US export controls. Through its chip design arm, HiSilicon, the company has played a key role in developing advanced semiconductors domestically. For much of its history, however, Huawei operated primarily in telecommunications infrastructure, a sector in which it eventually grew into a global giant, helping position China as a leader in 5G and electronics. This success, however, came at a cost: its expanding global footprint turned it into a focal point of geopolitical tensions, as several countries began to assert that Huawei posed national security risks.

The rise of Huawei and China's broader electronics ecosystem would not have been possible without sustained access to foreign semiconductors. Here, we trace China's transition from a mass importer of semiconductors towards its own path of chip independence.

Setting up the scene: China's chip origins (1980s-2000s)

China has been the largest single market for semiconductor exports since the country underwent its technological revolution in the 1980s. Under Deng Xiaoping's program to modernize the economy via science and technology, China transitioned from a closed, centrally planned system toward a market-oriented economy, laying the groundwork for a domestic technology sector. From this period of reform emerged firms such as Huawei, which was founded in 1987 to supply China's growing demand for telecommunications equipment.

Throughout the 1990s and 2000s, China relied heavily on imports from the US, Japan, and increasingly Taiwan to power its industrial expansion. As China entered the 2000s, the government began to make early moves toward domestic semiconductor manufacturing, most notably through the establishment of Semiconductor Manufacturing International Corporation (SMIC) in 2000. Progress, however, remained limited.

Early moves: China's move towards self-reliance amidst heightening geopolitical tensions (2010s-2020s)

By the 2010s, China accounted for ~60% of global semiconductor demand but only ~13% of global supply, underscoring a deep structural dependence on foreign technology¹². This gap became a central vulnerability in China's economic model.

Recognizing this vulnerability, the government began to scale up state support. In 2014, Xi Jinping launched the National Integrated Circuit Industry Investment Fund (the "Big Fund") to finance domestic foundries, design firms, and equipment suppliers. With support from both central and local governments, capital flowed into fabrication plants and upstream supply chains, enabling firms like SMIC to reach 14nm production by 2019¹³.

China's ambition to become a global leader in high-tech manufacturing was further formalized through its "Made in China 2025" strategy. This industrial policy, unveiled in 2015, sought to reduce reliance on foreign technology and help the nation avoid the "middle-income development trap", setting an aggressive target of 70% semiconductor self-sufficiency by 2025.

Unlike the earlier semiconductor success stories of Japan, South Korea and Taiwan – which were deeply integrated into the US-led ecosystem – China pursued, and continues to pursue, a more state-driven and increasingly independent approach aimed at building domestic capabilities across the full semiconductor value chain. Thus, China's scope and

¹² James McBride and Andrew Chatzky, "Is 'Made in China 2025' a Threat to Global Trade?" Council on Foreign Relations, August 2, 2018, <https://www.cfr.org/backgrounders/made-china-2025-threat-global-trade>.

¹³ Anton Shilov, "SMIC Mass Produces 14nm Nodes, Advances to 5nm, 7nm," *Tom's Hardware*, September 16, 2022, <https://www.tomshardware.com/news/smic-mass-produces-14nm-nodes-advances-to-5nm-7nm>.



scale prompted concerns among US officials, with some members of Congress describing Beijing's methods for independence as ones that "violate, circumvent, or otherwise challenge current global trade rules."¹⁴

During this period, several geopolitical episodes also exposed the risks of China's reliance on foreign semiconductor technology, contributing to a broader deterioration in bilateral relations and serving as a wakeup call for the government to accelerate its push for technological self-reliance.

The first was the 2013 Snowden revelations, which highlighted the extent of US surveillance capabilities and amplified Chinese concerns over dependence on foreign digital infrastructure¹⁵. Then in 2018, the US imposed export restrictions on Zhongxing Telecommunication Equipment (ZTE) Corporation, a major Chinese telecommunications equipment and network systems company, for violating US sanctions by supplying equipment to Iran and North Korea. During this period, US semiconductor firms like Qualcomm and Intel were prohibited from supplying critical semiconductor components to ZTE, forcing the company to suspend major operations within weeks. Although the ban was lifted later that year, the episode was a powerful demonstration of how quickly access to semiconductors could become a binding constraint – effectively translating technological dependence into operational paralysis.

Together, these events heightened strategic mistrust on both sides, and reinforced Chinese officials' imperative to achieve technological self-reliance. As a result, the country intensified its efforts to localize semiconductor production and reduce its dependence on US technology, accelerating a broader trend toward supply chain decoupling. We will delve more into the specifics of China's present strategy later in the chapter.

¹⁴ [Semiconductors and the CHIPS Act: The Global Context," Congress.gov, Library of Congress.](#)

¹⁵ The Snowden revelations refer to the disclosure of thousands of classified documents from the U.S. National Security Agency (NSA) by Edward Snowden, a former NSA contractor. Beginning in June 2013, newspapers like The Guardian and The Washington Post published articles based on these documents, revealing the vast scale of global surveillance programs conducted by the NSA and its international partners, most notably the UK's Government Communications Headquarters (GCHQ).



2. The semiconductor supply chain: Complexity and chokepoints

Overcoming chokepoints in the semiconductor industry is more difficult than in the critical minerals sector. This is because the semiconductor ecosystem, while vast, is characterized by extreme specialization and a highly fragmented global value chain, particularly at the advanced end of the market.

Producing a single semiconductor is an extraordinarily complex process involving billions of transistors. This complexity is often explained by Moore's Law, first articulated by Intel co-founder Gordon Moore in 1965. Moore's Law states that the number of transistors on a chip doubles approximately every two years. At the same time, a complementary dynamic, often referred to as Moore's Second Law, dictates that the cost of manufacturing these chips rises exponentially as their components shrink.¹⁶ As transistors become smaller, fabrication becomes significantly more complex and capital-intensive. The combined effect of these two laws has driven a relentless cycle of innovation that requires ever-increasing R&D and capital expenditure.

This is reflected in the economics of production. A leading-edge fabrication plant (fab) can cost \$15–20bn to build, while industry leaders such as TSMC and Samsung spend \$30–40bn annually on capital expenditure alone. As one study notes for the semiconductor race, “the technological clock speeds up while the financial ante keeps rising.”¹⁷

2.1 Breaking down the global supply chain

The semiconductor supply chain can be divided into three main parts: (i) design, (ii) fabrication, and (iii) assembly, testing, and packaging (ATP). Each represents a distinct layer of value creation and a potential geopolitical chokepoint.

At the upstream end, the design stage involves creating the architecture and logic of a chip using highly specialized software and intellectual property (IP). This segment is dominated by US firms, particularly in electronic design automation (EDA) software, chip architecture, and core IP. As a result, the US maintains a critical advantage at the earliest stage of the value chain.

Midstream, the fabrication stage refers to the manufacturing of chips in foundries. This is the most capital-intensive and technologically complex part of the supply chain, with production highly concentrated geographically, particularly in Taiwan and South Korea. Taiwan dominates leading-edge fabrication—largely through TSMC—with a dense concentration of foundries producing chips for GPUs, CPUs, mobile devices, and connectivity systems. This dominance reflects decades of industrial execution, as well as powerful economies of scale inherent in advanced manufacturing, where costs and complexity favour a small number of highly specialized players.

However, even this dominance is constrained by a chokepoint in semiconductor manufacturing equipment (SME), particularly lithography. The Netherlands, through ASML, maintains a near-monopoly on extreme ultraviolet (EUV) lithography machines, which are required to produce leading-edge chips at 5nm and below. As a result, leading foundries such as TSMC depend on ASML's EUV systems to manufacture their most advanced semiconductors. Without these machines – which have direct implications for smartphones, data centers, and AI systems – production at the technological frontier would not be possible. This was demonstrated in 2021, when high demand exceeded ASML's production, leading to significant chip shortages and extended delivery times for advanced chips.

¹⁶ Brian Albrecht, Geoffrey Manne, David Teece, and Mario Zúñiga, “From Moore's Law to Market Rivalry: The Economic Forces That Shape the Semiconductor Manufacturing Industry,” International Center for Law & Economics, November 2025, <https://laweconcenter.org/wp-content/uploads/2025/11/ICLE-Moores-Law-FINAL-2025-11-12-1.pdf>.

¹⁷ Albrecht, Manne, Teece, and Zúñiga. “From Moore's Law to Market Rivalry”.



The durability of this chokepoint reflects immense technological and economic barriers to entry. Each EUV system costs over \$200mn and consists of highly complex, precision-engineered components, with no commercially credible alternative supplier. This dynamic reinforces the logic of Moore's Law: as performance improves, the cost and complexity of production rise sharply, making replication increasingly difficult. Even leading firms struggle to remain competitive—Intel's delays at the 10nm node in the late 2010s illustrate the difficulty of scaling advanced manufacturing processes and transitioning to next-generation lithography. Yet, at the same time, ASML's annual production cap of roughly 40-50 EUV systems effectively dictates the maximum possible scale of global AI compute, as approximately 3.5 EUV tools are required to produce one gigawatt of AI data center capacity.

At the downstream end, the assembly, testing, and packaging stage involves preparing finished chips for integration into final products. While less technologically complex, this stage is essential to the functioning of the broader ecosystem. Geographically, this stage is heavily concentrated in Asia, particularly in China and Southeast Asia. China alone accounts for roughly 30% of global assembly, testing, and packaging capacity, compared to just 4% in the US, according to the US-China Economic and Security Review Commission (USCC).¹⁸

Because these stages are rarely vertically integrated, production is typically split across firms. A fabless company designs the chip, a foundry manufactures it, and third-party providers handle the assembly, testing, and packaging. This has created a system in which different countries control different chokepoints: the US leads in design and software, Taiwan and South Korea in fabrication, and China in downstream manufacturing. Each stage is essential and difficult to replicate, making the system highly interdependent yet strategically fragile.

Figure 2: The global chip supply chain

Design	Stage 1: Intellectual property Design and license pre-designed components (IP cores) like processors, memory controllers, and interface protocols	Dominant players Arm Holdings, Synopsys, Cadence Design Systems, Imagination Technologies
	Stage 2: Electronic design automation EDA companies provide software tools, techniques, and methodologies for designing and producing electronic systems before physical manufacturing	Dominant players Cadence, Synopsys, Siemens, ANSYS, Keysight Technologies
Manufacturing	Stage 3: Fabless Fabless companies focus solely on the design and development of semiconductor chips, outsourcing the actual manufacturing to foundries	Dominant players NVIDIA, Qualcomm, Broadcom, AMD, Apple
	Stage 4: Foundries Specialised factories that manufacture chips for design firms. They turn the designs into physical chips and are a crucial part of the technology supply chain	Dominant players TSMC, UMC, Samsung, GlobalFoundries, SMIC
	Stage 5: Semi manufacturing equipment Produce the equipment crucial for the fabrication and testing of semiconductor devices	Dominant players Applied Materials, ASML, Lam Research, Tokyo Electron, KLA
Assembly, testing & packaging	Stage 6: Integrated device manufacturer Handle both the design and production of chips in-house, owning and operating their manufacturing facilities (fabs)	Dominant players Intel, Samsung Electronics, SK Hynix, STM, Micron
	Stage 7: Outsourced semiconductor assembly & test Specialise in providing third-party packaging and testing services for semiconductor manufacturers	Dominant players ASE Tech, Amkor Tech, JCET Group, Powertech, Tongfu Microelectronics

China US Taiwan Europe South Korea Japan

Source: Deutsche Bank Research.

¹⁸ "Chapter 9: Chained to China: Beijing's Weaponization of Supply Chains," in 2025 *Annual Report to Congress*, U.S.-China Economic and Security Review Commission, n.d.



2.2 A bifurcated industry: Advanced vs. legacy chips

Adding another layer to an already complex supply chain, semiconductors are also differentiated by their manufacturing process nodes, measured in nanometers (nm). These nodes reflect a chip's performance, efficiency, and technological sophistication. As the industry evolves, these nodes have effectively split the market into two parallel segments:¹⁹

- Advanced chips ($\leq 7\text{nm}$) → high-value, technologically complex, geopolitically sensitive
- Legacy chips ($\geq 22\text{nm}$) → high-volume, industrially essential, more geographically diffuse

In this age of infrastructure realism, control over advanced nodes provides technological leadership while dominance in legacy nodes secures a critical position in global manufacturing supply chains. At present, the US leads in advanced chip design and innovation, while China has built scale and dominance in legacy manufacturing. We delve into how the two countries have attained advantages below.

Advanced chips ($\leq 7\text{nm}$): The AI frontier

Advanced and leading-edge nodes—roughly 7nm and below—represent the technological frontier of semiconductor manufacturing. These chips power high-performance computing applications, including data centres, premium consumer devices, and artificial intelligence workloads, forming the backbone of modern digital infrastructure and cloud-based services.

The advent of AI has elevated the importance of advanced chips, transforming them from a market input into a strategic asset and a geopolitical chokepoint in recent years. Despite representing a small share of total semiconductor volumes, AI chips now drive a disproportionate share of industry growth, accounting for approximately 55% of the total chip revenue market by 2030 according to TSMC.²⁰

Thanks to its superiority in advanced compute, the US maintains a clear lead in AI. However, DeepSeek's launch in January 2025 challenged this perception and triggered a sharp ~17% decline in Nvidia's share price.²¹ This is because production of advanced and leading-edge chips is extremely capital-intensive and technologically complex. Thus, China's reveal of a high-performing and lower-cost model reportedly trained on less advanced hardware was enough to spook investors that demand for their increasingly more costly hardware would taper off.

Despite the DeepSeek scare, China's AI capabilities remain constrained by access to advanced semiconductors. DeepSeek itself relied on Nvidia H800 chips—which are subject to US export controls but still based on US architecture—to train its models, reportedly using 2,788,000 GPU-hours (equivalent to around \$5.6mn, assuming \$2 per GPU-hour).²² Investigations by CSIS also suggest that restricted H100 chips may also have been accessed indirectly by DeepSeek to train its model.²³ According to Epoch AI, Chinese large language models (LLMs) lag their US counterparts by approximately seven months.²⁴

¹⁹ These categories (referring to chip nodes/process nodes) are a subjective definition and reflect our views, not a standard defined by the industry.

²⁰ Company, Manufacturing. 1970. "2026 Technology Symposium Highlights." Tsmc.com. Taiwan Semiconductor Manufacturing Company. 1970. https://www.tsmc.com/english/symposium_highlights/2026.

²¹ According to Nvidia CEO Jensen Huang, AI competition is a "five-layer cake": power, chips, infrastructure, models, applications. The US leads in chip design (Nvidia), infrastructure (hyperscalers) and frontier models (Anthropic, OpenAI), while China leads in efficiency and open-source.

²² GPU rental prices were based off CSIS calculations from 2025; rental prices for GPUs per hour have since grown exponentially as of May 2026, ranging from \$3.50-\$10 GPU/hour. Gregory C. Allen, "DeepSeek, Huawei, Export Controls, and the Future of the U.S.-China AI Race," Center for Strategic and International Studies, 2025, <https://www.csis.org/analysis/deepseek-huawei-export-controls-and-future-us-china-ai-race>.

²³ Allen, "DeepSeek, Huawei, Export Controls."

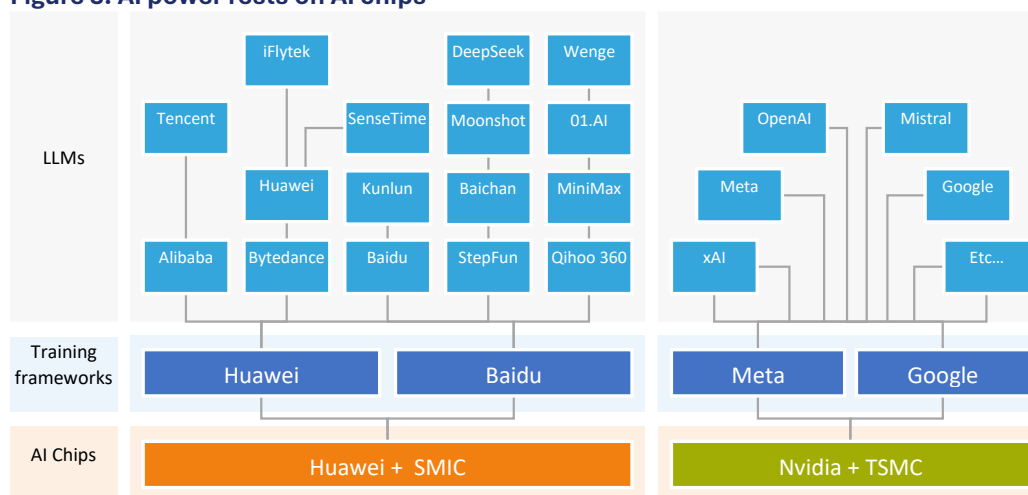
²⁴ Luke Emberson, "Chinese AI Models Have Lagged the US Frontier by 7 Months on Average since 2023," Epoch AI, 2023, <https://epoch.ai/data-insights/us-vs-china-eci/>.

The broader issue is that China remains compute-constrained to produce any of its own leading-edge chips. A key bottleneck lies in high-bandwidth memory (HBM), which provides the storage within AI chips that can help train and serve large AI models. HBM requires 4x the wafer space of regular memory chips, and according to Epoch AI, AI chips already consumed more than 90% of the HBM industry's output in terms of revenue in 2025.²⁵ And feeding the AI boom will require even greater dependence on ASML, which China does not have access to, as HBM is built from DRAM wafers produced with its EUV tools – which, as mentioned earlier, have a finite annual production capacity. While China's ChangXin Memory Technologies (CXMT) is making progress, global leaders like SK Hynix, Samsung, and Micron remain more than two generations ahead.

In addition, Chinese chip ecosystems are plagued by weaker software integration and stability issues. Domestic alternatives from firms like Huawei and Cambricon are less competitive, with some estimates suggesting performance levels of around 80% of Nvidia's A100.²⁶ As a result, China's AI development continues to depend on Western hardware and architectures. In contrast, the US is buoyed by massive capital investment from its hyperscalers, absolute dominance in frontier software models, and a top-tier global AI research talent pool.

Finally, export controls implemented by the US and its allies on advanced semiconductor equipment—particularly EUV lithography tools—also continue to limit China's ability to scale leading-edge production. Under Moore's Law, the compute requirements for frontier AI models continue to rise rapidly, further widening this gap.

Figure 3: AI power rests on AI chips



Source: Mercator Institute for China Studies, Deutsche Bank Research.²⁷

Legacy / mature chips ($\geq 22\text{nm}$): Scale and industrial resilience

In contrast, legacy and mature nodes—typically $\geq 22\text{nm}$ —form the backbone of the physical economy. These chips are widely used in automotive systems, power management, sensors, microcontrollers, and industrial machinery, as well as consumer

²⁵ "Total AI Chip Memory Bandwidth Has Grown 4.1x per Year, Now Reaching 70 Million Terabytes per Second," Epoch AI, March 24, 2026, <https://epoch.ai/data-insights/hbm-shipped/>.

²⁶ Chris McGuire, "China's AI Chip Deficit: Why Huawei Can't Catch Nvidia and U.S. Export Controls Should Remain," Council on Foreign Relations, December 15, 2025, <https://www.cfr.org/articles/chinas-ai-chip-deficit-why-huawei-cant-catch-nvidia-and-us-export-controls-should-remain>.

²⁷ Wendy Chang, Rebecca Arcesati, and Antonia Hmaid, "MERICS Report: China's Drive Toward Self-Reliance in Artificial Intelligence from Chips to Large Language Models," July 2025, https://merics.org/sites/default/files/2025-07/MERICS%20Report-AI_Stack_final.pdf.

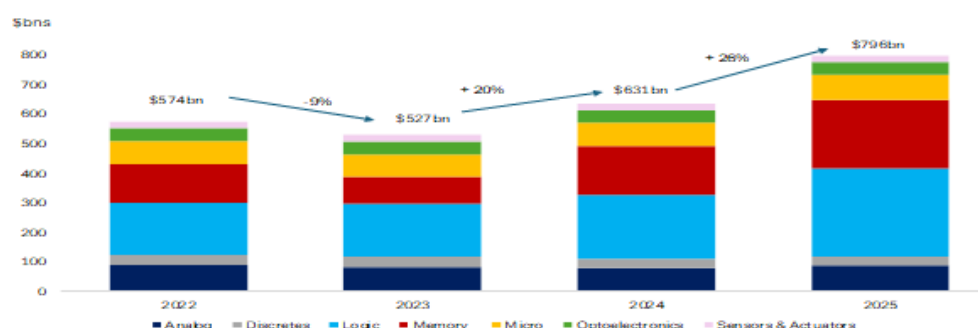
electronics and defence applications.²⁸ They are characterized by lower production costs, high reliability, and long product life cycles.

Within this category, mature nodes represent a critical middle layer, balancing performance and cost. These chips are prevalent across automotive and industrial applications, as well as mainstream consumer electronics. Production is widely distributed across global players such as TSMC, UMC, SMIC, and GlobalFoundries.

China has emerged as a dominant player in this segment. Since 2014, it has been able to significantly expand its manufacturing capacity through sustained state subsidies, particularly via firms such as SMIC, Hua Hong, and Nexchip, and by focusing on mature-node manufacturing production, where it possesses the necessary expertise and equipment.²⁹ According to the Semiconductor Industry Association (SIA), China is projected to account for roughly 35–40% of global mature-node capacity by 2027.³⁰

This expansion is already reshaping global markets. Increased Chinese supply has contributed to pricing pressure on international foundries, and some global firms are increasingly sourcing from Chinese manufacturers due to cost advantages—for example, in PC supply chains involving companies such as HP, Dell, and ASUS.³¹

Figure 4: Semiconductor shipments by segment (\$ bn)



Source: Deutsche Bank Research, World Semiconductor Trade statistics.

3. The geopolitical chessboard: Taiwan's central role

No discussion of the US-China chip competition is complete without spotlighting Taiwan's pivotal role in the race. Taiwan produces over 90% of the world's most advanced semiconductors and roughly around 55% of the world's semiconductors, making it a critical chokepoint in the global technology supply chain. US Commerce Secretary Howard Lutnick has vowed to reduce dependence on Taiwan's semiconductors as quickly and as much as possible, calling for up to 40% of Taiwan's-linked semiconductor supply chains to eventually be reshored to the US.³²

3.1 Global dependence on a single island

Taiwan is a high-tech economy deeply embedded in global supply chains. Exports account for roughly 70% of its GDP, with semiconductors acting as the primary engine of growth.³³

²⁸ Process nodes refer to the feature size of the transistors on a semiconductor. Advanced logic chips are made below the 22-nm threshold, where node shrinkage generally yields faster and more efficient chips but requires more complex manufacturing processes. Within advanced, cutting-edge chips use 3-nm and 5-nm process modes.

²⁹ "Public Report on the Use of Mature-Node Semiconductors," Bureau of Industry and Security, December 2024, <https://www.bis.gov/media/documents/public-report-use-mature-node-semiconductors-december-2024.pdf>.

³⁰ "Winning the Chip Race," Semiconductor Industry Association, 2025.

³¹ Lauly Li and Cheng Ting-Fang, "HP, Dell, Acer and Asus Mull Using Chinese Memory Chips amid Supply Crunch," *Nikkei Asia*, February 5, 2026, https://asia.nikkei.com/spotlight/supply-chain/hp-dell-acer-and-asus-mull-using-chinese-memory-chips-amid-supply-crunch?utm_source=substack&utm_medium=email.

³² Dylan Butts, "Impossible": Taiwan Pushes Back against Washington's 40% Chip Supply Relocation Goal," *CNBC*, February 10, 2026, <https://www.cnbc.com/2026/02/10/taiwan-chips-us-supply-chain-lutnick-trade-deal.html>.

³³ Calculated using Haver data.



At the centre of this ecosystem is Taiwan Semiconductor Manufacturing Company (TSMC), which alone produces 90% of the world's most advanced chips.

The US remains heavily reliant on Taiwan for chips, despite its ongoing efforts to build domestic capacity through the 2022 CHIPS and Science Act and by applying political pressure on TSMC to expand its US footprint. In 2025, 70% of TSMC's revenue came from the US, compared to 10-15% from China.³⁴

Trade flows underscore this dependence. In 2025, Taiwan was the US's fifth-largest trading partner and fifth-largest source of imports, resulting in a US trade deficit with Taiwan of \$147bn.³⁵ This reliance is particularly acute in the AI sector. Nvidia – now the world's most valuable company with a market cap that exceeds the entire stock market value of several G7 nations – depends almost entirely on TSMC to produce its advanced GPU and AI chips. These chips account for most of Nvidia's data centre revenue, its fastest-growing segment.

At the same time, Taiwan remains intrinsically linked to China. The country is Taiwan's largest investment and trading partner, even though exports to China have halved in the last few years, from 42% in 2021 to 23% in 2025.³⁶ While Taiwan has sought to decouple from China, the latter remains the largest single destination for Taiwan's chip exports by value. This is because China serves as the world's final-assembly hub, integrating these components into electronics that are then re-exported globally, tying China's manufacturing ecosystem inextricably to Taiwan's foundries.

3.2 How did Taiwan (and TSMC) become so powerful?

Today, TSMC is Taiwan's prized industrial jewel, owing much of its success to an early-mover advantage in semiconductor manufacturing in the late 1980s. Its rise was supported by close ties to Silicon Valley – many of its first engineers were trained in the US – as well as significant Taiwan government funding, which provided almost half of its initial capital.³⁷ This occurred as US power reached its post-Cold War peak, championing a globalized order built on the belief that expanding trade and integrated supply chains promoted peace and economic prosperity. The resulting mutual interdependence between TSMC and Silicon Valley was the best example of this. As a result, officials from both the public and private sectors saw no issue in increasingly outsourcing their chip designs to fab companies abroad such as TSMC.

Crucially, TSMC pioneered the pure-play foundry model. By manufacturing chips designed by external customers, it enabled the rise of the US's "fabless" semiconductor giants, such as Qualcomm, Nvidia and Apple. This allowed these companies to focus on high-value design while outsourcing capital-intensive manufacturing, a division of labor that fundamentally reshaped the global semiconductor industry.

TSMC also distinguished itself through a willingness to take calculated technological risks. Its pivotal partnership with Apple in the early 2000s – after other firms had declined to manufacture chips for the first iPhone – helped cement its position as a leading-edge producer. TSMC was also an early and committed adopter of extreme ultraviolet (EUV) lithography in the 2010s, despite significant uncertainty surrounding the technology at the time. More recently, it correctly anticipated the need to expand advanced packaging capacity to support Nvidia's rapid rise in the AI chip market.

The ability to "anticipate technological inflection points" has been central to TSMC's success.³⁸ Its premier customer base has enabled it to continuously invest in increasingly advanced nodes, despite the exponentially rising costs associated with each new

³⁴ "TSMC Annual Report 2024," Taiwan Semiconductor Manufacturing Company, 2025, <https://investor.tsmc.com/sites/ir/annualreport/2024/2024%20Annual%20Report-E.pdf>.

³⁵ Karen M. Sutter, "U.S.-Taiwan Trade and Economic Relations," CRS In Focus IF10256 (Washington, DC: Congressional Research Service, 2026), <https://www.congress.gov/crs-product/IF10256?h=semiconductors&s=1&r=1>.

³⁶ Sutter, "U.S.-Taiwan Trade."

³⁷ Chris Miller, *Chip War: The Fight for the World's Most Critical Technology* (New York: Simon and Schuster, 2022).

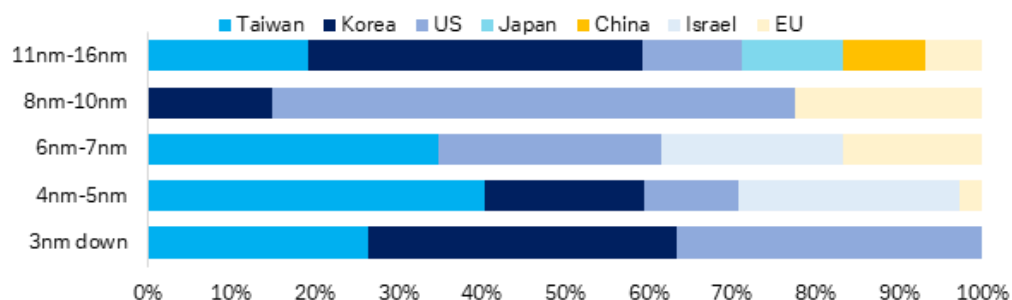
³⁸ Albrecht, Manne, Teece, and Zúñiga, "From Moore's Law to Market Rivalry."



generation of semiconductor manufacturing. As a result, TSMC recently joined the elite group of nine companies globally that enjoy a market cap above one trillion dollars – and this despite the tariff-related uncertainty stemming from the US's 15% tariff levied recently against Taiwan.

Finally, Taiwan's semiconductor strength extends beyond TSMC. Other major local players include United Microelectronics Corporation (UMC), Nanya Technology Corporation, and Winbond Electronics, which rank among the world's leading manufacturers of memory and specialty chips.

Figure 5: High-end node fabrication capacity by region/island (nanometres)



Source: Deutsche Bank Research, Bloomberg Finance LP (2021 data). Estimates are based on the machinery installed, and refer to theoretical capacity, not what has been commercially produced. The data also refers to all semiconductor types.

3.3 Mutually assured dependence

Despite its market dominance, TSMC does not operate independently of the broader semiconductor ecosystem. Instead, it is deeply embedded in a system of mutual reliance. The TSMC-Apple relationship illustrates this dynamic best: TSMC depends on large, predictable orders from Apple to justify its massive capital expenditures, while Apple depends on TSMC's cutting edge manufacturing capabilities to maintain its technological edge. TSMC has replicated this model with other major clients, including Nvidia, AMD, and Broadcom. This symbiotic relationship underscores the Taiwan and US relationship of mutually assured dependence, where the fortunes of both are so closely intertwined that the relationship itself has increasingly become a source of strategic vulnerability.³⁹

Historically, TSMC has also acted as a bottleneck that has kept the China/US competition asymmetric. Until 2020, Huawei was TSMC's second-largest customer, relying on it for advanced chip fabrication. However, this shifted following the US expansion of the Foreign Direct Product Rule (FDPR) in May 2020 – an extraterritorial measure that restricted any company using US-origin technology from supplying Huawei. Given TSMC's reliance on US semiconductor equipment and design software, the company determined it was required to comply with the new regulation, leading it to halt shipments to Huawei by September 2020. As a result, Huawei's sales collapsed, declining 30% in H1 2021, forcing it to stockpile aggressively and sell off divisions to conserve its chips for strategic priorities.⁴⁰

The rapid growth of AI has only reinforced this dependence, increasing demand for advanced chips and further embedding Taiwan within US supply chains. With TSMC holding a monopoly on the most advanced sub-3nm nodes, Taiwan effectively dictates the extent of the US's AI progress and, by extension, its national security strategies. Despite the more isolationist rhetoric of the second Trump administration, Taiwan's share of US high-tech imports has continued to rise, underscoring the structural nature of this reliance.

³⁹ Albrecht, Manne, Teece, and Zúñiga, "From Moore's Law to Market Rivalry."

⁴⁰ BBC News. 2021. "Huawei Revenue Sees Biggest Ever Fall," August 6, 2021, sec. Technology. <https://www.bbc.co.uk/news/technology-58104614>.



TSMC's early conservatism in scaling up for AI growth also created a new bottleneck, with forecasts suggesting advanced compute may not meet AI-driven demand until 2028 or 2029.⁴¹ Notably, the company chose to keep its capex relatively flat during 2023 and 2024, just as the AI boom was accelerating post-ChatGPT.⁴² Given the 2-3 year lead time required to build a new semiconductor fab, TSMC's conservatism has created a significant supply/demand gap that is not expected to narrow significantly over the next couple of years. This constraint has been noted in recent earnings calls by leading hyperscalers and has incentivized both the US government and the private sector to invest in domestic foundry competition, particularly through Intel and Samsung in an attempt to diversify.

Finally, TSMC remains embedded in China's supply chains. Although the company has attempted to move away from its reliance on mainland China, it still reports that 9-10% of its revenue comes from Chinese customers. Furthermore, exposure remains high, as overall non-US revenue is around 80%.⁴³

Ultimately, while the US still leads the global technology race, the TSMC chokepoint means that its leadership is defined more by resource management than absolute abundance. This dynamic critically impacts US strategy toward China.

3.4 Protecting the silicon shield

TSMC also acts as a bottleneck to the US's AI progress. This is because AI has a critical physical dependency – specifically, on the fabrication plants of TSMC in Taiwan.⁴⁴ This deep reliance on Taiwan's semiconductor supply suggests that the US might hesitate to relinquish its security commitment to Taiwan, even amid efforts to stabilize relations with China.

The economic consequence of any disruption to Taiwan's semiconductor supply could be severe. A blockade or conflict scenario could trigger a global shortage of advanced chips, effectively paralyzing large parts of the global economy. Bloomberg Economics estimates that a Chinese blockade of Taiwan could result in \$5trn in economic losses in the first year alone.⁴⁵ Similarly, the Semiconductor Industry Association warns that a complete disruption could lead to the "largest economic crisis since the Great Depression," with US GDP potentially declining by 11% and output decreasing by 16%.⁴⁶

Given these stakes, the ongoing tensions between China and Taiwan regarding Taiwan's sovereignty have heightened global concerns about supply chain security. In response, the US, Japan, and the EU have sought to diversify semiconductor manufacturing capacity. However, replicating Taiwan's ecosystem would take years, if not decades, and would come at a significantly higher cost.

Ultimately, as the sole provider of the most advanced nodes, TSMC's manufacturing capacity and its geographic safety in Taiwan plays a key role in the limits of US national security power and AI progress.

⁴¹ The extent of TSMC's hegemony in the chip ecosystem was spotlighted in August 2025, when news broke that several employees had stolen critical information on TSMC's 2-nanometre chip technology, marking the first-ever case involving national core sensitive technologies to be investigated under Taiwan's National Security Act, enacted in 2022 to protect its chip industry from foreign threats. See: Cheng Ting-Fang and Lauly Li, "Taiwan Indicts Trio for Alleged Theft of TSMC's 2-Nm Trade Secrets," *Nikkei Asia*, August 28, 2025, <https://asia.nikkei.com/business/technology/taiwan-indicts-trio-for-alleged-theft-of-tsmc-s-2-nm-trade-secrets>.

⁴² Ben Thompson, "TSMC Earnings, the TSMC Brake Revisited, Why AI Needs Foundry Competition," *Stratechery*, January 21, 2026, <https://stratechery.com/2026/tsmc-earnings-the-tsmc-brake-revisited-why-ai-needs-foundry-competition/>.

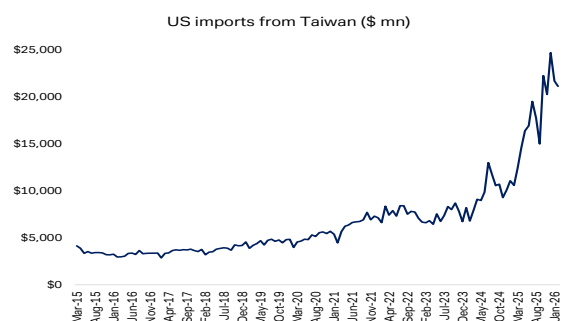
⁴³ TSMC 2025 annual report.

⁴⁴ Thompson, "TSMC Earnings."

⁴⁵ Chris Anstey, "Bloomberg New Economy: China 'Blockade Simulation' Raises \$5 Trillion Risk," Bloomberg.com, May 25, 2024, <https://www.bloomberg.com/news/newsletters/2024-05-25/bloomberg-new-economy-china-blockade-simulation-raises-5-trillion-risk>.

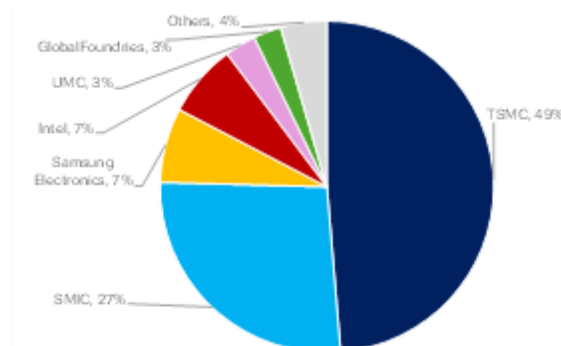
⁴⁶ Mickle, "The Looming Taiwan Chip Disaster."

Figure 6: The AI race has fuelled chip demand, pushing up Taiwan's share of US imports over the past few years



Source: Census Bureau, Haver Analytics.

Figure 7: TSMC continues to dominate in foundry capacity, keeping a cap on global AI progress



Source: Bloomberg Finance LP, Deutsche Bank Research.

4. US vs. China: A shifting balance

The strategic vulnerability posed by TSMC illustrates a broader structural weakness in the US semiconductor ecosystem. Once the undisputed pioneer in the chip race, today's US companies are dependent on global partners for manufacturing, assembly and packaging. The decline is stark: in 1990, the US accounted for nearly 40% of global chip production but by 2019, that share had plummeted to just ~11%.⁴⁷

4.1 The US response: Rebuilding domestic capacity

Since 2022, successive US administrations have sought to strengthen their semiconductor supply chains. The most significant effort thus far has been the ~\$278bn CHIPS and Science Act, a landmark piece of legislation designed to bolster domestic chip capabilities via direct subsidies and tax credits for manufacturing. As of December 2025, the Department of Commerce has awarded ~\$30.9bn across 40 projects to 19 companies.⁴⁸ These include massive, multi-billion-dollar packages to foreign companies like TSMC (\$6.6bn), Samsung (\$4.75bn) and Intel (\$7.86bn), all part of the US government's strategy to onshore advanced chip manufacturing.

Under CHIPS, advanced semiconductors are now being made in the US for the first time in over a decade at TSMC's new Arizona fab. The core theory behind this is that if the US could onshore anchor tenants like TSMC, Samsung in Austin, Intel in Arizona, and SK Hynix in Indiana, a domestic ecosystem of suppliers would naturally follow. While the CHIPS Act represented the first clear strategic vision and effort at rebuilding the US's domestic chip supply chains, its success in weakening China's long-term industrial power and strengthening its own supply-chain has been mixed.

First, the \$278bn funding, while substantial, could not outcompete Moore's Law, and proved insufficient for total de-risking. Within the overall funding for example, the \$39bn allocated to revive domestic semiconductor manufacturing pales in comparison to China's own state-led spending, which totals around \$142bn – roughly 3x the total funding the US has allocated towards direct industrial support.⁴⁹ Second, the Act's limited resources led to the neglect of several critical downstream segments of the US supply

⁴⁷ Blevins, Platzer, and Sutter, "Semiconductors."

⁴⁸ "Semiconductors: Information on Projects Funded to Strengthen U.S. Supply Chain," U.S. Government Accountability Office, GAO-26-107882, December 11, 2025, <https://www.gao.gov/products/gao-26-107882>.

⁴⁹ [China has spent 3.6 times more than the US on chipmaking subsidies over the past decade — \\$142 billion and counting, easily outweighs CHIPS Act | Tom's Hardware](#)



chain system like conventional packaging and materials. Third, CHIPS has faced difficulty attracting suppliers. Outside of the giants like TSMC and Intel, smaller Asian suppliers have been hesitant to onshore operations until more fabs or sufficient volume exist to justify the shift from exporting from Taiwan.⁵⁰ Finally, the initiative suffered from a lack of formal coordination amongst officials to protect and promote the US chip supply chain, with few clear metrics for what “self-sufficiency” would look like.

The limited success of the US CHIPS Act’s implementation suggests the US can no longer scale industrial projects as rapidly as it did during the 20th century.⁵¹ Under the second Trump administration, the US has reportedly revisited some of the multi-billion-dollar contracts, although the CHIPS program office has been transferred to the newly created “US Investment Accelerator” within the Department of Commerce, tasked with fulfilling pre-existing awards. Interestingly, the administration has also expanded on what the CHIPS Act money is directed towards, with new investments in areas like critical minerals and materials now being considered.⁵²

How has the second Trump administration tried to solve US chip dependency on Taiwan? The slew of trade deals negotiated in 2025 all contain provisions requiring partner nations – including Japan, Malaysia, and Taiwan –to invest in core US industries such as semiconductors. The January 2026 US-Taiwan Agreement on Trade and Investment is particularly notable for its focus on semiconductors.⁵³ Among the provisions, Taiwanese chip producers that invest in the US will get a cut on tariff imports of up to 2.5x the planned capacity. US Secretary of Commerce Howard Lutnick has said that the administration’s objective is to bring 40% of Taiwan’s chip supply chain and production onshore.⁵⁴

This strategy has been backed by some dramatic interventions regarding tariffs. For example, in 2025 President Trump threatened TSMC with tariffs of up to 100% if it did not expand its US presence.⁵⁵ In return, the company pledged a total \$165bn investment in the US as it sought to quell accusations that Taiwan was undermining the US semiconductor business.⁵⁶

Leading US companies are also attempting to address Taiwan’s bottleneck in AI chips. Nvidia, for instance, is transitioning from a monolithic GPU to a “multi-architectural” approach, designing multiple specialized chips to handle different AI workloads.⁵⁷ In doing so, Nvidia aims to ensure that no component of the AI compute stack becomes a bottleneck that leaves expensive GPUs idle.

4.2 Cutting off the chokepoints

In addition to the domestic measures taken to strengthen its chip resilience, since 2010 the US has also implemented a series of export controls designed to manage the flow of semiconductor technology to China. Its measures on China were initially focused on specific companies like Huawei and SMIC. As trade tensions and allegations of espionage and intellectual property theft grew, the Department of Commerce added Huawei to its

⁵⁰ Jordan Schneider and Phoebe Chow, “The Future of Economic Security with Dan Kim and Chris Miller,” *ChinaTalk* (podcast), January 21, 2026, <https://www.chinatalk.media/p/the-future-of-economic-security-with->

⁵¹ Trump has also turned to domestic efforts like The Pax Silicon Fund, a US-led multilateral coalition that includes an investment fund targeting \$4trn in funding for energy projects, critical minerals and semiconductor supply chains. The US has announced it will contribute \$250mn toward the fund.

⁵² “Tracking CHIPS and Science Act Awards,” *Manufacturing Dive*, December 5, 2024, updated March 6 2026, <https://www.manufacturingdive.com/news/chips-and-science-act-tracker-semiconductor-manufacturing/734039/>.

⁵³ “Fact Sheet: Restoring American Semiconductor Manufacturing Leadership through an Agreement on Trade & Investment with Taiwan,” U.S. Department of Commerce, Office of Public Affairs, January 15, 2026. <https://www.commerce.gov/news/fact-sheets/2026/01/fact-sheet-restoring-american-semiconductor-manufacturing-leadership>.

⁵⁴ “Howard Lutnick Breaks down Recent \$250B Trade Deal with Taiwan,” CNBC Television, YouTube, January 15, 2026, <https://www.youtube.com/watch?v=cSoPNbdLLTg>.

⁵⁵ “Trump Says He Told TSMC It Would Pay 100% Tax If It Doesn’t Build in US,” *Reuters*, April 9, 2025, <https://www.reuters.com/world/us/trump-says-he-told-tsmc-it-would-pay-100-tax-if-it-doesnt-build-us-2025-04-09/>.

⁵⁶ “TSMC Intends to Expand Its Investment in the United States to US\$165 Billion to Power the Future of AI,” Taiwan Semiconductor Manufacturing Company Limited, March 4, 2025, <https://prtsmc.com/english/news/3210>.

⁵⁷ Pratik Kumar Patel, “Optimize Data Center Efficiency for AI and HPC Workloads with Power Profiles,” NVIDIA Technical Blog, December 4, 2025, <https://developer.nvidia.com/blog/optimize-data-center-efficiency-for-ai-and-hpc-workloads-with-power-profiles/>.



entity list in 2019, which prevented the company from obtaining advanced chips developed or produced in the US. In 2020, Commerce added SMIC to the entity list.⁵⁸ The regulations concerning Huawei were further expanded that year when the US modified its Foreign Direct Product Rule (FDPR) to prevent foreign fabs like TSMC from supplying certain chips to Huawei and other listed entities without US authorization. The focus on specific firms like Huawei was intended to limit China's influence at the end-user stage of the supply chain, where the country held a leading role in global markets.

Washington has been able to exert this pressure because, as we noted earlier, US companies dominate upstream segments of the chip supply chain, including fabless design, advanced electronic design automation (EDA) software, and critical IP. This dominance underpins the “extraterritorial” reach of US export controls, where any chip made with US tools or IP can be legally restricted from being sold to Chinese companies like Huawei.⁵⁹

The FDPR's 'extraterritorial' reach created a clear choice for global foundries in 2020: they could either continue using US technology or continue supplying designated entities like Huawei, but not both. As TSMC's operations relied on US software and machine tools/equipment to power its foundries (illustrating the “mutually assured dependence” relationship introduced earlier), the company suspended production of Huawei's Kirin chips, and related services from companies like Google were withdrawn. As a result, Huawei's smartphone business—once shipping >200m phones/year—collapsed in the high-end segment.

Washington has also implemented measures directed at the manufacturing segment of China's semis supply chain. In 2019, the US put restrictions in place to prevent ASML from selling advanced EUV equipment to China, further tightening these in 2023 to include the selling of advanced EUV and related components to China. The US also added SMIC to the entity list and restricted its access to tools that could create 10nm and below chips in 2020.

As the global chip industry recuperated from the pandemic, US export control policy shifted more broadly to address China's development of cutting-edge chips. In a landmark ruling in October 2022, the US Department of Commerce imposed controls on exports to China of advanced logic AI chips below defined performance thresholds of 16nm/14nm.⁶⁰

Commerce's rules also restricted the country's chief chip designer Nvidia from selling its flagship A100 and H100 GPUs to China. In response, Nvidia developed modified A800 and H800 chips for the Chinese market, but these too were later controlled by the Department of Commerce's Bureau of Industry and Security (BIS) in November 2023.⁶¹ Nvidia's subsequent H20 chip, designed to comply with the updated control thresholds, was a source of contention between the company and the US government until August 2025, when the government approved its sale to China in return for 15% of sales revenue from Nvidia.

As the AI race gathered steam, Japan and the Netherlands joined the US in restricting the export of advanced equipment to China in January 2023.⁶² Other successive actions during this period include the addition of 140 more Chinese companies to the Entity List in

⁵⁸ “Commerce Department Further Restricts Huawei Access to U.S. Technology and Adds Another 38 Affiliates to the Entity List,” U.S. Department of Commerce, August 17, 2020, <https://2017-2021.commerce.gov/news/press-releases/2020/08/commerce-department-further-restricts-huawei-access-us-technology-and.html>.

⁵⁹ Marion Labouré and Cassidy-Ainsworth Grace, “Thematic Research: CHARTBOOK - Tangled Wires: The Geopolitical Landscape of Semiconductors & Rare Earth Metals,” Deutsche Bank Research, June 2023, https://research.db.com/research/Article?rid=e5a16334_9484_490f_9217_7ab2649de709_604&id=RP0001&wt_cc1=IND-3146993-0000.

⁶⁰ Demetri Sevastopulo and Kathrin Hille, “US Hits China with Sweeping Tech Export Controls,” *Financial Times*, October 7, 2022, <https://www.ft.com/content/6825bee4-52a7-4c86-b1aa-31c100708c3e?syn-25a6b1a6=1>.

⁶¹ Karen M. Sutter, 2025, “U.S. Export Controls and China: Advanced Semiconductors,” CRS Report R48642 (Washington, DC: Congressional Research Service, 2025), <https://www.congress.gov/crs-product/R48642>.

⁶² Demetri Sevastopulo and Sam Fleming, “Netherlands and Japan Join US in Restricting Chip Exports to China,” *Financial Times*, January 28, 2023, <https://www.ft.com/content/baa27f42-0557-4377-839b-a4f4524cfa20?syn-25a6b1a6=1>.



December 2024, the expansion of the FDPR to include High-Bandwidth Memory (HBM), and further controls on advanced SME equipment for leading-edge logic and memory nodes.

Figure 8: Nvidia's modified chips for China have evolved along with the US government's policies

Date	US Bureau of Industry and Security (BIS) export controls actions	Nvidia's modified chips and related responses
Oct-2022	A100 and H100 chip export controls	Announced A800 and H800 modified chips for China
Nov-2023	A800 and H800 export controls	Announced H20, L20 and L2 modified chips for China
May-2025	Controlled the H20	Announced efforts for B30 and B40 for China
Jul-2025	Grants export licenses for H20 sales in China	China's Cybersecurity administration expressed concerns regarding H20
Dec-2025	Approved export licenses on a case-by-case basis for Nvidia's H200 and AMD's M1235X AI logic devices	Chinese customs officials initially said H200 chips were "not permitted" to enter China, later loosened to accept limited imports. Restrictions by both governments (the Trump admin has also placed a 25% tariff on H200 chip imports, which is deterring Nvidia

Source: Deutsche Bank Research, Congressional US reports, CSIS.

4.3 The Trump administration and the turning point

The current US policy towards China on semiconductors has become more uncertain and, at times, appears contradictory. The Trump administration has not sought to unilaterally deny China access to technology in the same manner as the 2022–24 framework under the Biden administration. Instead, its approach has involved a partial relaxation of chip controls, seemingly in favour of monetizing and re-politicizing the chokepoints.

The biggest demonstration of this shift occurred in December 2025, when the Trump administration granted permission for Nvidia to sell its second-best AI chip, the H200, to China in exchange for a 25% share in revenue. This was a notable step away from the 15% cut in the less-advanced H20 chip deal in August. Still, the precise mechanism – whether a security check before being shipped to China, followed by a tariff or another process to collect the 25% – remains unclear. Following this decision, in early 2026, Trump imposed a 25% tariff on advanced chips with total processing performance of less than 21,000 and total DRAM bandwidth of less than 6,500 GBs, and their spin-off products used in AI applications. Nevertheless, these moves mark a departure from the hardline stance of previous administrations and are a dial-down from the 100% rate on chip imports that Trump first indicated the US would impose in August 2025.⁶³

The administration's more accommodative stance was further demonstrated in February, when the White House overruled a Department of Defense (DoD) recommendation to blacklist major Chinese firms, subsequently removing companies like Alibaba and Baidu from the sanctions list.⁶⁴

The Trump administration's seemingly contradictory policy can be understood as the result of several competing pressures. On one hand, some experts believe the policy change was driven by concerns over the declining effectiveness of export controls – evidenced by the rise of competitors like Huawei.⁶⁵ Others are of the opinion that President Trump is seeking diplomatic stability with Beijing in 2026, focusing on cooperation on issues like fentanyl and soybean purchases. A more prevalent view is that the administration is divided internally between a very hawkish camp that advocates for strict controls on China, versus the diffusionist camp, led by White House AI Czar David Sacks and the lobbying influence of Nvidia, who argue that, in order to promote US AI leadership, US tech standards should be diffused as quickly as possible before a Chinese

⁶³ Anthony Rapa and Rachel D Evans, "Trade Rules: Trump Administration Revises Export Restrictions on Advanced Semiconductors While Levying Additional Tariffs," *Reuters*, March 13, 2026, <https://www.reuters.com/legal/legalindustry/trade-rules-trump-administration-revises-export-restrictions-advanced-pracin-2026-03-13/>.

⁶⁴ Edwin Chan, "Alibaba Leads Tech Slide after Pentagon Briefly Shows Blacklist," *Bloomberg.com*, February 16, 2026, <https://www.bloomberg.com/news/articles/2026-02-16/alibaba-leads-tech-slide-after-pentagon-briefly-shows-blacklist>.

⁶⁵ Chris McGuire, "China's AI Chip Deficit."



competitor like Huawei can catch up and lead the world to build AI capabilities on the Huawei stack.⁶⁶

Once again, Taiwan stands as the central chokepoint. As the sole provider of the most advanced nodes, TSMC's manufacturing capacity and its geographic safety in Taiwan ultimately dictate the limits of US national security power and AI progress. The US cannot completely isolate China from the advanced chip supply when it does not control the primary manufacturing gateway. To maintain its lead, the US must ensure its tech stack diffuses globally to gain a competitive advantage before it can effectively exercise control, particularly as China excels in model training (like DeepSeek) and dominates the open-source landscape.

A final consideration in this strategic calculus is China's control over critical minerals, which provides the central government with significant counter-leverage. The retaliatory dynamic between the US and China on critical minerals and semiconductors has led some experts to conclude that China has a greater willingness to absorb economic pain to achieve strategic objectives than the US. By implementing its own export restrictions on rare earths – a move that was not without cost to its own economy – China demonstrated a capacity to retaliate, potentially influencing the US to de-escalate its own measures.⁶⁷

In this context, perhaps the only short-term solution is for the US to continue loosening its export controls on advanced chips to China, which the Trump administration is already doing. This decision has split stakeholders, with some corporate officials arguing that doing so is necessary to keep China dependent on the US AI stack, while other public and government officials warn that this approach provides Beijing with the tools to eventually outcompete the US in the global race for AI hegemony. For example, quantitative analysis from the Council on Foreign Relations (CFR) projects that if the US exports 3mn H200 chips to China in 2026, it would give China more AI computing power than it could produce domestically until 2028 or 2029 earliest.⁶⁸

As a result, this policy has faced backlash from congressional hawks who oppose the new AI chip licenses to China, with the Financial Times reporting that several bipartisan officials called on the Department of Commerce to suspend Nvidia's export licenses to China.⁶⁹ Congressional lawmakers are also looking to close loopholes in Trump's more relaxed posture towards Chinese tech via the proposed Remote Access Security Act, which will extend chip export controls to cover Chinese access to restricted compute via overseas cloud services.

A major proposal, the MATCH (Multilateral Alignment of Technology Controls on Hardware) Act, would allow the US to apply FDPR rule against allies if they do not match US chip export controls. This Act specifically targets technologies like ASML's EUV equipment and, if passed, could significantly increase the risk of Chinese retaliation. However, many investors we have spoken to see this as more of a strategic move than a real threat, fearing China would retaliate with rare earth export controls, as it did in May 2025 over US EDA restrictions.

⁶⁶ "Nvidia's Jensen Huang: 'China Is Going to Win the AI Race,' FT Reports," *Reuters*, November 6, 2025, <https://www.reuters.com/world/asia-pacific/nvidias-jensen-huang-says-china-will-win-ai-race-with-us-ft-reports-2025-11-05/>.

⁶⁷ Schneider and Chow, "The Future of Economic Security."

⁶⁸ McGuire, "China's AI Chip Deficit."

⁶⁹ Demetri Sevastopulo, "US Must Suspend Nvidia AI Chip Exports to China, Senators Say," *Financial Times*, March 24, 2026, <https://www.ft.com/content/556e534d-bbc5-46e0-8965-ec3a13a8871a?syn-25a6b1a6=1>.



5. China's response

5.1 Beijing's counter-leverage

Despite Western efforts to constrain China's chip self-sufficiency, China maintains sizable access to global computing power. While the precise amount is difficult to quantify, a bottom-up analysis from the think tank ChinaTalk estimates that the country has access to a computing capacity equivalent to more than one-eighth of the world's total (a median estimate of 2.8mn H100-equivalent GPUs).⁷⁰

Stockpiling and chip smuggling

Even before the US administration partially relaxed export controls on Nvidia's H200 chips, China was able to maintain some access to US advanced chips through various means.⁷¹ Until the end of 2023, China had powerful access to foreign AI chips, with estimates suggesting that legal purchases from foreign providers accounted for over 460,000 H100-equivalent GPUs.⁷² After the full crackdown on Nvidia's chip exports took effect in 2024, reports indicate that China began to acquire chips through unofficial channels, with the Center for a New American Security estimating that approximately 140,000 chips were smuggled into China that year alone, a figure that likely increased in 2025.⁷³ Significant stockpiling represents another strategy: CISI reported that Huawei was able to procure enough HBM chips, primarily via Samsung in South Korea, to support at least a full year's worth of production.⁷⁴

Finally, a substantial portion of China's computing capacity is reportedly accessed remotely via the cloud. According to ChinaTalk estimates, the median of this access is roughly 1,026,000 H100e, though it notes that the variability is high due to weak controls against remote access.⁷⁵ This would mean that the bulk of China's compute comes from its remote access.

These factors demonstrate that, even before the US government's approval of Nvidia H200 shipments in December 2025, China has proven its ability to grow, adapt and innovate using globally available technologies, a resilience underscored by Huawei achieving a record year in revenue in 2025, despite US restrictions.

Materials & market power

China has also leveraged its own structural advantages in the chip supply chain to maintain some concessions. As the world's largest electronics manufacturing hub, China consumed 31% of US chip sales in 2023.⁷⁶ Its dominance in the end-user market has made global chipmaking companies like Qualcomm, Micron, and TSMC reliant on China to buy their chips and assemble them into electronic end goods. With the country representing an estimated 35–40% of global end-demand for chips, the loss of access to this market can wipe out tens of billions of dollars from a chipmaker's valuation.

This dynamic was illustrated following the 2019 export controls on Huawei. As Huawei was forced to curtail its smartphone production, companies that supplied chip components to Huawei were hit by the decline in demand. The public backlash from industry leaders, combined with diplomatic negotiations between Trump and Xi, led the US to place a 90-

⁷⁰ "Compute" here means the number of chips within China plus the number of chips China can remotely access abroad. Aqib Zakaria, "How Much Compute Does China Have?" *Chinatalk*, April 7, 2026, <https://www.chinatalk.media/p/how-many-chips-does-china-have>.

⁷¹ One famous example being a former Micron employee, who stole intellectual property for China's UMC in 2020. See "Taiwan Company Pleads Guilty to Trade Secret Theft in Criminal Case Involving PRC State-Owned Company," US Department of Justice, October 28, 2020, <https://www.justice.gov/usao-ndca/pr/taiwan-company-pleads-guilty-trade-secret-theft-criminal-case-involving-prc-state-owned>.

⁷² Zakaria, "How Much Compute Does China Have?"

⁷³ Erich Grunewald and Tim Fist, "Countering AI Chip Smuggling Has Become a National Security Priority," Center for a New American Security, June 11, 2025, <https://www.cnas.org/publications/reports/countering-ai-chip-smuggling-has-become-a-national-security-priority>.

⁷⁴ McGuire, "China's AI Chip Deficit."

⁷⁵ Zakaria, "How Much Compute Does China Have?"

⁷⁶ "Winning the Chip Race," Semiconductor Industry Association, 2025.



day reprieve on Huawei.⁷⁷ At the same time, while Huawei's international phone sales were impacted by the controls, the company's 5G business was relatively unaffected as it was able to stockpile a year's worth of integrated circuits and other components. The Huawei incident demonstrates the difficulty of isolating a major Chinese firm so deeply integrated into the global supply chain ecosystem.

In another example of how deeply entwined one segment of the supply chain process is with another, Nvidia's "China-compliant" A800/H800 product line exists solely to keep some access to Chinese demand under US rules—an example of how corporate strategies are now being shaped by geopolitics. So, while the US seeks to keep China lagging, at the same time, it cannot leave it too far behind.

US export controls have also galvanized China's pursuit of domestic self-sufficiency. In response to the export controls announced in 2019, the CEO of Huawei's chip development arm, HiSilicon, noted that technologies long developed as "spare tires" had "become the main tires" overnight. This reflects a broader trend: increased US pressure has strengthened China's resolve to vertically integrate its domestic semiconductor industry. This ambition is apparent in the PRC's draft of its 15th five-year plan (2026-2030), which calls for "extraordinary measures" to achieve "full-chain decisive breakthroughs" across EDA software, manufacturing equipment, and advanced materials. The plan also acknowledges that, while China is good at quickly scaling up existing tech, it needs to improve domestic innovation in basic research and discovery. To support this, the government plans on establishing a national M&A fund to improve the exit environment for venture capital.

5.2 China remains compute-constrained

Despite its measurable success in countering certain US chip controls, China still lacks the critical technologies required to shore up its own domestic supply chains. The reason is that the country faces two significant constraints: access to top-tier talent and access to advanced logic compute.

The first constraint is human capital. As our China analysts point out [here](#), while China is the leading source of top-tier AI researchers, a majority of them ultimately end up working in the United States. The recent success of DeepSeek's team, which was overwhelmingly domestically educated, suggests that if Chinese universities and companies can offer conditions and job opportunities comparable to the US for AI talent, the country can better retain its talent pool.⁷⁸ Until then, however, China is likely to continue to experience a net outflow of its top AI researchers to the US.

The more serious barrier is access to advanced logic compute. An analysis from the Council on Foreign Relations (CFR) finds that the leading US AI chips are approximately 5x more powerful than China's current domestic alternatives from firms like Huawei.⁷⁹ The same analysis projects that this performance gap will continue to grow and could widen to as much as 17x.

Unless China can produce its own advanced chips, it will continue to rely on Taiwan for manufacturing. And it cannot build its own advanced chip supply without critical EUV lithography tools, a market monopolized by the Dutch firm ASML. In alignment with US policy, ASML has restricted the sale of its most advanced equipment to Chinese chipmakers, preventing them from building fabs capable of producing leading-edge chips and reducing their reliance on Taiwan. As a result, Chinese manufacturers are effectively stalled at the 7nm node.

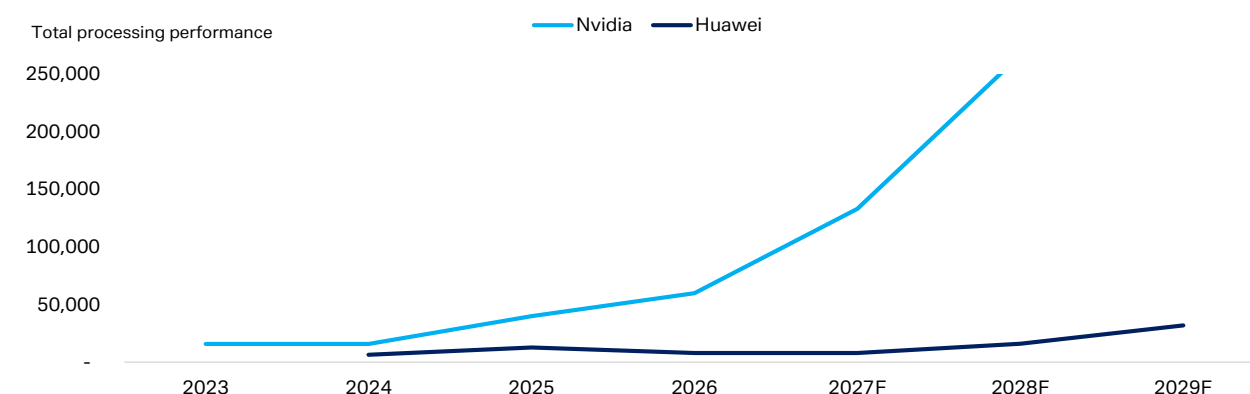
⁷⁷ For more on sanctions and chokepoints, read Edward Fishman.

⁷⁸ Yi Xiong, "China Macro: The Upcoming AI Boom in China," Deutsche Bank Research.

⁷⁹ McGuire, "China's AI Chip Deficit"

While Chinese companies like Xiaomi and Huawei are designing and developing their own semiconductors, the process is very costly and expensive. Although SMIC is reportedly moving toward mass production of 7nm and potentially 5nm advanced chips within the next one to two years, significant doubts persist about its ability to achieve the necessary scale and yield.⁸⁰ Even if successful, the technological frontier continues to advance, with the US and Taiwan already producing at the 2nm node, and some companies reportedly already planning for 1.4nm technologies. The cumulative result of these constraints is that China's frontier AI models have lagged their US counterparts by an estimated seven months since 2023.⁸¹

Figure 9: Nvidia vs Huawei's AI chips – total processing performance (TPP) comparison



Note: Based on the best chip available in H1 of each year.

Source: Deutsche Bank, Council on Foreign Relations, Nvidia Roadmap, Huawei Roadmap, SemiAnalysis.⁸²

5.3 China is establishing a competitive advantage in a US blind spot

While China lags the leading-edge competition, it is rapidly establishing global dominance in legacy chips. This development has been enabled, in part, by a US policy framework that has not cohesively addressed either domestic legacy chip production or China's growing strength in this segment. Thus, US export controls have left China's access to legacy chip technology largely unaffected, and China's overall chip imports remain in line with pre-pandemic trends.⁸³ This has led to warnings from bodies like the US National Security Council that, if unaddressed, the US could become dependent on China for components used in even the most "mundane" electronic devices, creating a new point of strategic leverage for Beijing.⁸⁴

In fact, Beijing has made massive investments in recent years to build China's dominance in mature-node chips, utilizing strategies ranging from accelerated fab production to import-substitution policies that guarantee domestic demand for its chip companies.

The US China Security Council predicts that China's share of global production could rise to 47% by 2027 from 34% in 2024.⁸⁵ In contrast, Taiwan's share is expected to decline from 43% to 36% over the same period. Should China establish hegemony over foundational chips, it could compel US manufacturers to rely on Chinese suppliers for essential components as non-Chinese competitors are displaced from the market. Non-Chinese entities, like Global Foundries (for chip manufacturing) and Texas Instruments (for integrated devices) are increasing their share in the semiconductor supply chain to reduce this risk.

⁸⁰ "China Reportedly Aims to Boost 7nm, 5nm Output Fivefold in Two Years, Driven by SMIC, Hua Hong," TrendForce, February 25, 2026, <https://www.trendforce.com/news/2026/02/25/news-china-reportedly-aims-to-boost-7nm-5nm-output-fivefold-in-two-years-driven-by-smic-and-hua-hong/>.

⁸¹ Luke Emberson, "Chinese AI Models Have Lagged the US Frontier by 7 Months on Average since 2023," Epoch AI, 2023, <https://epoch.ai/data-insights/us-vs-china-eci/>.

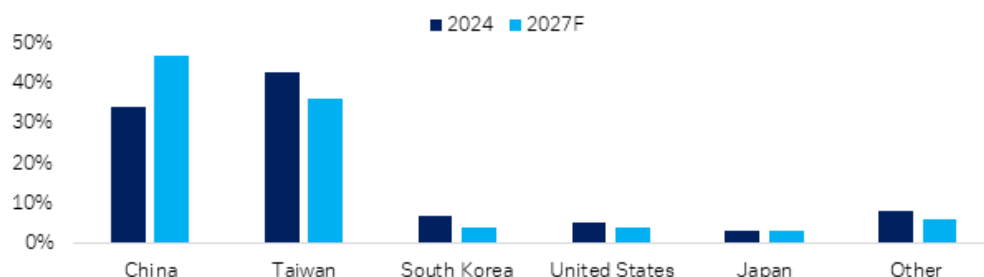
⁸² McGuire, "China's AI Chip Deficit."

⁸³ "Chapter 9: Chained to China," U.S.-China Economic and Security Review Commission.

⁸⁴ "Chapter 9: Chained to China," U.S.-China Economic and Security Review Commission.

⁸⁵ "Chapter 9: Chained to China," U.S.-China Economic and Security Review Commission.

Figure 10: Foundational foundry capacity by location, 2024 and 2027 (forecast)



Source: Based on USCC analysis of TrendForce Data, Deutsche Bank Research.

Finally, the global focus on high-bandwidth memory (HBM) has also created an opportunity for China's semiconductor exports. According to Chinese customs data, memory chips accounted for 52.8% of the country's total semiconductor exports in December 2025, as electronics companies have turned to Chinese suppliers amidst the memory supply glut.⁸⁶ US opposition aside to Chinese firms (e.g. CXMT and Yangtze Memory Technologies Corp) Western companies rely on China's expansion in this segment to support conventional memory, not HBM.

If this trend persists, China might end up being a winner in this market. If memory chip supply is unable to meet demand, average selling prices could continue to rise through 2026, potentially benefiting Chinese producers. However, some analysts believe it is still too early to tell if China can sufficiently fill these supply gaps.⁸⁷ This is why algorithmic developments like Google's TurboQuant could have major implications for China; if efficiency gains stemming from the algorithm reduce the physical memory capacity required by data centers, future demand may not be as high as currently projected.⁸⁸ Nevertheless, multiple industry sources suggest that a surplus in the market for HBM seems unlikely in the near term.

6. Global implications & industry response

6.1 Impact on industry

The trade wars and retaliatory strategies between China and the US in 2025 created arguably peak volatility for the semiconductor market. This was followed by a V-shaped recovery in chip equities (post a bottoming in April 2025), as investors were reassured by company-specific mitigation strategies and global renegotiation. The Philadelphia Stock Exchange Semiconductor Index (SOX) recovered to end 2025 up 41% YTD – roughly 25pp ahead of the S&P 500 and more than 20pp ahead of the Nasdaq 100 in 2025.⁸⁹ Even with the geopolitical tensions of 2025, seasonal semiconductor sales were not meaningfully impacted, suggesting continued fundamental strength in chip demand.⁹⁰

⁸⁶ General Administration of Customs, People's Republic of China.

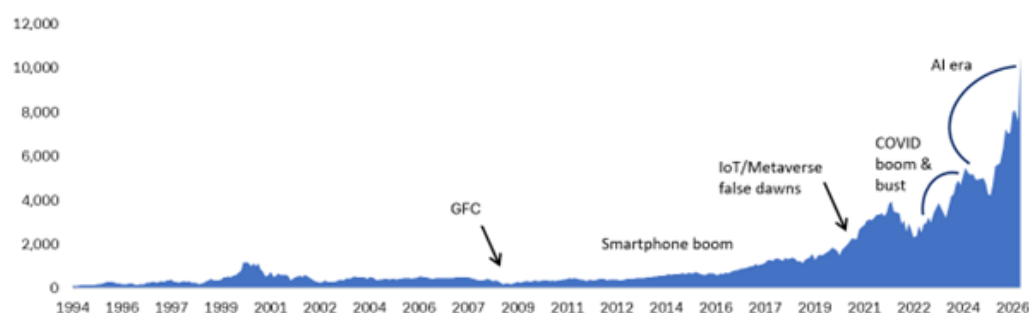
⁸⁷ Irene Zhang and Nick Corvino, "Review of Chinese AI Rings in the Year of the Horse," *ChinaTalk* (blog), February 18, 2026, https://www.chinatalk.media/p/chinese-ai-rings-in-the-year-of-the?utm_campaign=email-half-post&r=6q6sor&utm_source=substack&utm_medium=email.

⁸⁸ Tim Bradshaw and George Steer, "Memory Chip Stocks Shed \$100bn as AI-Driven Shortage Trade Unwinds," *Financial Times*, March 27, 2026, <https://www.ft.com/content/e4e15692-187e-4466-832e-ec267e792292?syn-25a6b1a6=1>.

⁸⁹ Bloomberg Finance LP.

⁹⁰ Ross Seymore, Apoorva Kumar, DJ Sebastian, and Melissa Weathers, "Semiconductors: 2026 Outlook: History Doesn't Repeat, but It Rhymes (Again)," Deutsche Bank Research.

Figure 11: Semiconductor index (SOX)



Source: Bloomberg Finance LP, Deutsche Bank

6.2 The AI race and its effect on the chip industry

As AI infrastructure investment continues to be driven by Chinese and US companies, we believe the industry should expect to see a continued bifurcation between AI leaders and broad-based laggards. As our equity analysts point out in their note [here](#), investor focus on AI is broadening beyond GPUs to related segments such as ASICs, networking, memory, and power management solutions.

Looking to 2026 and beyond, the rate of hyperscaler capex will continue to impact chip prices. While key hyperscalers maintain their air of confidence and continue to announce AI-related spending deals (Sovereign AI, OpenAI, Anthropic, etc.), many of these deals are not yet set in stone and are more akin to agreements or MOUs rather than finalized contracts. Investors are increasingly worried about the return on investment on these massive investments, with concerns about the "monetization" of AI beginning to overtake the focus on continued acceleration. For example, Microsoft's inability to meet growth expectations for its Azure cloud division – attributed to a lack of advanced chips – was the precipitating factor in the company's \$357bn single-day market value loss earlier this year. Still, Q1 2026 earnings from Big Tech revealed more spending by large cloud providers and strong cloud / AI income than anticipated, suggesting the AI trend will drive significant growth for chip companies. Our equity analysts expect the total chip market to reach \$969bn (+25% y/y) in 2026, driven by capital expenditures from these large cloud providers.⁹¹

Looking ahead, market participants should continue to brace themselves for continuous AI-related disruption. The most recent example is Google's TurboQuant algorithm announcement, which suggests that companies may require fewer GPUs/TPUs to serve large language models (LLMs). The announcement caused a sharp decline in the share price of memory providers like Samsung (-23%), Micron (-18%) and SK Hynix (-24%) in March.⁹² However, whether the TurboQuant technique will create a structural shift in demand remains to be seen. A key indicator to watch is actual order behaviour; unless top suppliers report significant HBM order cancellations, it is difficult for us to envision a substantial reduction in memory demand.

6.3 Impact on memory chip demand

Notwithstanding disruptions like the TurboQuant announcement, the risk of memory chip shortage in everyday standard DRAM and NAND – the integrated circuits used for phones, cars, computers – is becoming more acute. This concern has arisen as memory giants like Samsung turn their attention and resources toward developing HBM for AI instead, which as we have said, requires vast amounts of high-performance memory. For example, Micron, a key memory producer, is described as "supply-constrained" and only able to meet "50-66% of customer demand in the medium term" despite efforts to increase

⁹¹ Seymore et al., "Semiconductors: 2026 Outlook."

⁹² Bloomberg Finance LP.



supply.⁹³ The broader memory market is now expected to remain "supply-constrained through 2027".⁹⁴

As a consequence, companies in consumer-facing markets are warning of potential slowdowns in PC and handset sales due to the rising cost of memory, which makes the final products more expensive. Qualcomm, for instance, is seeing "significant headwinds" in its Handsets segment due to elevated memory pricing.⁹⁵ Other industry leaders now describe memory as a more critical constraint than even semiconductor manufacturing capacity, with the CEO of ARM noting in a recent interview that he is now less concerned about TSMC fab allocation than he is about memory.⁹⁶

6.4 The role of Europe, Japan, South Korea

Europe, South Korea and Japan play critical roles in the semiconductor supply chain but find themselves caught in the middle of the complex strategic competition between the US, Taiwan and China.

For Europe, the EU's Chips Act has been in force since September 2023 with limited initial results. The legislation, introduced in response to the supply chain disruptions caused by the COVID pandemic, set a goal for the EU to achieve a 20% share of global chip production by 2030. However, a recent audit by the European Court of Auditors concluded that the initiative is unlikely to meet its target in time, and is insufficient to stimulate the level of investment needed for Europe's chips industry.⁹⁷ As a result, the EU's overall share of the global value chain is projected to reach only 12% by 2030, despite its significant increase in manufacturing capacity.

ASML, a Dutch chip equipment maker, is Europe's key player in this industry. The company plans to produce at least 60 EUV machines this year and at least 80 in 2027.⁹⁸ However, some critics argue that Europe is not leveraging this asset effectively. The Dutch government's decision to align with US policy and restrict exports of both EUV and DUV (deep ultraviolet) machines to China has impacted ASML's earnings and demonstrated Europe's dependence on US strategic direction, turning Europe's strongest card into just another constrained asset.

In South Korea, the country's primary advantage has been its leadership in high-end memory. Until late 2024, when US policy changes required Seoul to halt the manufacturing of 7nm and more advanced chips for Chinese AI firms, a considerable portion of Chinese-designed AI chips were manufactured by Samsung. Since then, the South Korean government has pursued a dual strategy: while aiming to closely align itself with the US policies around semiconductors, it has also increasingly focused on boosting its domestic material sourcing and expanding into the logic and AI chip markets. These efforts are backed by significant government support, including a March 2023 law providing tax credits of 15-35% for semiconductor investments. The country's aim is to locally source 50% of its semiconductor materials by 2030, up from the current 30%.

Meanwhile, Japan has also played a key role in the effort to manage China's technological advancement while bolstering its own industry. Under Sanae Takaichi, Japan's government has committed to boosting its domestic semiconductor industry to reach \$254bn in annual sales by 2040. The country's flagship project, Rapidus, has received over \$11 bn in

⁹³ Joseph Wilkins, "Micron CEO Says It Can't Deliver Enough Memory to Key Customers after Blowout Earnings," CNBC, March 19, 2026, <https://www.cnbc.com/2026/03/19/micron-falls-blockbuster-earnings-chips-ai.html>.

⁹⁴ Ross Seymore, Apoorva Kumar, and Melissa Weathers, "Global Memory: Why this cycle is different; Updating Demand-Supply model; MU T/P to \$500," Deutsche Bank Research, Feb 10 2026.

⁹⁵ Ross Seymore, DJ Sebastian and Melissa Weathers, Feb 5 2026. "Qualcomm: F1Q26 (Dec) Results: Memory loss," Deutsche Bank Research.

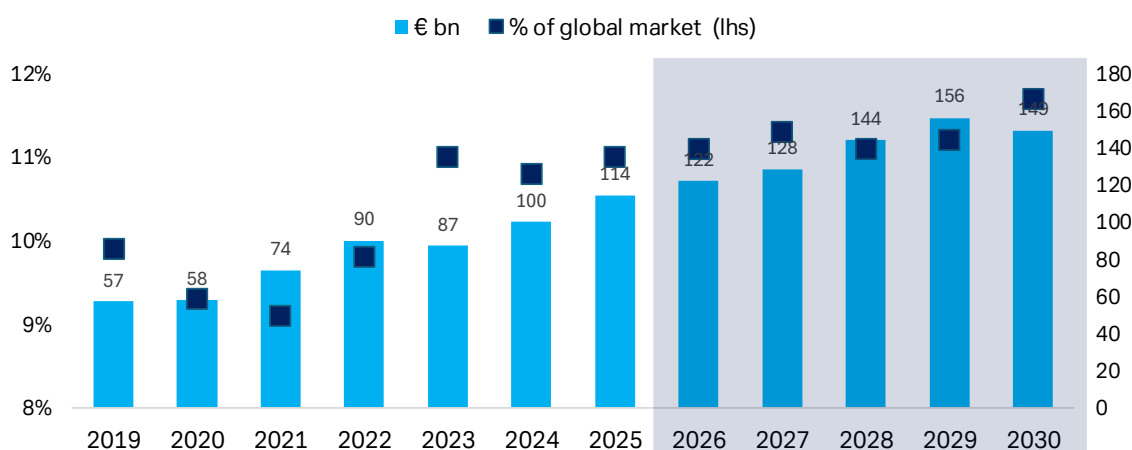
⁹⁶ Ben Thompson, "An Interview with Arm CEO Rene Haas about Selling Chips," Stratechery (podcast), 2026, <https://stratechery.com/2026/an-interview-with-arm-ceo-rene-haas-about-selling-ch>.

⁹⁷ "Special Report 12/2025: The EU's Strategy for Microchips – Reasonable Progress in Its Implementation but the Chips Act Is Very Unlikely to Be Sufficient to Reach the Overly Ambitious Digital Decade Target," (Luxembourg: European Court of Auditors, 2025).

⁹⁸ Mackrael, Kim. 2026. "The Race to Make the World's Most In-Demand Machine." *The Wall Street Journal*. April 25, 2026. <https://www.wsj.com/business/the-race-to-make-the-worlds-most-in-demand-machine-092e8cea>.

state investment and aims to reach 2nm production (currently from 40nm) by the late 2020s. The latest technical milestone was reached in July 2025, when Rapidus announced it had successfully prototyped a 2nm-generation GAA transistor at its IIM-1 manufacturing facility in Chitose, Hokkaido. And in another win for Japan this year, TSMC unveiled plans to produce 3nm chips in Japan, following heavy negotiations with the Japanese government to secure funding for its expansion into the country.⁹⁹

Figure 12: 2030 forecast of EU-27 share of global market value chain



Source: European Court of Auditors, COM (2024), based on the International Data Corporation's study in 2022.

7. Conclusion: Silicon sovereignty unlikely in the near to mid-term

The U.S.-China Economic and Security Review Commission has framed critical minerals as the “immediate” challenge in the bilateral relationship between the US and China, whilst viewing semiconductors as a “future chokepoint.”¹⁰⁰ This suggests that in the short-term, China can exert leverage at the bargaining table because its control over critical minerals represents a more immediate global bottleneck, whereas the semiconductor dynamic has a longer time horizon.

While this assessment has merit, this chapter has sought to demonstrate that the semiconductor supply chain is already filled with chokepoints, making chips a latent but escalating threat. The US will likely need 15 more years to develop the necessary capacity and skills to establish a chip manufacturing and packaging industry that does not rely on Taiwan, not to mention the additional costs involved.

The result is a state of complex interdependence: while China continues to struggle to produce chips on par with those from the US, Europe, Japan, South Korea and Taiwan, these allied nations also face significant hurdles in achieving full industrial sovereignty and breaking their reliance on one another. At the same time, China is growing increasingly more competitive in areas where US policy is less focused, such as legacy and foundational memory chips.

This complex reality helps explain why the current US administration's partial relaxation of advanced AI chip controls has split stakeholders and perplexed some analysts. At Davos in January, Anthropic's CEO likened the sale of advanced US chips to China to “selling nuclear weapons to North Korea.”¹⁰¹ In contrast, others argue that because the US relies on TSMC in Taiwan for its most advanced AI chips, a complete technological cutoff of

⁹⁹ Harry Dempsey and Kathrin Hille, “TSMC to Produce Advanced Chips in Japan,” *Financial Times*, February 5, 2026, <https://www.ft.com/content/83e0fd2a-1fa4-4004-9ffb-ef929075a497?syn-25a6b1a6=1>.

¹⁰⁰ “Chapter 9: Chained to China,” U.S.-China Economic and Security Review Commission.

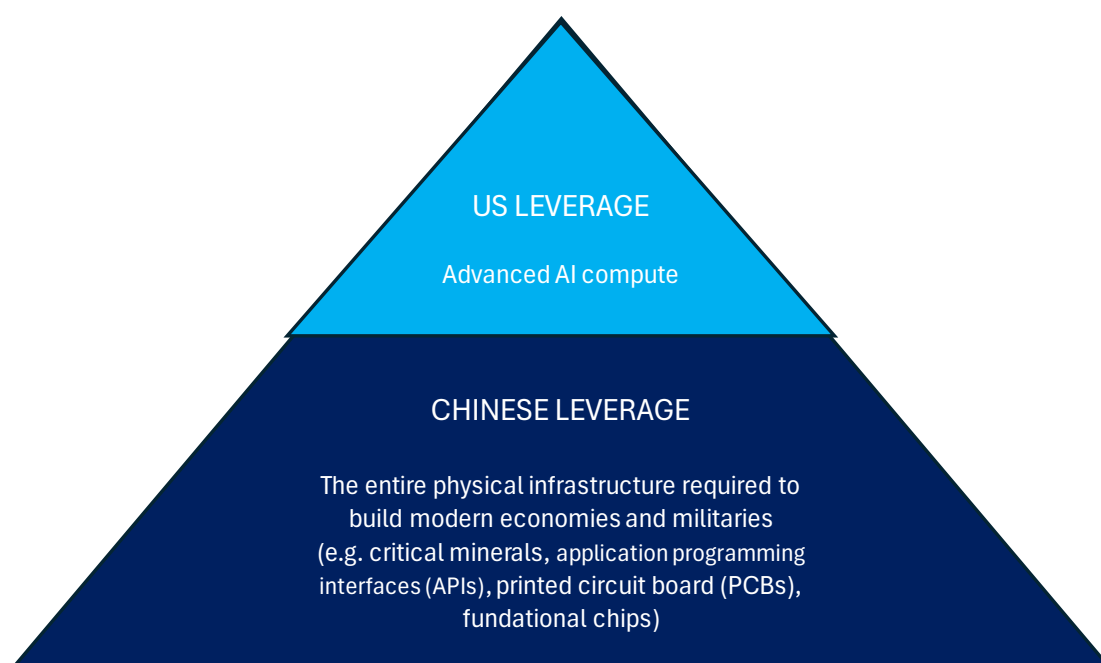
¹⁰¹ Christina Kyriasoglou, “Anthropic CEO Says AI Chip Sales to China like Selling Nukes to North Korea,” *Bloomberg.com*, January 20, 2026, <https://www.bloomberg.com/news/articles/2026-01-20/anthropic-ceo-says-selling-advanced-ai-chips-to-china-is-crazy>.



China would incentivize Beijing to adopt a “scorched-earth” strategy towards Taiwan’s foundries.¹⁰² This has led some experts to advocate for a policy of keeping China dependent on the US-based ecosystem, while simultaneously denying it the advanced EUV equipment needed to ultimately create its own leading-edge fabs.

Looking ahead to the next few chapters, one of the wider themes emerging from this series is that the chip competition is increasingly about the race for AI dominance. The outcome of this race may depend on its timeline: a rapid evolution of AI likely favors the US due to its massive capital and compute advantages, whereas a slower, more industrial-focused timeline could favour China due to its superior physical supply chains and manufacturing scale. In other words, China could arguably win from an industrialization standpoint, while the US can lead by diffusing its AI capabilities. But the limitations of US industrial capacity, as highlighted by the mixed results of the CHIPS Act and its latest export controls, have created an opening for private corporations to tackle the production problem.¹⁰³ The influence of firms like Nvidia, for example, was a significant factor in the administration’s decision regarding H200 chip sales to China. Thus, in this era of “infrastructure realism” driven by the pursuit of AI supremacy, perhaps the more accurate way of surveying the strategic actors will be via companies rather than countries or governments. We will explore this theme further in our forthcoming chapter on cloud computing.

Figure 13: The US maintains a lead in advanced AI, but China holds the industrial capacity



Source: Deutsche Bank Research. An API is a set of defined methods of communication among various software components. In the context of semiconductors or design, it could refer to interfaces for software tools or IP blocks. A PCB is a board used to mechanically support and electrically connect electronic components using conductive pathways, tracks, or signal traces etched from copper sheets laminated onto a non-conductive substrate. They are fundamental in nearly all electronic devices.

¹⁰² Leading tech analyst Ben Thompson instead advocates for keeping China dependent on a US-based ecosystem.

¹⁰³ Tesla CEO Elon Musk announced the construction of a massive factory called “TerraFab” capable of producing AI chips at an annual scale of 1 terawatt (TW). This decision stems from the assessment that current AI chip production volumes are insufficient to meet demand. The project will involve all three of Musk’s companies: Tesla, SpaceX, and xAI.



Appendix 1

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Sizing Up GLP-1 Impacts: Survey Shows Durable Shift in Pharma, Diets, and Dining

May 11, 2026

We recently conducted our 2026 survey of current and former GLP-1 weight-loss medication users and the results are clear: GLP-1 adoption is shifting long-term behaviors affecting pharmaceuticals, food and beverage companies, and restaurants.

These themes were already indicated in our initial [2024 survey](#) of GLP-1s, but today's results are even more striking. Now several years into GLP-1 usage, the survey shows sustained changes in dietary and restaurant behavior that stick even after patients have stopped taking GLP-1s. Following the recent introduction of oral pill versions of GLP-1 medications, the survey showed there is strong interest in orals, largely due to the convenience they offer. However, it also shows there are limits related to affordability and long-term commitment.

The takeaway is that, for pharma, there is a plausible pathway for new oral GLP1s to accelerate market penetration over the coming years. Long-run market expansion is likely, although questions remain about how much consumers will be prepared to pay for orals and how long they will stick with them. Meanwhile, regarding food and beverages, consumers are shifting away from large portions and snacking towards protein-forward, functional, and nutrient-dense choices.

For restaurants, the survey points to declines in the number of times people dine outside the home, partially offset by higher spending each time they eat out. Unsurprisingly, brands with differentiated, healthier offerings or more occasion-based demand are likely to be more competitive.

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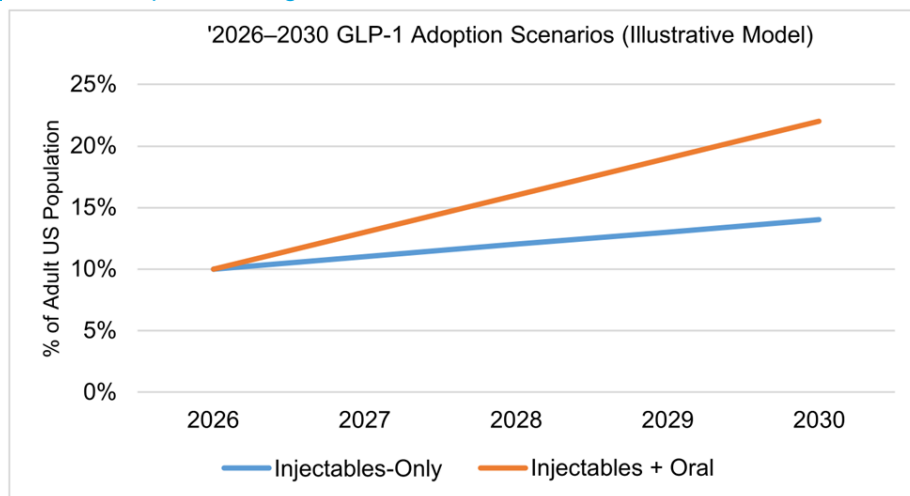
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Figure 1: GLP-1 adoption represents a durable reallocation of consumer demand, in our view; behaviors likely to broaden under accelerated oral GLP-1 adoption through 2030



Source: Deutsche Bank Research

Implications for US and European Pharma

The survey points toward a sustainable promise of oral GLP-1s, but also a cost ceiling, a durability problem, and potential for better risk-benefit to be disruptive. The picture it portrays is of a highly cost-sensitive user base that (still) clearly prefers orals and struggles with adherence to therapy, while mostly intending to return to it at some point and clearly not viewing diet and exercise as a viable alternative.

What that implies for the market outlook is somewhat uncertain: with the up-front caveat that this survey addresses only those trying GLP-1 within the past year and hence perhaps not the key demographic from a market-growth-outlook perspective (i.e., those that have never tried it), for the existing installed user base it implies some risk of injectable cannibalization (though we note this is not particularly apparent yet in prescription data) and also some risk that volume demand may plateau without further price degradation (a minor negative, in our view). However, it also implies that a return to therapy is likely to be high, and improved safety profiles plus oral convenience could help to build out the market opportunity over time. It also suggests the amylin class could help optimize risk-benefit and thus help address the persistent durability problem.

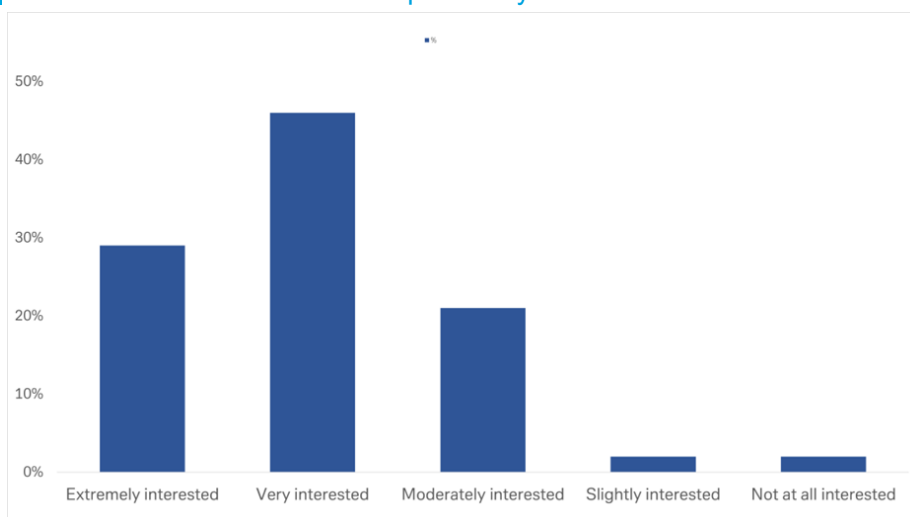
As with our last survey, there was a bias to Ozempic as a preferred brand with a fairly even distribution between Wegovy, Zepbound, and Mounjaro beyond that. Interestingly, the main consequence of increasing use of GLP-1 therapeutics to manage weight loss appears to be a reduction in diet/exercise approaches to helping manage weight loss (rather than the reverse). Beyond that, consistency of adherence appeared static with our last survey (about 70% within a 12-month window), whilst duration of therapy also appeared relatively stable at a mean of 6-9 months. This is despite most patients not being at their target weight loss/implied BMI (current average about 30 versus average target closer to about



25) and many continuing to experience a frequent (20%) rate of GI toxicity of some kind.

Payor coverage did appear to have slightly improved (to about 40%), with most patients still paying up to \$200 per month out of pocket, but interestingly willingness to self-pay has declined to less than half of patients (with most unwilling to pay more than \$200 per month in that scenario). Almost half continue to access therapy via retail pharmacies, with only a modest increase in online channel use. On therapeutic options, interest in oral formulations was very high at about 2/3 (and 80% in younger age groups), with 50% expressing clear preference for orals over injectables all else equal. Interestingly, the greatest sensitivity on the question of switch intent related to cost, with early experience pointing toward a double-digit percentage lower cost for orals versus injectables and most citing this as sufficient to drive switching behavior.

Figure 2: Oral GLP-1 medications are becoming available. With this in mind, please indicate your level of interest in starting to use an oral form should it become an available option for you.



Source: dbDIG Primary Research

The most common reason provided for discontinuation among the approximately 30% who stopped therapy within the past year was cost, though side effects were a close second, and a variety of other reasons followed closely behind. Only about 20% of patients who had discontinued treatment cited "reaching weight loss goal" as the reason. In this subgroup, the advent of oral options also seemed to strike a particular chord (75% extremely or very interested versus 64% in the earlier group of patients still on therapy).

Implications for US and European Consumer Staples

What stood out in our 2026 data compared to our 2024 survey is that GLP-1 adoption is here to stay, with an enduring influence on consumer behavior. The introduction of the oral version of GLP-1s will only reinforce these trends. For packaged food and select beverage categories, the growing prevalence of GLP-1 usage represents a sustained structural pressure that operates through lower



portion sizes, reduced frequency, and selective category substitution rather than outright elimination of consumption.

While many companies have begun to pivot their portfolios in response—introducing protein-fortified snacks, functional beverages, and smaller portion formats—we question whether these innovations can fully offset the net impact of structurally lower intake and frequency. The survey data suggest that many of these shifts improve portfolio relevance and defend share, but do not fully restore lost volume, particularly in indulgent, frequency-dependent categories.

The survey data suggest that protein formats primarily benefit from share reallocation within smaller meals rather than incremental meal creation. Moreover, with traditional categories also chasing protein, we would argue that protein itself is becoming increasingly mainstreamed across everyday food products, diluting the relative advantage of specialized nutrition SKUs and capping category-level growth. Elevated competitive intensity and rising A&P needs increase the risk that theoretical demand alignment does not fully translate into a bigger bottom line.

Outside of performance nutrition, fresh and clean-label portfolios benefit from broader, yet seemingly durable “eat better” behavior rather than calorie avoidance. Survey responses show consistently higher relative consumption of fresh and minimally processed foods during GLP-1 usage, though absolute consumption gains appear incremental rather than transformative.

Flavor-centric and ingredient-led categories appear structurally more resilient than volume-dependent staples. Conversely, sugar-dense, portion-dependent categories face persistent headwinds. Energy drinks, though, stand out as one of the few categories showing net consumption stability or increases among active GLP-1 users. Water and hydration also tend to see benefits, with coffee and tea more neutral.

Overall, GLP-1 adoption increasingly favors goods aligned with satiety, functionality, and quality over calories, but the survey underscores that alignment alone is insufficient. Company-specific execution, competitive position, reinvestment discipline, and cost structure remain decisive in determining whether theoretical portfolio alignment translates into realized financial outcomes.

From a European Food and HPC perspective, our 2026 survey reinforces the view that GLP-1s represent a persistent but comparatively modest influence on consumer demand, rather than a step-change in consumption behavior. In our view, GLP-1 also cannibalizes other forms of weight loss to some degree (though is likely to be more effective and lasting) and as such the net calorie drag per annum that many have worried about seems to be less than feared. It is fair to note though that rolling 12m US Nielsen aggregate “Food” volumes are running down about 1% year-over-year currently and the pace is accelerating. It’s difficult to disaggregate GLP-1 from macro, demographic, SNAP and immigration factors, however, its unlikely to be a non-zero impact.

Importantly, the 2026 cohort appears to have entered GLP-1 usage with healthier baseline habits than in 2023, resulting in a smaller incremental behavioral shift once on medication. In several categories, the change from “before” to “during” GLP-1 usage is less pronounced than previously observed, suggesting an



element of normalization rather than acceleration. That doesn't make the GLP-1 effect itself less valid in its net effect, just that it may not all be related to actual GLP-1 usage. We have also previously noted that the US has also gone through waves of weight loss previously, it's not linear and as such GLP-1 usage increased at a likely time of one of these waves persisting post COVID anyway and we have previously seen a pivot to categories like dairy during non-GLP-1 weight loss.

While some behavioral persistence is evident post-discontinuation, the data do not point to a broad-based collapse in category participation. In fact, select traditionally "indulgent" categories show stabilization—or modest re-engagement—among former users, complicating a simple winners/laggards framework. At an aggregate level, the data are also consistent with lower intensity rather than outright avoidance, implying moderation rather than displacement and an opportunity for companies to tailor their suite of SKUs to mitigate or indeed benefit. As of yet we have seen limited impact on demand for household and personal care categories; thus far, consumer health companies have seen a surprisingly limited number of innovations that support GLP-1 usage.

For our European Beverages coverage we view GLP-1 as a headwind to consumption overall but see some of the survey responses as providing conflicting data points to this potential risk. We view GLP-1 as one of a number of threats to beverage alcohol consumption including 'health', younger consumers, cannabis and, for spirits, smaller players and ready-to-drink. Companies with a higher exposure to the US and those geographies where GLP-1 is coming off patent in the near future (Canada, India, China, Brazil) likely present the greatest risk with beverage alcohol more 'at risk' than soft drinks.

The impact on nicotine consumption is not specifically covered in the survey results. We do recognize that there is growing body of scientific evidence to support the role GLP-1s may play in tackling smoking addiction, but efficacy varies by the specific drug used. The observational data also point to increased effectiveness in reducing nicotine consumption (cigarettes and vapes) although these studies remain relatively small in scale. The industry has long dealt with declining volumes, and so we see GLP-1's as another contributor to the volume contraction but one which, on current evidence, does not represent an existential threat to the industry.

We would also posit the view that the motivation for smoking cessation is typically health related. Increasingly, the industry is able to offer consumers a broad range of smoking alternatives via Next generation products, which significantly improve the health outcomes, compared with smoking.



Figure 3: Categories emerging as relative beneficiaries with the increasing use of weight loss medication

	Category	Description/Details
Relatively Advantaged Categories	Protein shakes and bars	Protein shakes and bars are clear winners. In 2026, 72% of current GLP-1 users consume more than before, up from 64% in 2023, reinforcing protein formats as efficient meal substitutes rather than incremental calories
	Fresh meat and fish	Fresh meat and fish benefit from satiety-led substitution. In 2026, 59% of current users report eating more than pre-medication, up from 45% in 2023, supporting increased prioritization of whole, minimally processed protein sources.
	Fresh produce	Fresh fruits and vegetables remain durable beneficiaries. In 2026, 73% of current GLP-1 users report eating more produce, similar to 2023, indicating sustained "eat better" behavior despite overall moderation in total food intake
	Nuts	Nuts show low downside and solid upside. In 2026, 56% of current users report eating more nuts than before, positioning the category as a functional, nutrient-dense snack aligned with satiety rather than indulgence.
	Condiments, sauces, and spices	Condiments are resilient but not growth-oriented. In 2026, "a lot less" consumption is only 12% while "neither more nor less" rises to 57%, suggesting stability as flavor intensity per bite matters in smaller meals. size.
	Energy drinks	Energy drinks stand out within beverages. In 2026, 47% of current users report drinking more than before, while only 33% report drinking less, supporting functional, zero-sugar positioning over indulgent beverage formats.

Source: Deutsche Bank Research

Figure 4: Categories impaired by GLP-1 usage but more resilient, especially with adaptation to remain competitive

	Category	Description/Details
Mixed / Execution-Dependent Categories	Packaged and frozen meals	Frozen and refrigerated meals sit at the execution frontier of GLP-1 impact: neutrality remains the modal outcome, with incremental—not structural—pressure, hinging on portion control, protein density, and clearer health cues
	Canned foods and traditional pantry staples	Pantry staples are neutral-to-slightly pressured, not broken: consumption skews toward "same or less," with limited upside absent reformulation, protein enrichment, freshness cues, and improved portion transparency
	Coffee	Coffee is compositionally resilient rather than volume-challenged: core consumption holds steady, while sugar- and syrup-heavy formats decline—shifting pressure to sweetened customization, not coffee occasions themselves
	Tea	Tea is one of the most resilient categories in the dataset: high neutrality, minimal cutbacks, and muted intent-to-reduce reinforce its role as a low-calorie, habitual utility beverage insulated from GLP-1 moderation dynamics

Source: Deutsche Bank Research



Figure 5: Categories that are more adversely impacted by the use of weight loss medication

	Category	Description/Details
More At-Risk Categories	Chocolate and candy	Chocolate and candy face the clearest structural pressure. Roughly half of current GLP-1 users consume less in 2026, and most former users fail to rebound, confirming sustained demand erosion in calorie-dense indulgence
	Cookies and biscuits	Cookies and biscuits remain highly exposed. One-third of current users report eating "a lot less" in 2026, while ~75% of former users continue to eat less or stop entirely, reflecting vulnerability of habitual, low-satiety snacks
	Ice cream	Ice cream participation is persistently impaired. About 70% of current users report same or lower consumption, and a similar share of former users continue to eat less or none, signaling weak post-GLP-1 recovery
	Salty snacks	Traditional salty snacks face uneven but meaningful pressure. Chips consumption declines, with "a lot less" rising to 17% in 2026; pretzels show limited rebound, reinforcing structural headwinds for refined-grain snack formats
	Cereals and refined pantry staples	Cereals show elevated "eat less" signals during GLP-1 usage and weak normalization afterward. Their association with refined carbohydrates and low satiety leaves cereals structurally disadvantaged versus protein-dense alternatives
	Alcohol (Beer, Wine, Spirits and RTD cocktails)	Alcohol faces moderate but notable headwinds. Participation declines materially (spirits, beer, wine), yet "drink more" exceeds "drink less" during treatment, suggesting reduced frequency rather than category exit and a nuanced, uneven risk profile

Source: Deutsche Bank Research

Implications for Restaurants

The increasing penetration of GLP-1 weight loss drugs is intuitively a headwind for total restaurant spend, and our survey results are in line with this thinking. Despite the fact that many users eventually stop taking GLP-1s, they continue to show sustained changes in behaviors. Our survey results indicate that former GLP-1 users do not meaningfully return to full-service restaurants after taking the medication, though some return to fast food and coffee shops at a rate above current GLP-1 users, and in reality, we believe restaurant trends rebound more than our survey suggests. This is based on anecdotal commentary and behaviors seen in other categories. The read-through on individual frequency was more positive, with current GLP-1 users reporting an increase in visits to full-service restaurants, and a modest decline in visits to fast food, coffee shops, and takeout relative to pre-GLP-1 levels. Moreover, average spending on meals per visit was higher than baseline, suggesting a more tempered impact on sales compared to traffic.

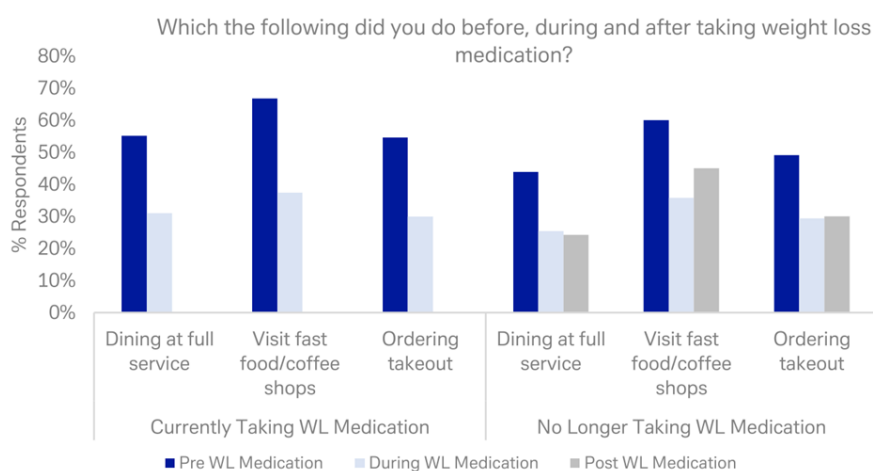
Restaurants have started to address behavioral shifts by offering smaller portion sizes, protein-forward options and functional beverages. We believe these innovations can help more effectively sustain participation in the restaurant industry; brands that are differentiated, screen healthy, and are more occasion based are relatively best positioned to be competitive.

It is clear that overall food consumption decreases when consumers are on GLP-1 medications, though the implications once consumers transition off are less clear. Based on our survey, 35% consumers report eating more on GLP-1 medications, 24% eat about the same, and 41% report eating less, for a net 6% eating less. When consumers were asked about current eating habits relative to pre-GLP-1 medication, among those currently taking GLP-1s, a net 10% reported eating less compared to pre-medication (38% more, 48% less). Notably, among consumers that are no longer taking GLP-1s, when compared to prior behaviors, a net 20% indicated they are actually eating more (45% more, 24% less), suggesting former users revert toward prior aggregate consumption levels.



That said, there is still a change in how consumers eat, most notably an increase in cooking. Based on our survey, among consumers currently taking GLP-1s, a net 54% of consumers are cooking more at home. Cooking usage also remains elevated among consumers no longer taking GLP-1s, with a net 47% of consumers cooking more at home than before medication.

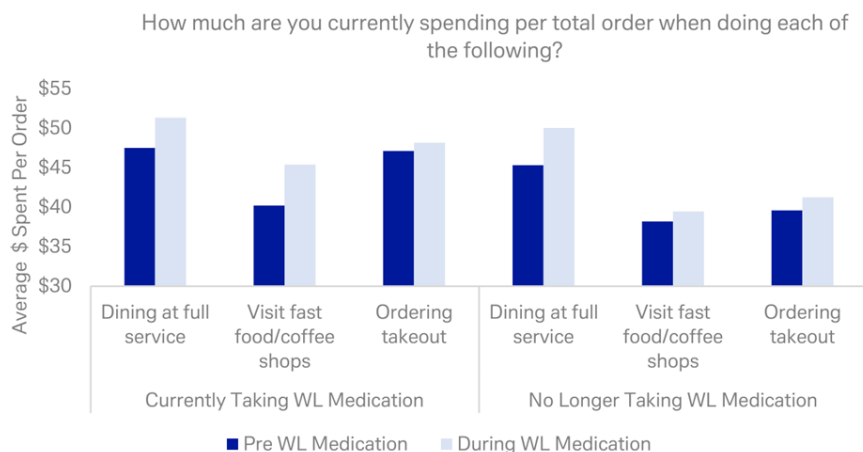
Figure 6: Our survey shows a meaningful and broad-based decline in restaurant usage when consumers take GLP-1s. Among current GLP-1 users, the percentage of respondents that visit restaurants drops by 25-30pp versus pre-weight loss behavior. Among former GLP-1 users, the percentage of respondents that visited restaurants while taking these medications versus prior behavior decreases at a relatively similar 20-25pp.



Source: dbDIG, Deutsche Bank Research

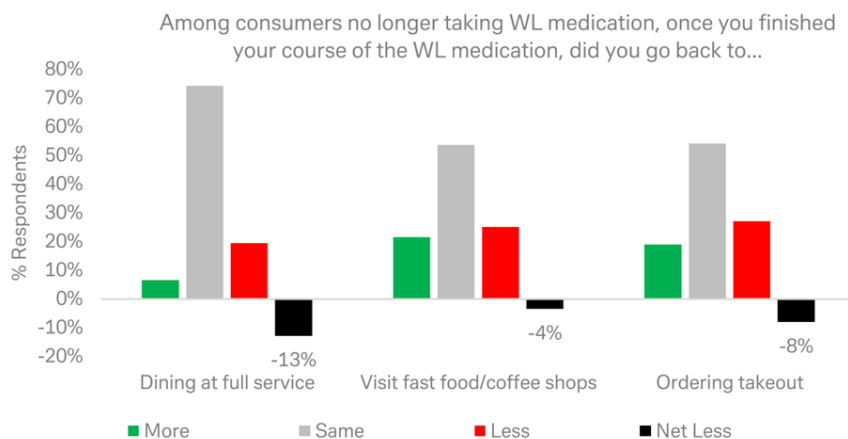


Figure 7: Among GLP-1 users that regularly visit restaurants, average spend per month did not take much of a hit, noting both current and former GLP-1 users report higher average spend per visit for full service, fast food/coffee shops and takeout. It is unclear whether the higher average spend per visit is a reflection of perceptions around higher prices or consumers ordering more/trading up.



Source: dbDIG, Deutsche Bank Research

Figure 8: Individual consumption for former GLP-1 users that are still eating at restaurants seems reasonably resilient (or at least manageable). On a net basis, 13% of consumers said they were dining less at full service compared to pre-GLP-1 behaviors, 4% said they were visiting fast food/coffee shops less, and 8% said they were ordering takeout less.



Source: dbDIG, Deutsche Bank Research



The Bottom Line: A Lot to Digest

Our survey implies that GLP-1s are starting to look less like a therapeutics cycle and more like an enabling platform. It is reshaping how consumers manage weight, what they eat and drink, and whether and how they choose to dine out. For pharma, orals offer an expanded market, offset to an extent by affordability and other constraints. For consumer staples and restaurants, brands that can focus on health-forward innovations will remain more competitive in this new environment.

Research Methodology

This study focuses on both current and past GLP-1 users, with demographic data collected via a dedicated set of questions in our US monthly survey. Compared to the 2023 wave, several demographic shifts were observed:

Gender: The user base is becoming more gender-balanced, with male GLP-1 users increasing from 40% to 45%.

Income: While the overall income distribution remained stable, there was a marginal increase in GLP-1 users within Income Tier 1 (households with an income up to \$25,000).

Overall, the survey was conducted through dbDIG from March 20-30, 2026, with 550 respondents. We would like to thank Anthony Chaimowitz for his significant contribution to the survey and this report.



Appendix 1

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The long road: Quantifying Europe's exposure to transport disruption

May 8, 2026

Heightened geopolitical risk in the Middle East has placed European transport networks under intense scrutiny, with a particular focus on the supply of jet fuel given the high share of the Gulf in the global supply of kerosene.

For the supply side of the European economy and overall input availability, the risk from a shortage of jet fuel is relatively small. The dominant fuel dependency for goods production is on diesel and electricity, reflecting the four-fifths share of input costs taken by land transport (road and rail).

The true vulnerability to a shortage of jet fuel lies squarely with the tourism sector. Our simulation shows a 10% drop in tourism flows would have a moderate impact on the Euro Area, on average (-0.2% of GDP), but a more significant impact on countries like Croatia (-1.5%), Greece, and Portugal (-0.7%).

Beyond the specific risk to jet fuel, a wider disruption to the global supply of transport would expose a more fundamental European vulnerability: the physical length of supply chains. To measure this vulnerability, we introduce a “physical distance travelled” metric, reflecting how far inputs travel for every industry. It reveals two critical points:

- A clear geographic divide: The economies with the longest supply chains are generally located in Southern Europe. Lacking deep integration with Europe's industrial core their value chains are inherently more global and reliant on long-distance sourcing. This exposure to transport frictions could create a vulnerability for some of the zone's high-debt member states. These countries are also those we previously found to have the highest energy import bill sensitivities.
- A dual "hidden" risk: From a sector perspective, the most-exposed sectors besides tourism are capital goods (electronics, air/spacecraft, machinery) and materials manufacturing (metals, refineries), whose supply chains stretch nearly 5,000 km, on average. Crucially, these are the same value chains we [previously identified](#) as having high "hidden" energy content, indicating a dual vulnerability to both transport shocks and embedded energy costs.

What does this mean for the ECB? Our baseline view is a 'measured' tightening with 50bp of hikes to 2.50%, the upper end of the range of neutral. With the labour market cooling, financial conditions tightening and this threat of fuel shortages, the growth outlook is not robust enough to think the ECB hikes much beyond 2.50%. The challenge for the ECB could be fragmentation if fuel shortages have a differential impact on member states.

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Concerns shifting from energy to transport. Aviation of particular concern. As we highlighted in a recent [note](#), Europe's economic exposure to the Gulf conflict is twofold, and is more complex than the singular energy shock of the Russia-Ukraine war. Risks of direct shortages in energy and key materials like petrochemicals or mineral fuels are now coupled with a more complex threat: second-order disruptions to global value chains.

Transport disruption is one of the most significant vectors of transmission for both these risks. As an input at every stage of production, the transport sector's resilience is once again at the forefront amid reports of looming [fuel shortages and logistical bottlenecks](#). In all this, jet fuel is of particular concern, as shown by recent statements by the Head of the International Energy Agency on April 16 that Europe has "[maybe six weeks of jet fuel left](#)", and our own [calculations](#) that the share of Gulf countries as a source for EU imports of kerosene reaches 56%.

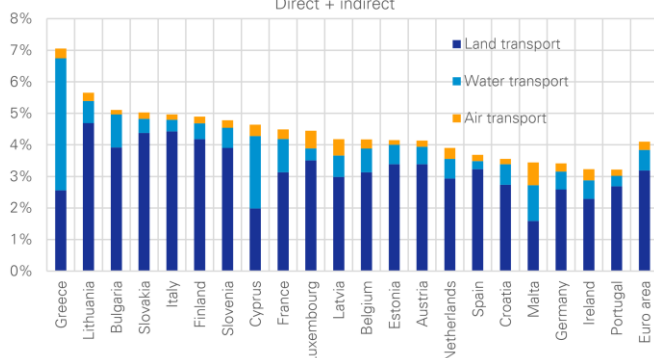
As our transportation sector specialists explain, the market is already under pressure. Jet fuel prices have doubled, and while replacement barrels from the US and Africa have helped Europe plug the gap for now, this fragile balance is unlikely to survive the summer demand surge. Airlines are already cancelling unprofitable flights and face difficult trade-offs between raising prices and eroding consumer confidence.

In this article, we think about the European economy's exposure to potential global transport disruption – in particular jet fuel – from two complementary perspectives. First, from a supply or production perspective, we look at the role of various forms of transport in European supply chains, including air transport, while introducing a distance metric reflecting how far European inputs travel and are exposed to transport disruptions. Second, from a demand or income perspective, we look at the exposure of European economies to tourism and in turn tourism's exposure to air travel.

How much transport in value chains? 3-5% of inputs, of which little air transport.
A traditional way to assess transport dependency is to measure its overall share in inputs along the value chain.

Figure 1: Cost share of transport services in Euro area input purchases, dominated by land. Air is tiny

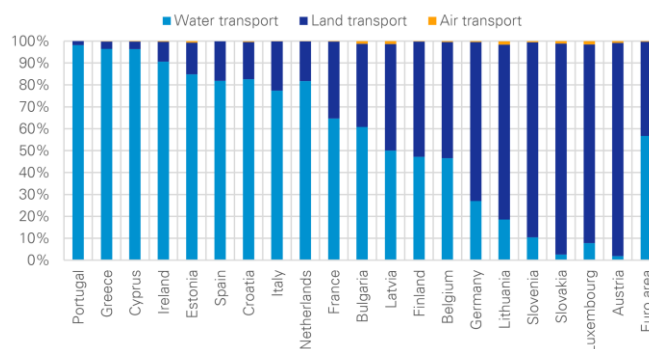
Euro area countries - Share of transport services in overall input costs - Direct + indirect



Source: Deutsche Bank Research, OECD ICIO. The figures include the absorption of all transport services at every stage of production, including that required to move inputs between two foreign suppliers (ie indirect absorption).

Figure 2: By physical weight, maritime shipping dominates freight transport

Euro area - Freight transport volume - By mode of transport



Source: Deutsche Bank Research, Eurostat (tran_hv_ms_frmod). The unit used for measuring the freight transport shares is the tonne-kilometre (tkm). One tonne-kilometre represents the movement of one tonne over a distance of one kilometre.



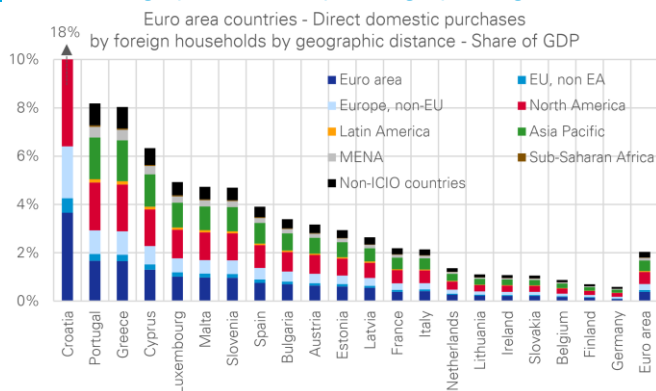
Our calculations, using a methodology [popularized by the OECD](#), suggest transport services account for 3-5% of total input purchases across the Euro area (Figure 1), with some notable outliers like Greece at 7%. Among the Big4, the role of transport is more pronounced in Italy and France (4.5-5%), less so in Germany and Spain (3-4%).

Land transport constitutes the bulk of this transport cost – approximately four-fifths. The share of water transport is significantly smaller, except for trade-reliant coastal nations like Greece, Malta, and Cyprus. Importantly, air transport represents only a minor share overall.

Implications for fuel usage. The dominance of land transport (i.e., road and rail) in the cost structure implies that the most significant fuel availability risk in the supply of goods and services comes from disruptions to diesel and electricity, not jet fuel. Given air freight's minimal share in both costs and physical volume of freight (as confirmed by both our analysis and Eurostat data in Figure 2), even a severe, un-substitutable 10% shortage of jet fuel may not cause much disruption to the flow of inputs and economic production in Europe.

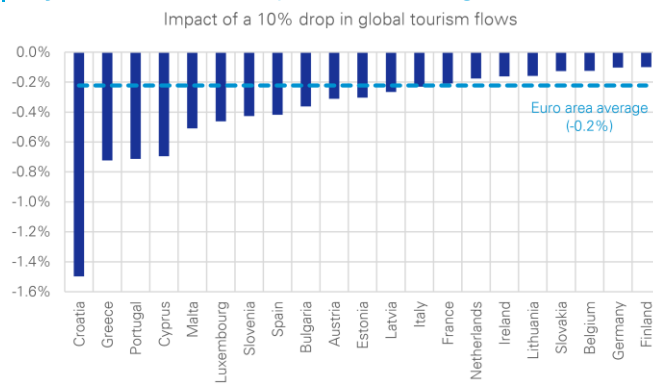
Tourism: a distinct, service-based vulnerability. The above numbers, focusing on freight and input purchases, do not capture the broader issue of the tourism economy: tourism involves moving individuals – not freight – and air transport is not counted as a formal input to the hotels, restaurants, and cultural activities that constitute the tourism economy (i.e., hospitality, recreation), but generally as a purchase by households.

Figure 3: The high share of long-distance travellers in direct foreign purchases (spending by foreign tourists)



Source: Deutsche Bank Research; OECD ICIO.

Figure 4: Tourism is where the Euro area's vulnerability to jet fuel and air transport becomes significant



Source: Deutsche Bank Research, OECD ICIO. The simulation focusses on the impact of spending by foreign tourists, and does not capture domestic tourism. As such, the numbers differ from the estimates of the sector's footprint published by Eurostat

In Figure 3, we calculate the share of GDP driven by direct purchases by foreigners (spending by foreign tourists). The result, unsurprisingly, is that the nations of Southern Europe are much more exposed, on average, but all nations in Europe derive income from a demand perspective from far-away visitors. We estimate that about three quarters of that spending is made by foreigners having travelled more than 2,000 km (50% of the spending by travellers more than 5,000 km away, often placing them on another continent).



A simulated 10% drop in global tourism flows reveals a manageable cost for the Euro Area, on average (a 0.2% hit to GDP – Figure 4). The impact is more pronounced in Croatia (GDP falling 1.5%), and Greece and Portugal (both seeing declines of 0.7%). On the Big4 countries, Spain would be faced with the most significant loss (0.4%), while France, Italy (0.2%) and Germany's (0.1%) exposures are more modest.

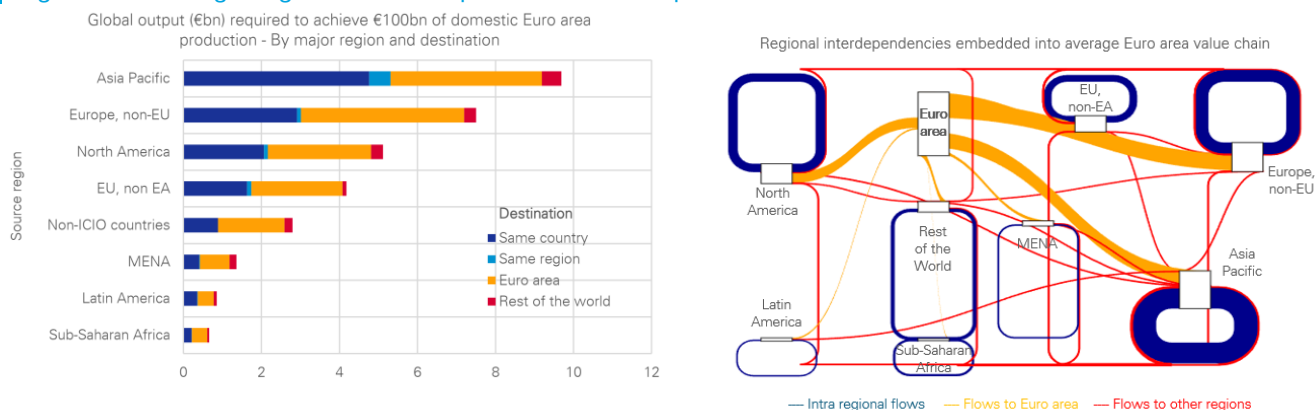
An important caveat to bear in mind: in cases of severe disruption of air travel, holidaymakers would likely travel more locally, which should reduce the size of a shock to the tourism industry. The overall impact will still depend on the size of the tourism sector relative to local demand.

Beyond jet fuel: Measuring exposures to transport friction. The focus on jet fuel, while critical for tourism, highlights a greater strategic risk: energy-driven transport disruptions may not stop at kerosene. A broader shock to transport fuels affecting diesel and maritime fuels would more certainly curtail the physical movement of goods. In such a scenario, what matters is not just the cost of transport, but a value chain's fundamental exposure to distance.

To assess this exposure, we introduce the “physical distance travelled” metric. This measure goes beyond a simple 'average import distance' to quantify the total physical journey embedded in the Euro Area's inputs to production, including domestic trade and all transactions that occur before a final good is imported. It reveals the true length – and vulnerability – of our supply chains. Deutsche Bank clients can access further details including the methodology on our [Deutsche Bank Research](#) client site.

Figure 5 illustrates the main input to the calculations. It shows the global transactions required to generate €100bn of production of an average good/service in the Euro Area, and where this output is directed. Both charts contain the same data, with the left-hand one showing how output is broken down (domestic consumption vs. exports), while the right-hand one illustrates the complex connections across regions to feed the Euro Area with inputs. The blue and red links represent the Euro Area's indirect exposure.

Figure 5: Picturing the global web of inputs for Euro area production



Source: Deutsche Bank Research, OECD ICIO. Note: our distance metric also includes the intra Euro area trade required in the value chain, not displayed in this chart.



We find a significant reliance on external partners, requiring approximately €10bn in output from the Asia-Pacific region alone. The charts also highlight the complexity of these value chains: a large portion of the Asian output is transacted and transformed internally (the blue segments) before being exported.

Tourism is clearly at risk, while the capital goods and materials value chains stand out on the goods-producing side. Southern Europe is more vulnerable, too.

The distance metric we calculate is simply the sum of the distances between the nodes in Figure 5 in kilometres, weighted by their values (it also includes intra euro area trade, not displayed in the chart). It shows that the longest, and, therefore, most transport-exposed, value chains in the Euro Area fall into three clear categories:

- First are tourism services, with most countries deriving a significant portion of revenue from travellers residing more than 5,000 km away (Figure 6).
- Second, Capital goods, where production is globally fragmented. Here, the computer and electronics sector stands out, with the average input traveling nearly 5,000 km, the longest of any European industry, after tourism. Other capital goods sectors like aircraft/railway equipment and electrical equipment (4,100 km) also exhibit noticeably long supply chains.
- The third category is materials manufacturing. Industries like non-ferrous metals (4,800 km), coke/refined petroleum (4,700 km), and iron/steel (4,000 km) all rely on sourcing raw materials from distant global locations. Notably, these are the same value chains we previously identified as having a high ["hidden energy"](#) imported content, indicating a dual vulnerability to both energy shocks and transport disruption.

Figure 6: Highest distance dependency for European tourism services, capital goods, and materials.

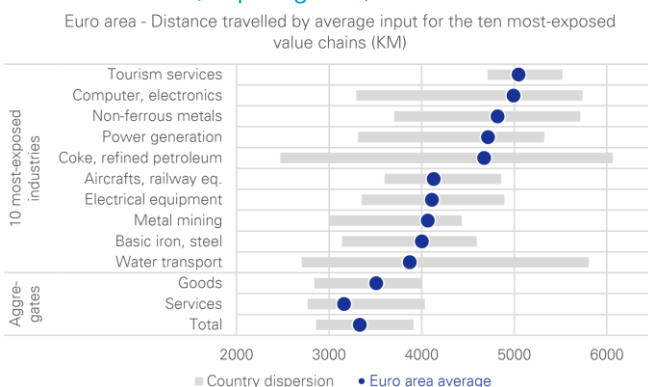
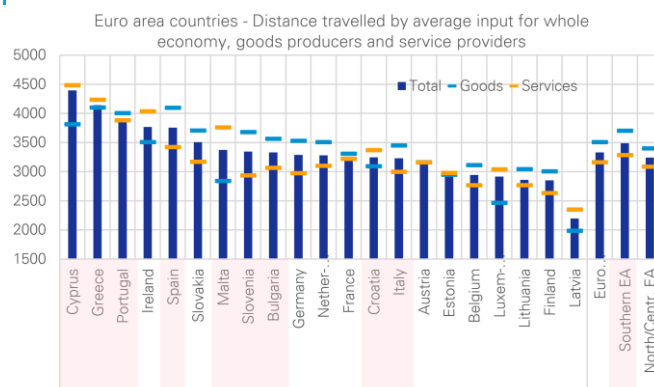


Figure 7: Longer distances in Southern European value chains



In terms of geographies, the simulations show that several Southern European countries have longer-than-average supply chains: a group including Cyprus, Greece, Portugal, and Spain are all positioned at the higher end of the spectrum (Figure 7). An average calculated for the nine Southern Euro area economies

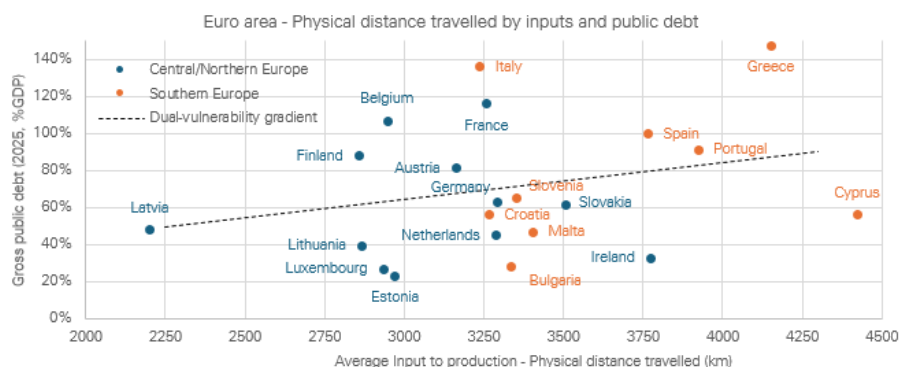


yields a kilometric distance of 3,490 km, a figure higher than the Euro Area average of 3,330 km. For goods producers within this southern bloc, the average distance is even longer, at 3,700 km.

The higher exposure to transport frictions could, therefore, create a vulnerability for some of the zone's high-debt member states (Figure 8). These countries are also those we previously found to have the highest energy import bill sensitivities.

Italy and Croatia are exceptions to this pattern, showing shorter supply chain distances more aligned with the industrial core of the Euro area, like Germany and France (3,200-3,300 km).

Figure 8: A vulnerability to transport disruptions for some of the highest-debt member States



Source: Deutsche Bank Research, OECD ICIO

What this means for the ECB: Measured tightening...at most

A negative supply shock pushes prices up and activity down. These opposing forces can neutralise the impact on monetary policy ('look through'). The ECB is nervous that the size and persistence of the upside inflation shock might need a tightening of monetary policy to ensure that inflation remains anchored at 2% in the medium term. However, the downside risks to growth are not irrelevant. The more significant the downside risks to growth, the less likely is a large or persistent inflation shock. Monetary policy needs to carefully weigh the inflation effects against the growth effects.

If Europe suffers from an outright shortage of transport fuels, then growth will undoubtedly weaken more than expected (assuming no substitutability of fuels or other sources of transport). Shortages could also push transport prices up even more, but only if firms believe demand can sustain even sharper price hikes, and this cannot be taken for granted – especially when the fiscal response to the crisis is limited and financial conditions are tightening.

In other words, if fuel supply shocks intensify to the point of outright shortages in Europe, it is not obvious that monetary policy would have to tighten even more than expected. It is even possible that monetary policy tightens less.



Our view is that the downside risks to growth are becoming more significant. One reason is the impact of the energy shock on supply chains. As discussed above, there are multiple aspects to this shock on supply chains.

First, we recently wrote about the hidden energy costs in Europe's imports, particular from Asia, a region where the risk of production disruptions from energy shortages is more acute.

Second, last year, we wrote about the high complexity (and low substitutability) of Europe's imports from Asia (China in particular). This increases the risk of an Asian shock compromising European supply chains.

Third, in this note, we offer a complementary perspective by showing the actual length ('physical distance travelled') of Europe's supply chains. The member states most at risk of an economic cost from a limited supply of transport fuels are those with longer supply chains. Generally speaking, they are also the ones with higher levels of public debt.

The bottom line is we see the net inflation/growth arguments favouring a measured ECB tightening at most. Our baseline is 50bp of hikes to 2.50%, with quarter point hikes in June and September. With the labour market cooling, financial conditions tightening and a threat of fuel shortages, the growth outlook is not robust enough to think the ECB hikes much beyond 2.50%. The challenge for the ECB could be fragmentation if fuel shortages have a differential impact on member states.



Appendix 1

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May 5, 2026

Q&A with Matthew Barnard

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As equity research adapts to rapid shifts in technology, markets and client behaviour, Matt Barnard, Head of US Equity Research, is helping steer how Deutsche Bank evolves its research offering. In this edition of our Q & A series, Matt shares his perspective on the new ways clients are consuming research, his outlook for US markets, and what it's like leading research teams through a period of significant change.

Q: You've spoken about the equity research landscape undergoing a fundamental shift. What has changed most in how clients discover, engage with and use research today?

AI is fundamentally changing how clients consume sell-side research. As information becomes more easily accessible through AI tools, the value of differentiated insights from trusted sources has moved higher up the value chain. Investors are increasingly using AI to discover new ideas and challenge consensus views more quickly and efficiently. This allows them to filter out noise and spend more time engaging in deeper, more meaningful conversations with our Deutsche Bank research analysts.

Q: Against a complex backdrop for US geopolitics and policy, how do you currently assess the outlook for US markets – and which themes matter most for clients as we move through 2026?

Geopolitical developments in the US and globally continue to be a significant driver of volatility across risk assets. Deutsche Bank research's macro and strategy teams have consistently taken bold, often counter-consensus views, which are especially valuable in this environment. While conflicts in the Middle East will likely continue to create ripple effects across markets, the underlying fundamentals for US equities remain relatively strong. The rapid acceleration of AI adoption – and the associated capital investment – remains a key thematic focus for clients, both in the US and globally.

Q: You now lead teams through a period of rapid change. What do you find most rewarding about working in research management at this point in the cycle?

At its core, the role of an equity research analyst is to form forward-looking views with imperfect information. The strongest analysts thrive in periods of uncertainty and rapid change, identifying emerging trends and uncovering new investment opportunities early. From a management perspective, the



Q&A with Matthew Barnard

most rewarding part of my role is seeing some of the brightest minds in research stay ahead of these shifts and deliver true alpha for our clients.



Q&A with Matthew Barnard



Q&A with Matthew Barnard

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Deutsche Bank
Research

The Great 2026 Reset...

May 2026

Jim Reid | Galina Pozdnyakova | Henry Allen | Rajsekhar Bhattacharyya

Deutsche Bank Research Institute

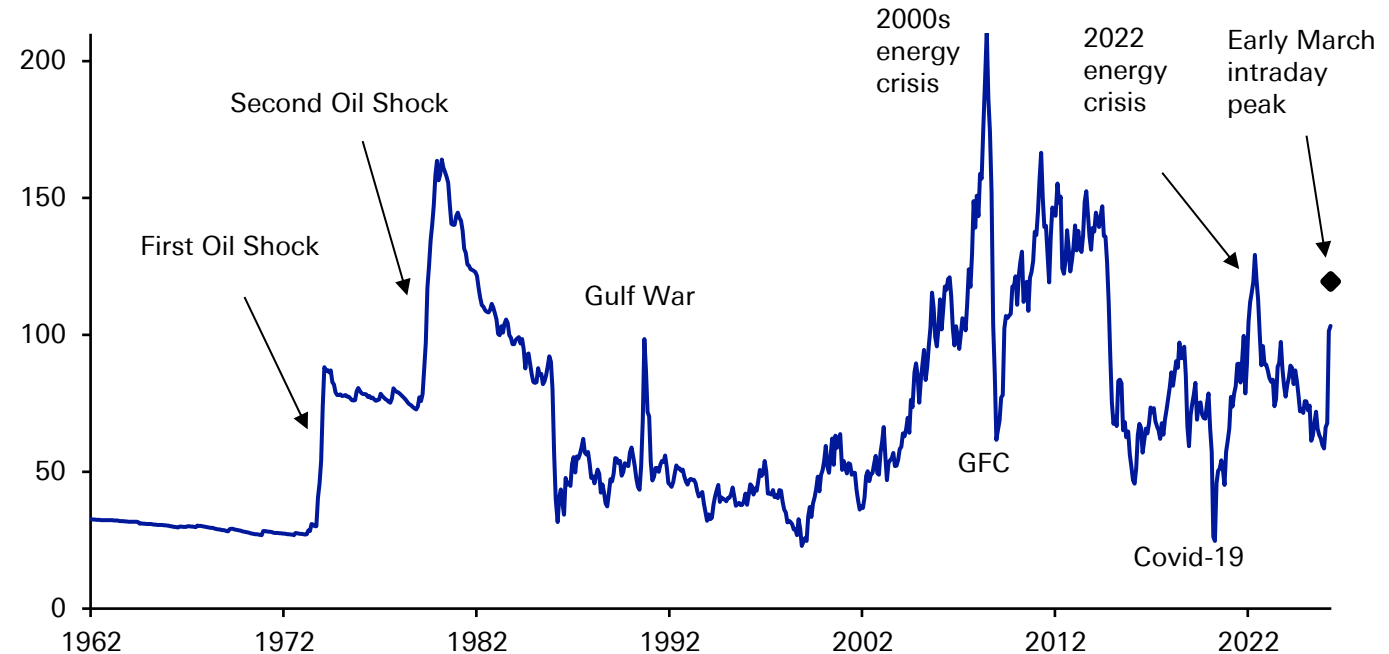
IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

Deutsche Bank
Research Institute



Today's oil shock is now becoming one of the bigger ones of recent decades... but despite the headlines, it's some way from being the most serious ever...

WTI Oil Prices in real terms (\$/bbl)

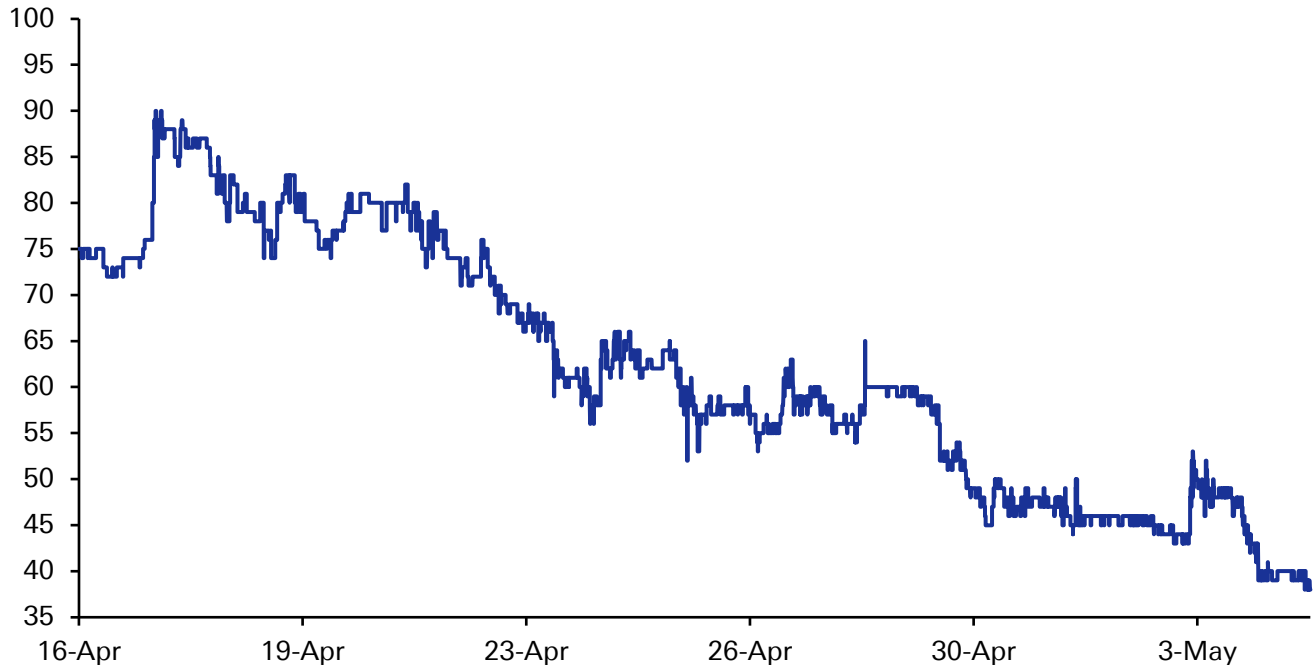


Source : Finaeon, Bloomberg Finance LP, Deutsche Bank.

Probability markets (and likely global investors) are becoming increasingly pessimistic about an imminent return to normality...



Polymarket probability (%) of the Strait of Hormuz traffic returning to normal by the end of June 2026.

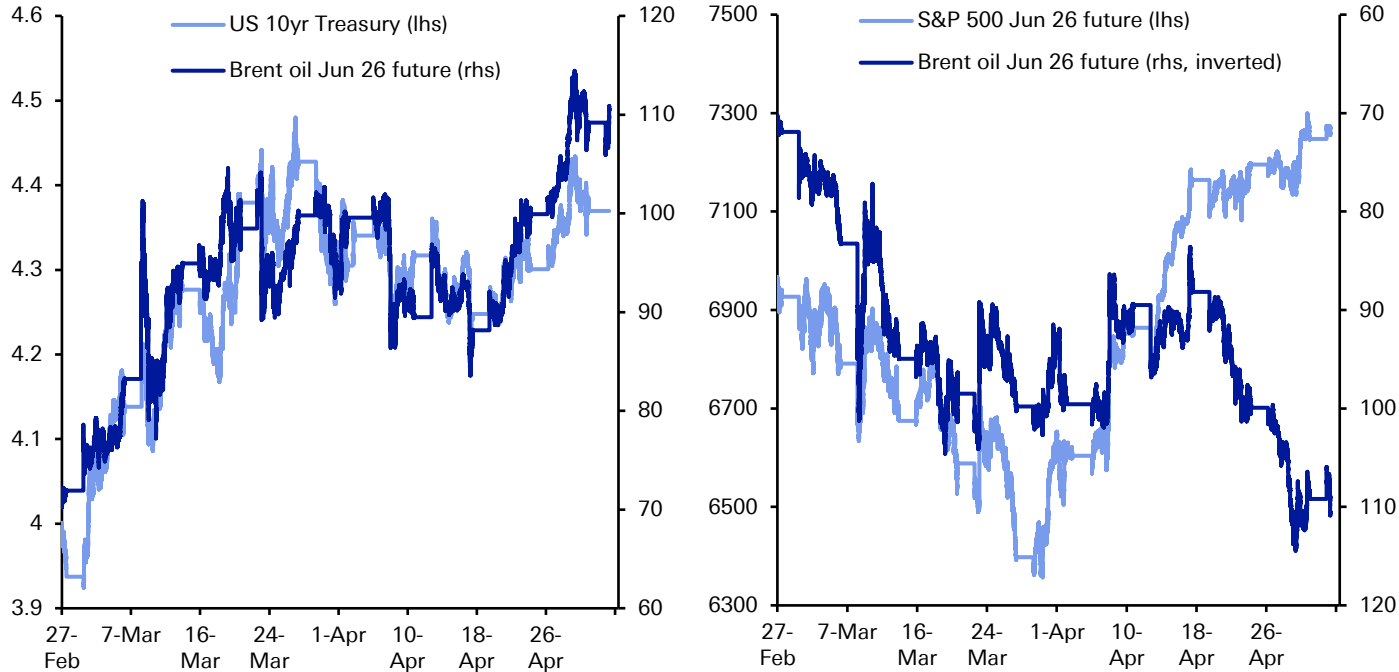


Source: Bloomberg Finance LP, Deutsche Bank, Last update: May 5



However, there's been an interesting divergence between equities and rates... the 10yr Treasury is still trading in a tight correlation with oil. US equities have now seen a big decoupling...

Correlation between 10yr Treasury yields, S&P 500 futures, and Brent crude oil

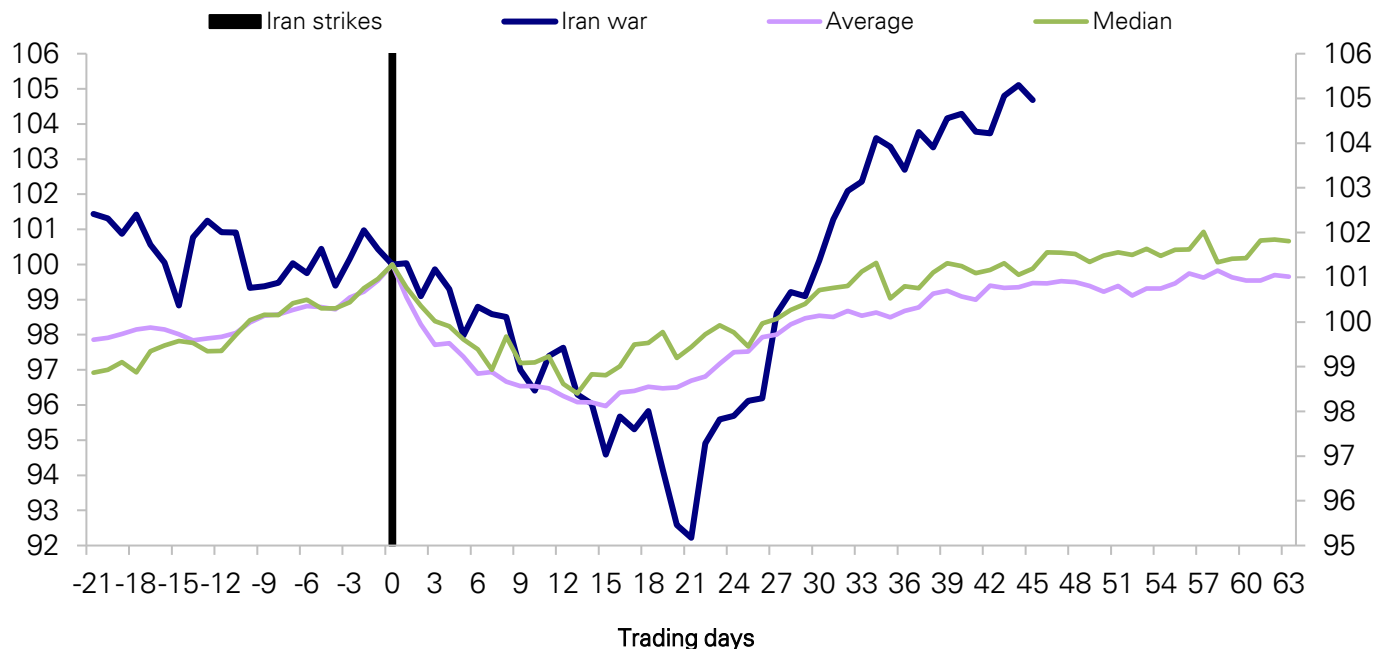


Source : Bloomberg Finance LP, Deutsche Bank.

... Indeed the historical playbook around geopolitical shocks and US stocks has (broadly) worked again...



S&P 500 around Geopolitical events since 1939

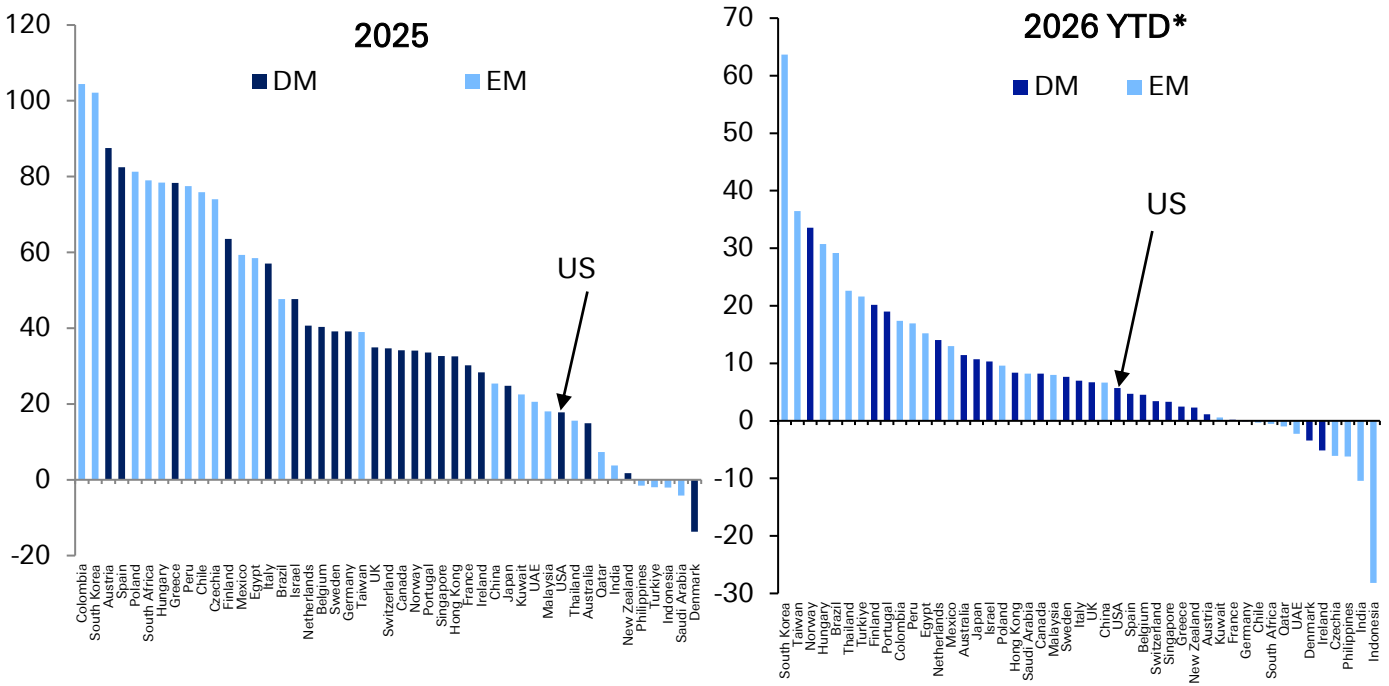


Source: Bloomberg Finance LP, Deutsche Bank, Last update: May 4



Even though the US equity market has rebounded strongly, it remains just below the middle of the global pack YTD 2026 even if it has risen up the leaderboard since the start of the war...

Bloomberg large/mid-cap country stock market indices: 2025 total return in USD (%) (left) and 2026 YTD (right)

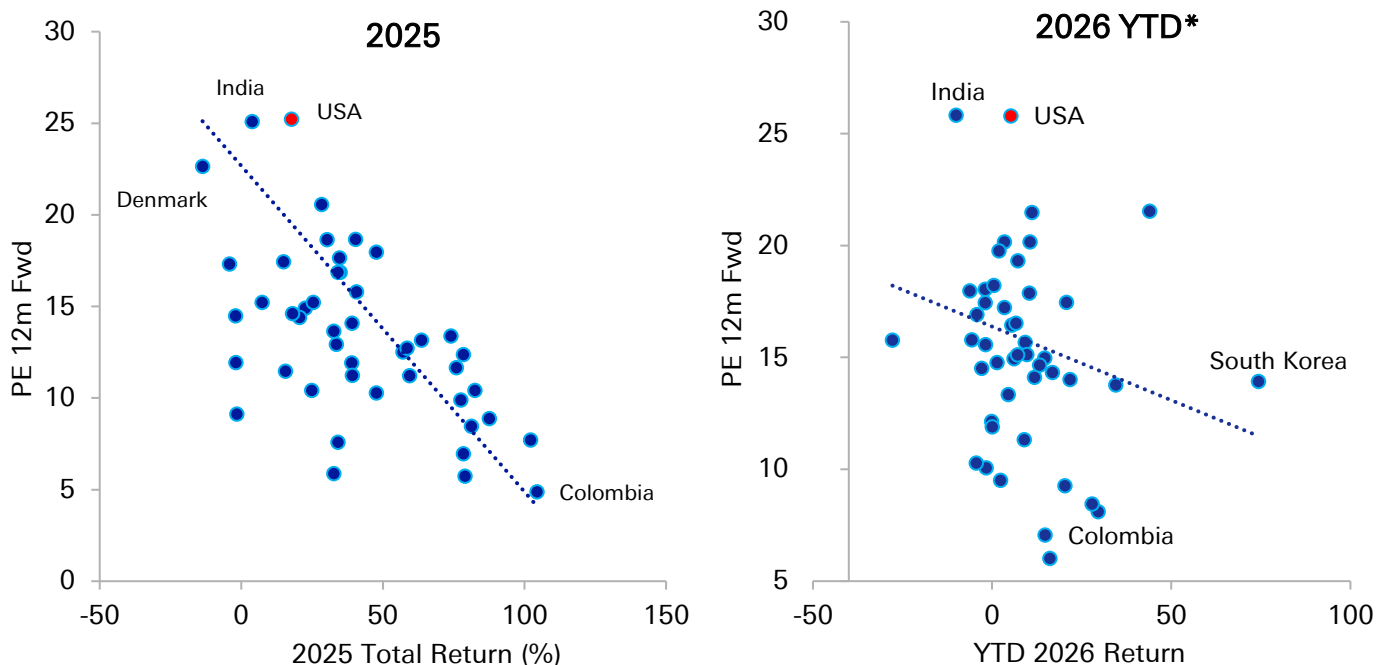


Source: Bloomberg Finance LP, Deutsche Bank, *Note: 2026 YTD is updated through May 1



2025 Global Equity returns saw a strong correlation to starting valuations. In 2026 there remains a slight bias that way but the war seems to have complicated returns a bit more...

Bloomberg large/mid-cap country stock market indices, 2025 total return in USD vs. 12m fwd PE estimates at the end of 2024 (left) & Bloomberg large/mid-cap country stock market indices, 2026 YTD return in USD vs. 12m fwd PE estimates at the start of 2026 (right)

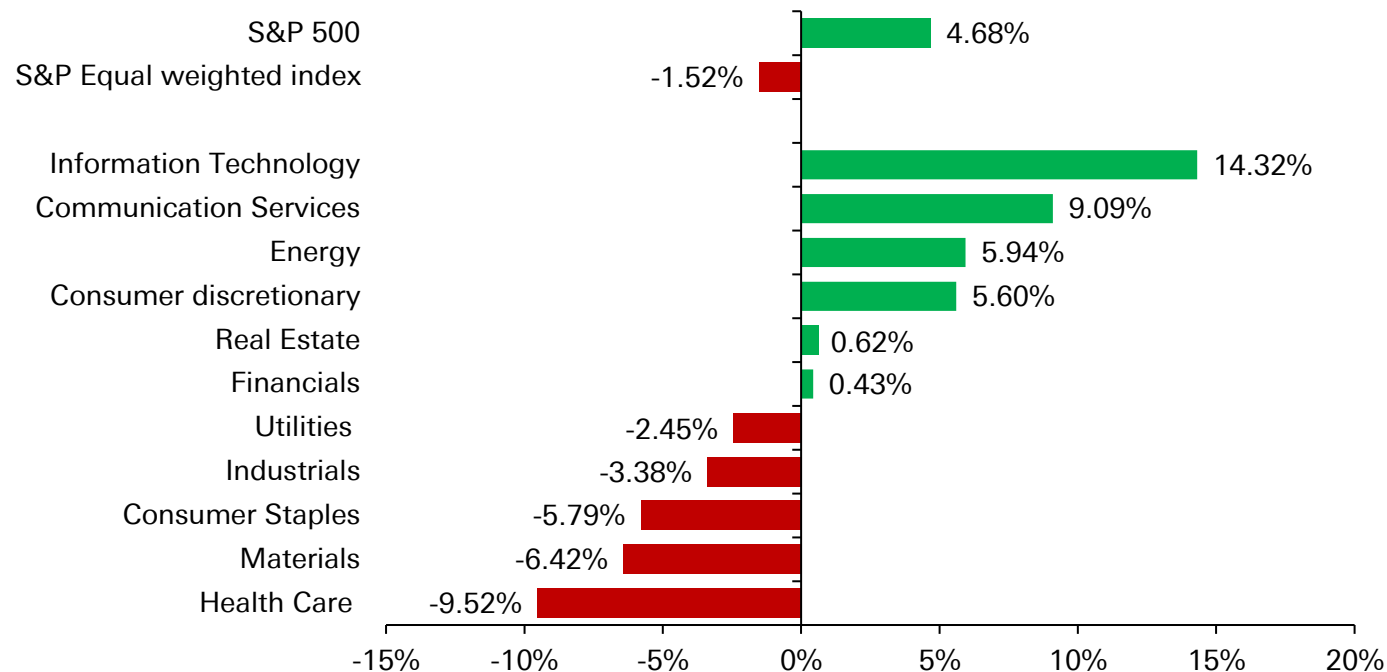


Source: Bloomberg Finance LP, Deutsche Bank *Note: 2026 YTD is updated through May 4

The US climb up the leaderboard since the war begun has been helped by a rebound in tech.



S&P Sectors return since Feb 28 : Middle East War



Source : Bloomberg Finance LP, Deutsche Bank; Updated through May 4



As an example of this tech outperformance, Intel's share price has finally exceeded its high back in 2000. A positive of course but is it a warning from the past as to how long it can take for stocks to recover after a valuation surge..

A Historic comeback: Intel's stock finally tops its 2000 high

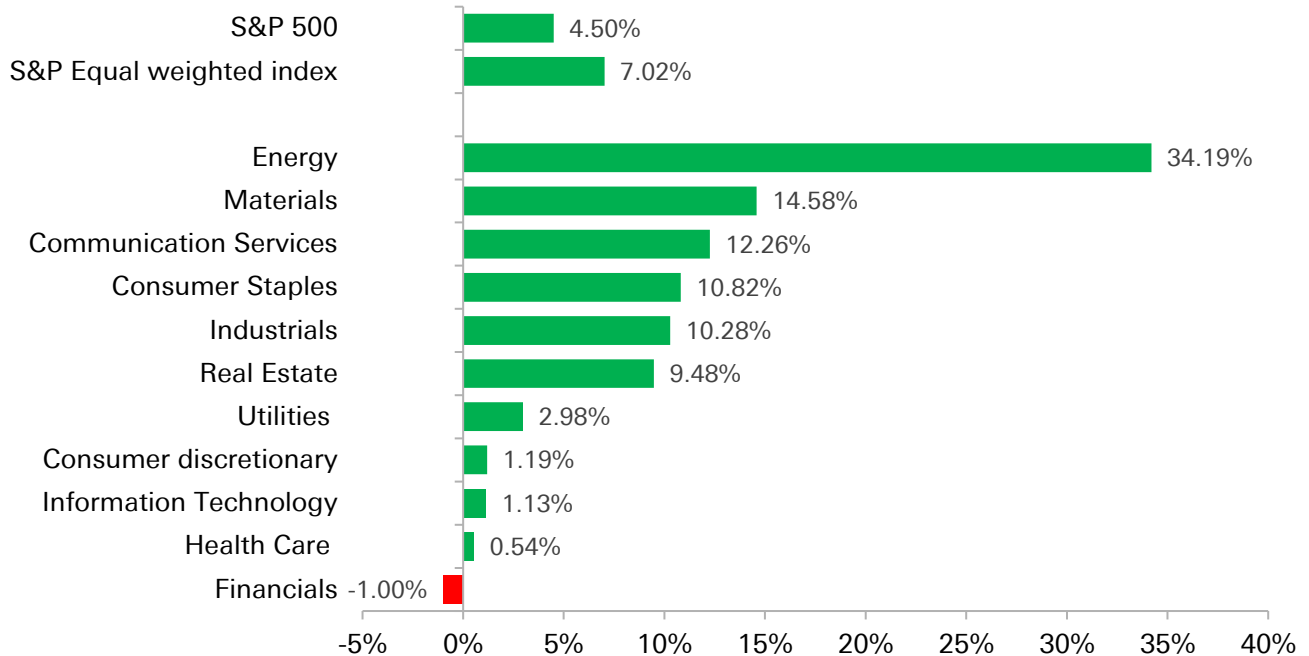


Source : Bloomberg Finance LP, Deutsche Bank, Last update: May 4



Zooming out, US tech is broadly flat in the 6 months since the end of October which is why other global markets have outperformed since this point...

S&P Sectors return since October 29, 2025

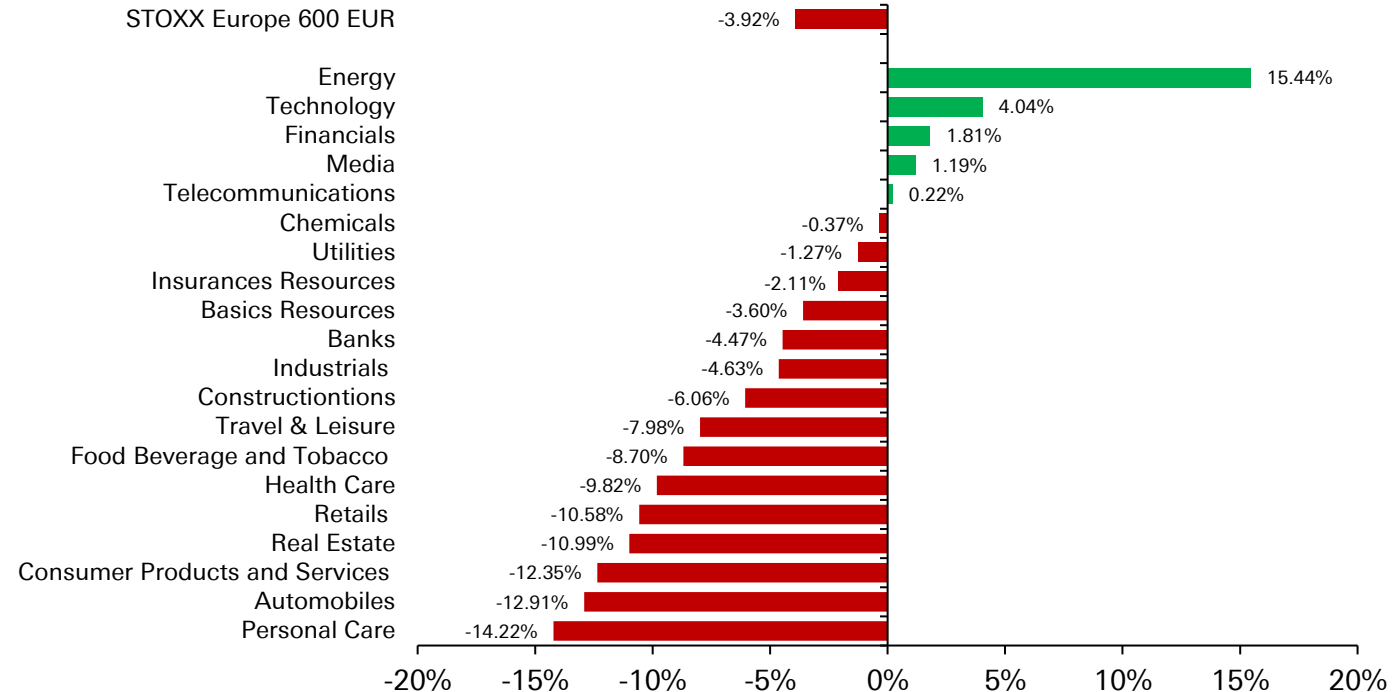


Source : Bloomberg Finance LP, Deutsche Bank., Last update : May 4



Given its exposure to the energy shock, Europe's STOXX 600 has struggled since the Iran conflict began... although energy stocks are the clear exception, with tech also outperforming...

STOXX Europe 600 Sectors return since Feb 28 : Middle East War

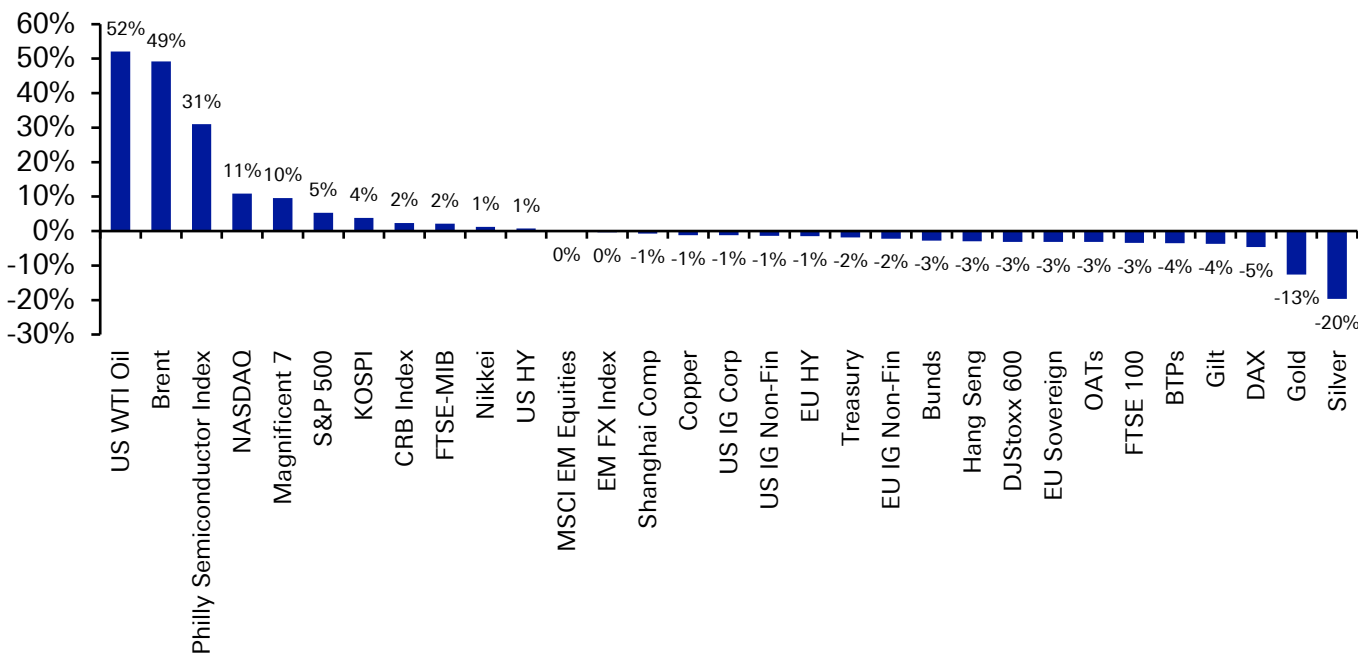


Source : Bloomberg Finance LP, Deutsche Bank., Last update : May 5



Looking at a global level, oil has unsurprisingly been the standout performer since the war began, followed up by US equities... at the other end are precious metals and European assets...

Total Return Performance of Selected Global Assets since the Iran conflict began (USD terms)

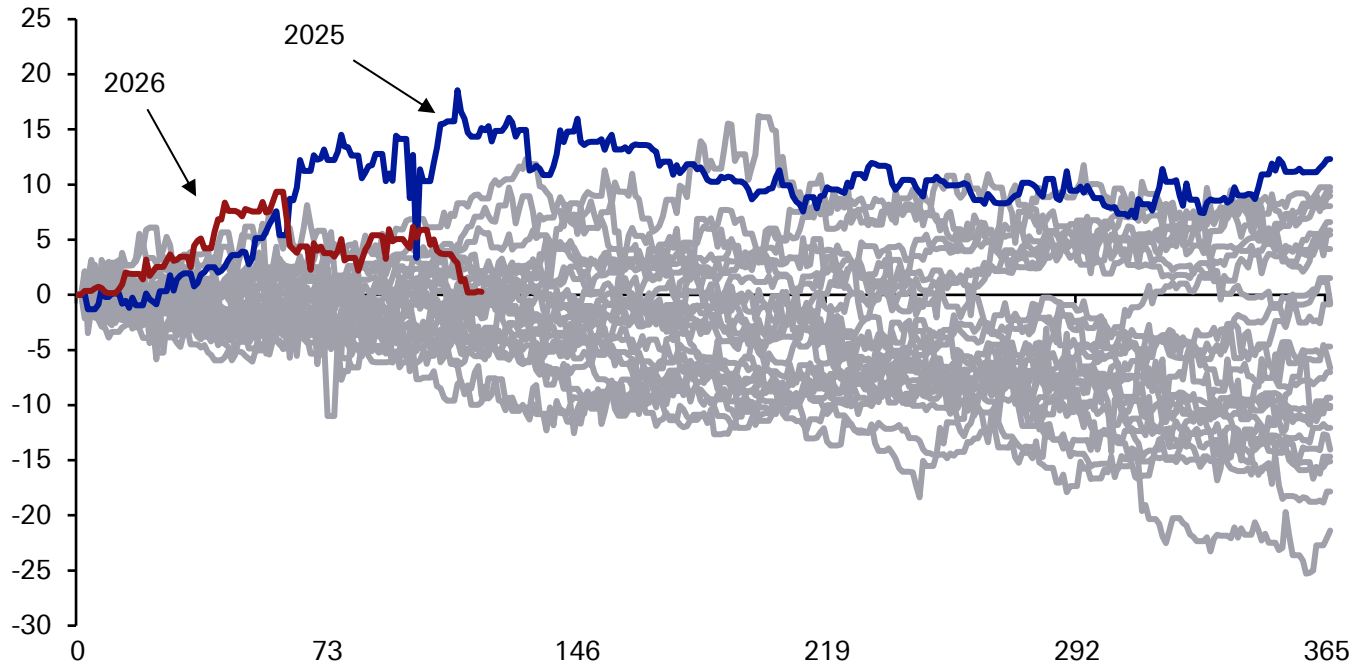


Source: Bloomberg Finance LP, Deutsche Bank, Note: Equities, credit and bonds shown on total return basis, FX and commodities shown on spot return basis



2026 had initially looked like another year of US equity underperformance until the Iran conflict, but Europe's exposure to the energy shock and the US tech outperformance has meant US equities are more in line with the rest of the world again...

Outperformance (percentage points) between the MSCI World ex US index and the MSCI US Index since 2000

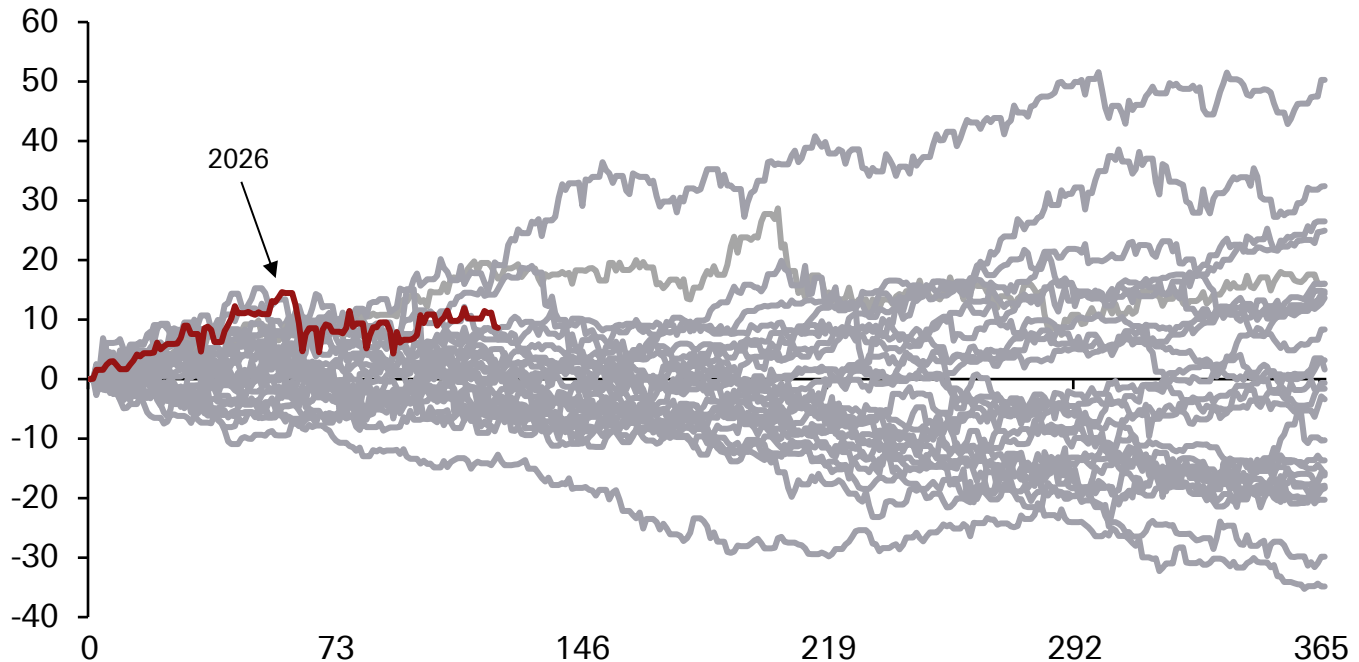


Source: Bloomberg Finance LP, Deutsche Bank

That said, the outperformance of EM equities this year has persisted, despite the Iran conflict...



Outperformance (percentage points) between the MSCI EM index and the MSCI US Index since 2000



Source: Bloomberg Finance LP, Deutsche Bank



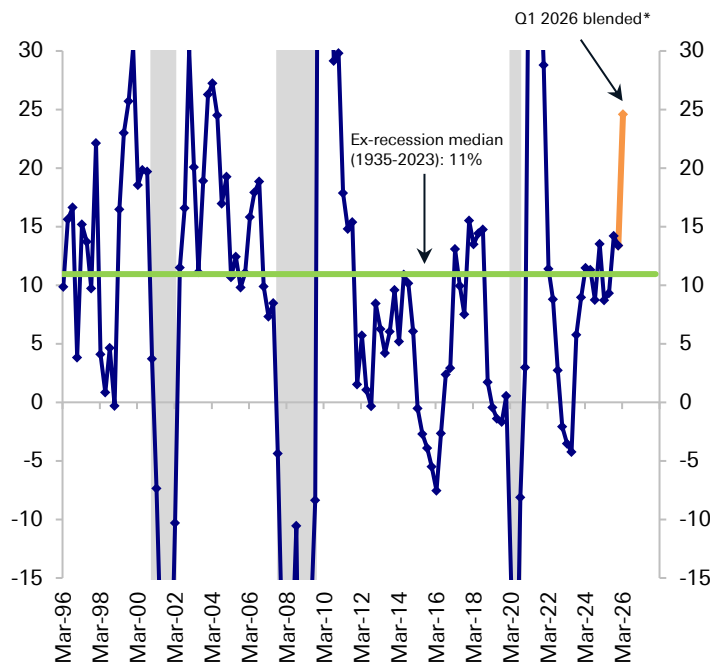
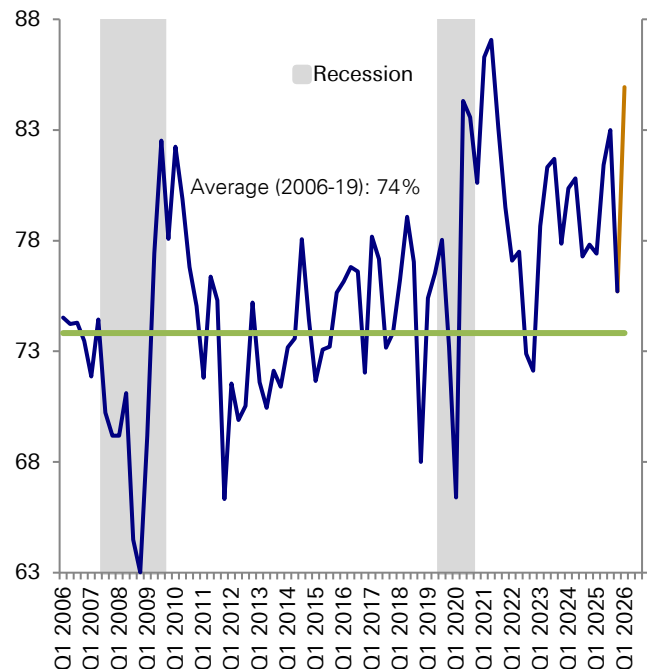
Q1 US earnings season has been great
though...

According to our equity analysts, this is one of the best earnings seasons in 20 years. See Parag Thatte and Binky Chadha's piece [here](#) for more...



Proportion of S&P 500 beating earnings estimate (%)

S&P 500 quarterly earnings growth* (% yoy)

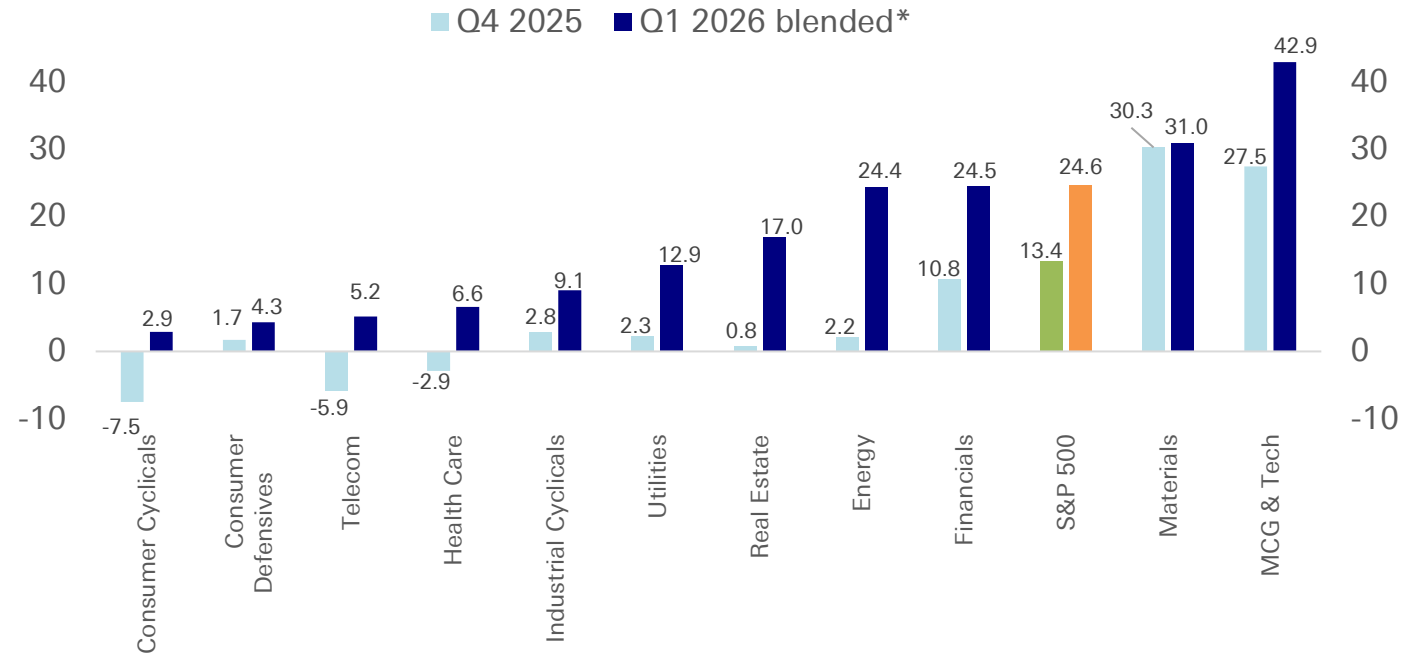


Source: Bloomberg Finance LP, Deutsche Bank Asset Allocation, *Excluding 2018 tax cut; blended assumes companies that report beat at the current rate



All 11 top level sectors are expected to show YoY earnings growth for the first time in 4 years... See Parag Thatte and Binky Chadha's piece [here](#) for more...

Earnings growth by S&P 500 sector group (% yoy)

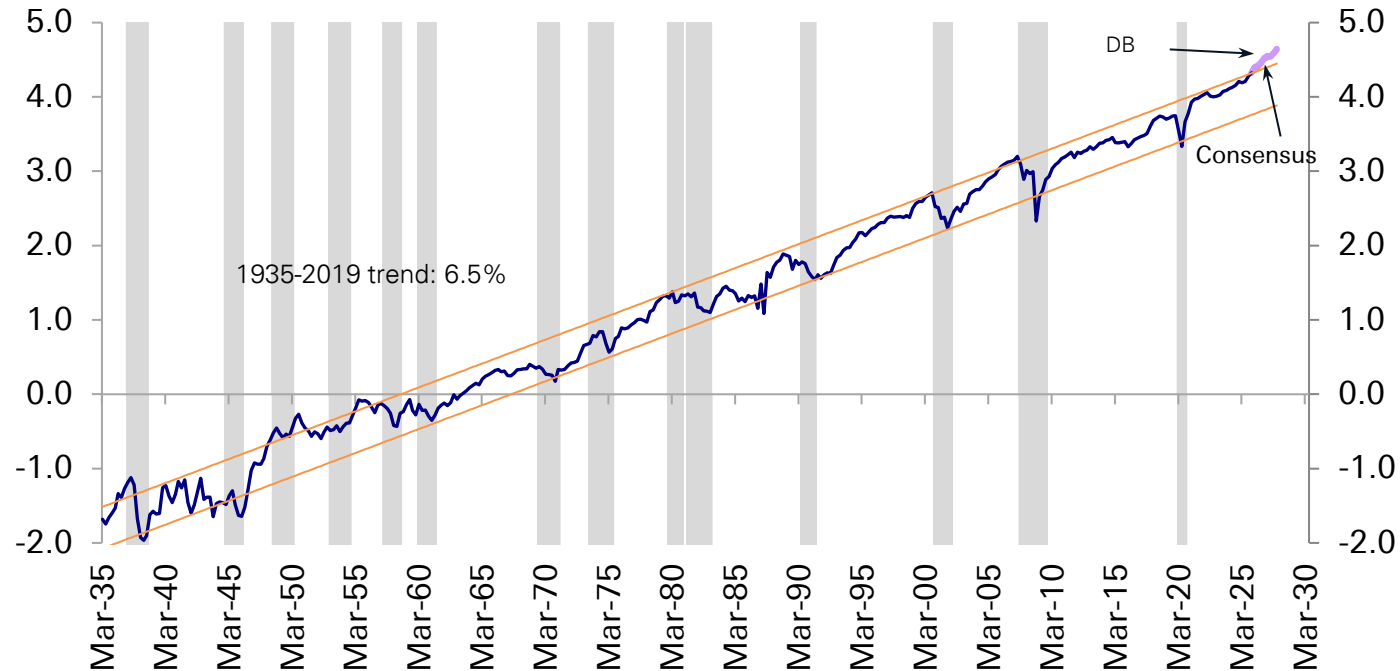


Source: Bloomberg Finance LP, Deutsche Bank Asset Allocation; * Blended assumes companies that report beat at the current beat rate



Remarkably, S&P 500 earnings growth is likely to break out on the upside of a 90-year channel even though real and nominal GDP, and productivity have been higher for long periods of time over this period. See Parag Thatte and Binky Chadha's piece [here](#) for more. Is this a new paradigm or unsustainable?

S&P 500 quarterly EPS seasonally adjusted (log)



Source: Bloomberg Finance LP, Deutsche Bank Asset Allocation

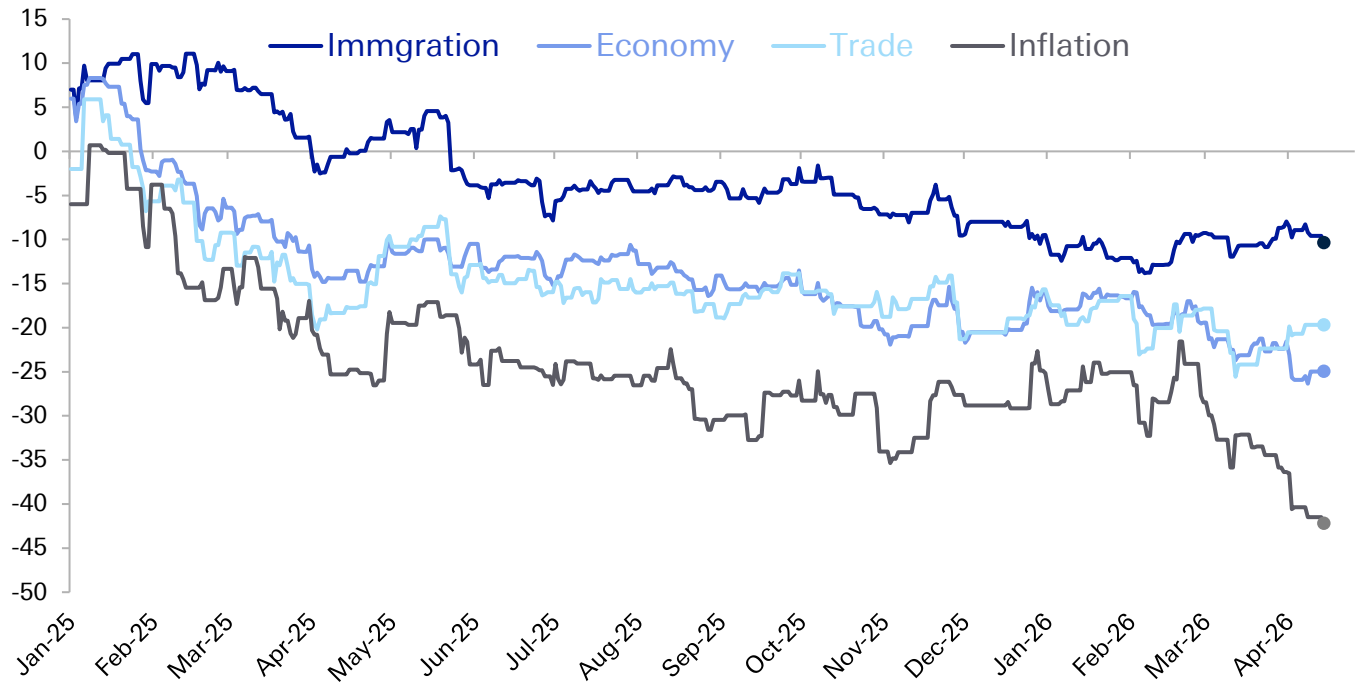


The longer the conflict/stand-off lasts the
tougher it is politically...



Concern over inflation and the cost of living is also a prominent political issue that's been hurting incumbents around the world in recent years...The War in Iran seems to have had a further impact. The US midterm elections are just 6 months away now as well...

An updating polling average of Donald Trump's net approval rating on immigration, the economy, trade, and inflation, accounting for each poll's quality, recency, sample size, and partisan lean

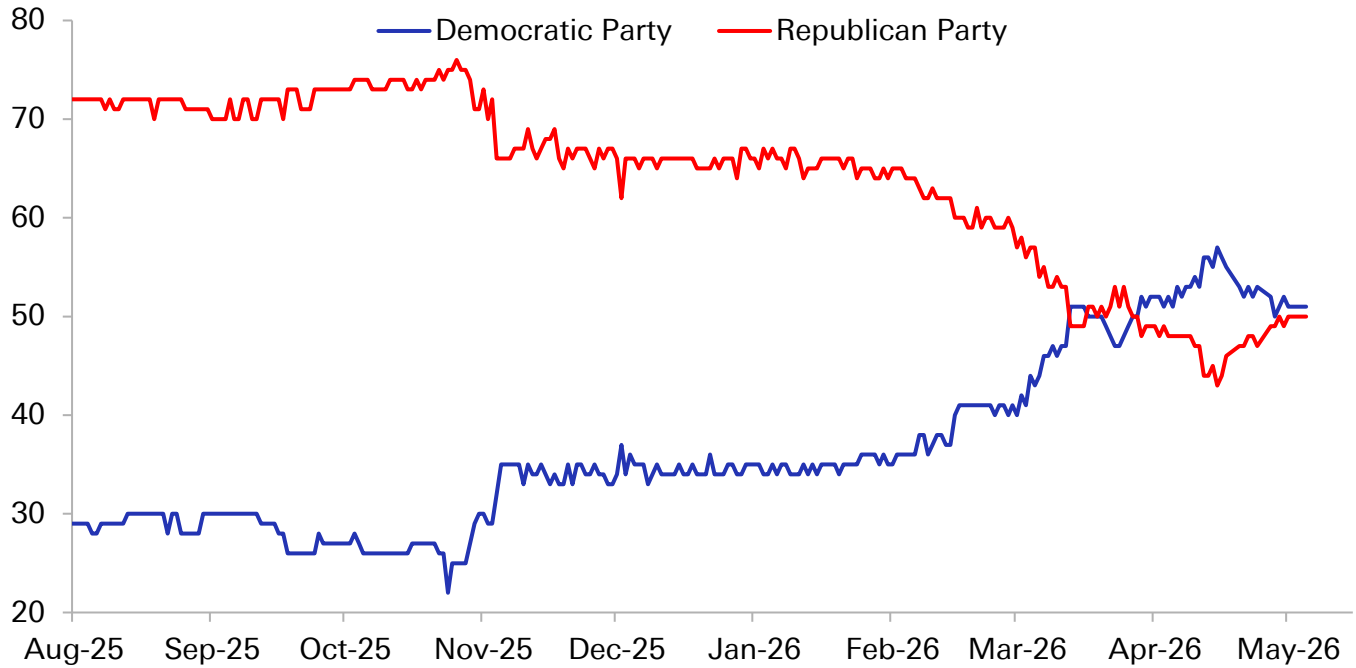


Source: Nate Silver, Deutsche Bank, updated through May 4



Since around the time the Iran strikes began, the Democrats have become the (narrow) favourites to retake control of the Senate... that would have a big impact on appointments like Fed Governors, cabinet members and Supreme Court justices, which all require Senate approval...

US Polymarket Party control Senate after 2026 Midterm election

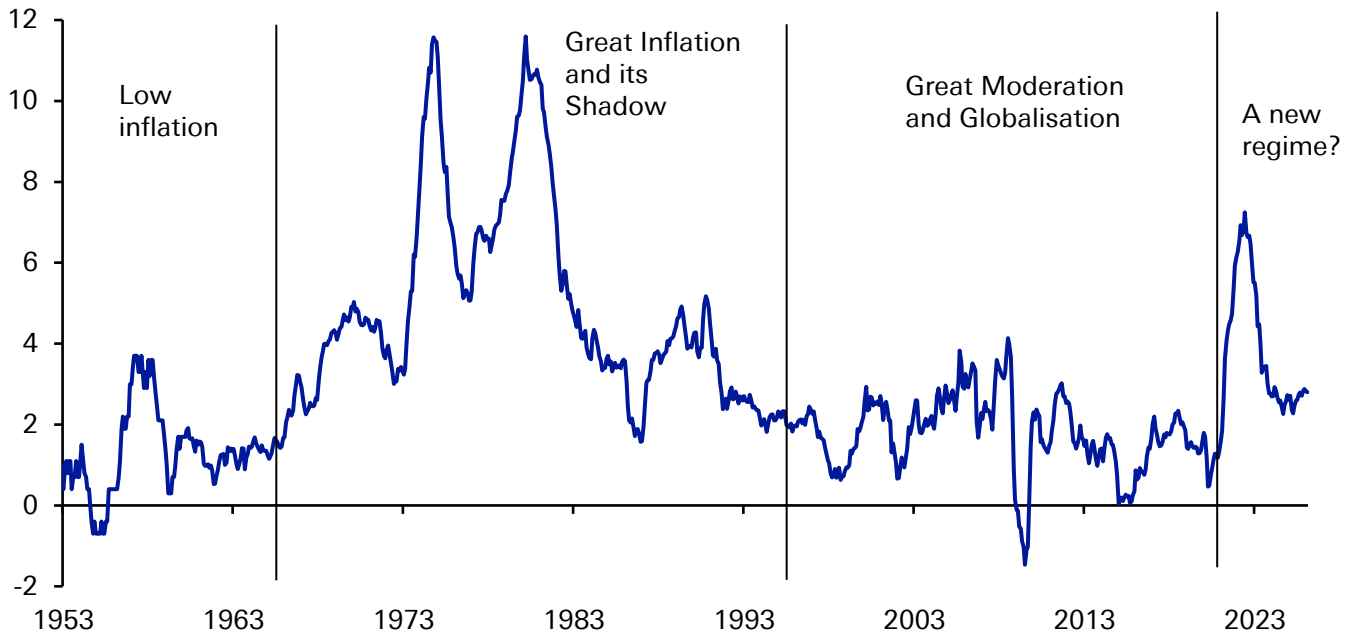


Source : Bloomberg Finance LP, Deutsche Bank, updated through May 5



Linked to the inflation approval rating, the Fed's target measure of PCE inflation has now been above the 2% target for 5 years... This is very different to the low-inflation era before the pandemic... could we be in a new regime?

US PCE Inflation (year-on-year % change, using CPI before PCE begins in 1959)

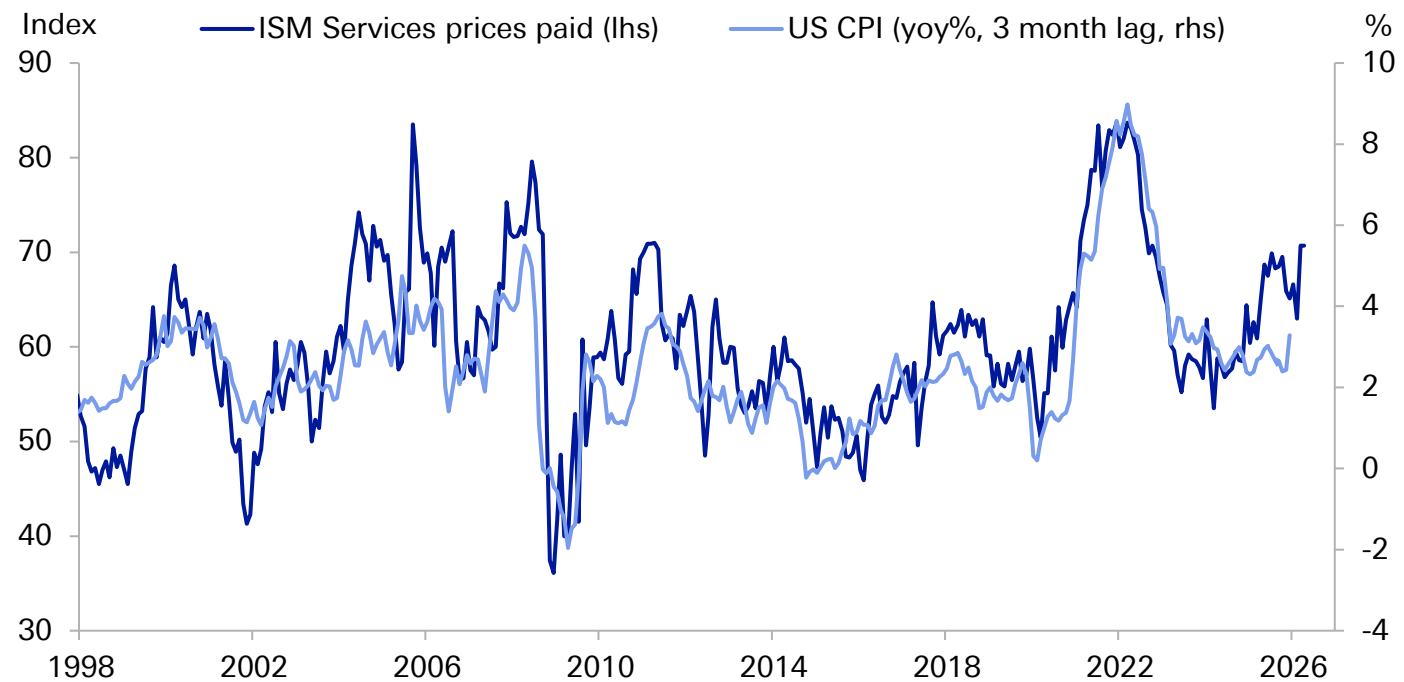


Source: Haver Analytics, Deutsche Bank

The prices paid component of the ISM Services is a strong leading indicator of inflation with a 3-month lag... It's just hit its highest since 2022...



ISM Services Prices Paid and US CPI (with a 3-month lag)

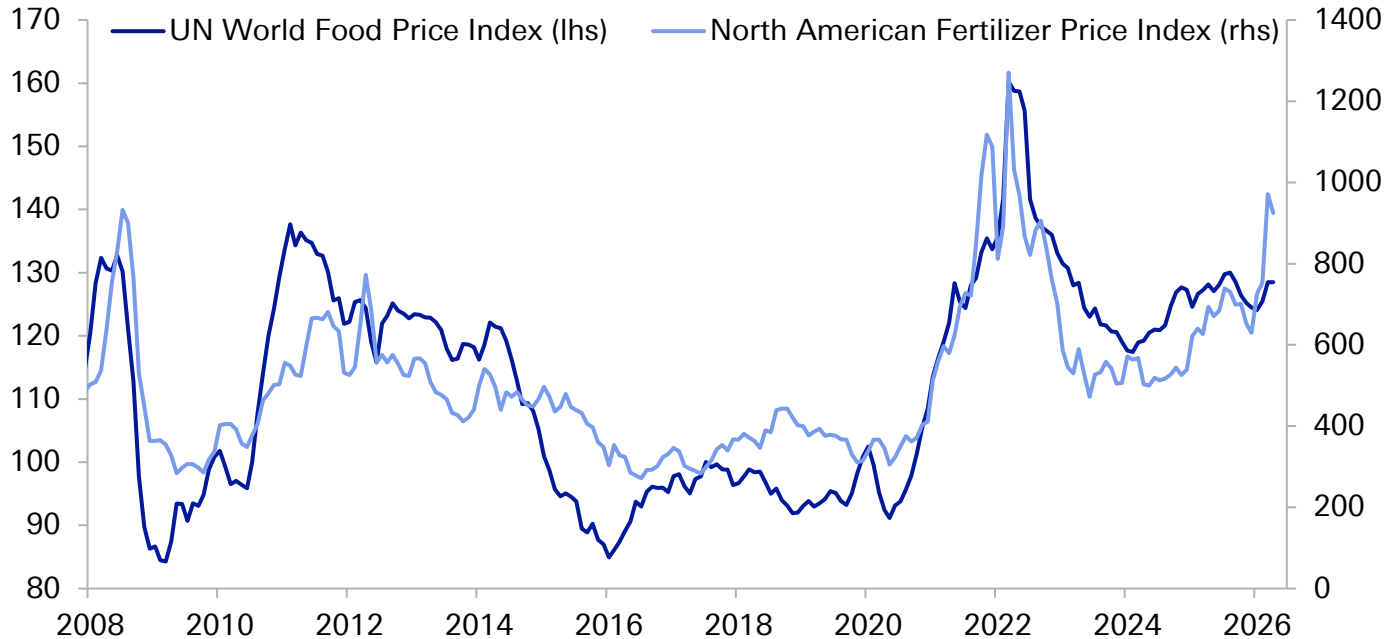


Source: ISM, BLS, Haver, Deutsche Bank



Another concern is that food price inflation will also start to rise, and we've already seen a rise in fertiliser prices... this could be particularly problematic as food is an essential good, so unlike other parts of the consumer basket, it's difficult to substitute away from...

Food prices and fertiliser prices

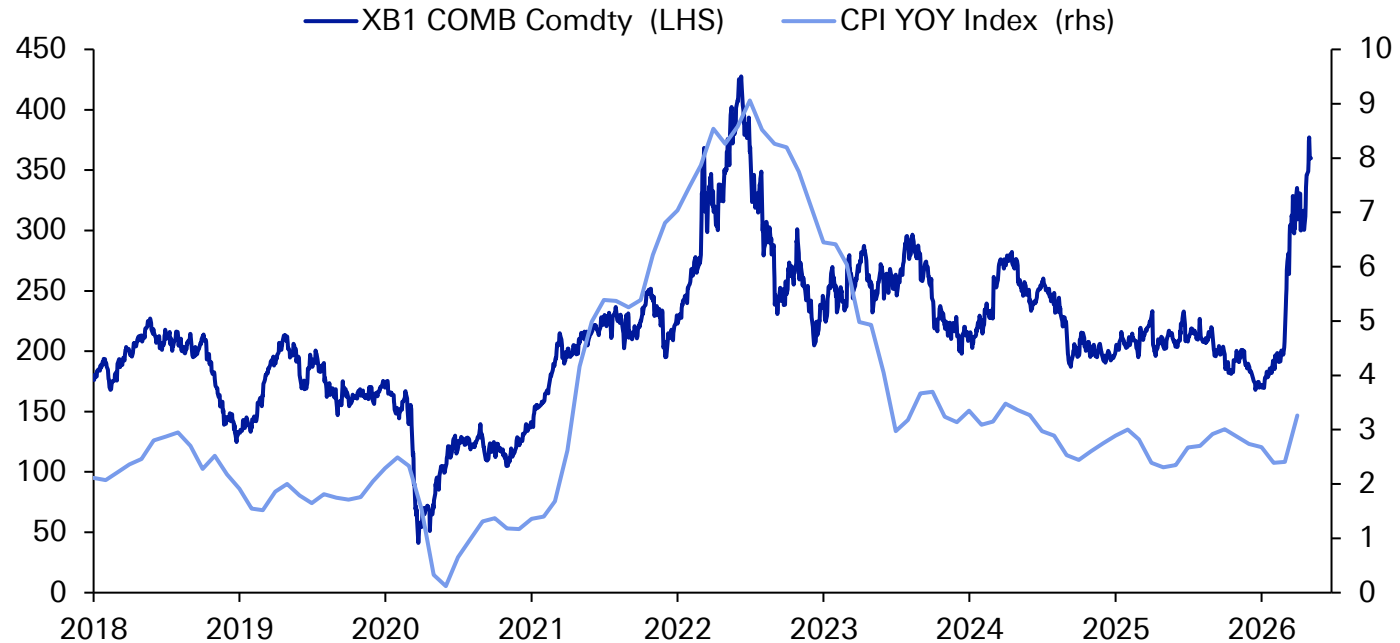


Source: Bloomberg Finance LP, Deutsche Bank

This inflation has been quite obvious to consumers, as higher gasoline prices have passed through quickly...



Gasoline prices (XB1) vs. CPI Inflation

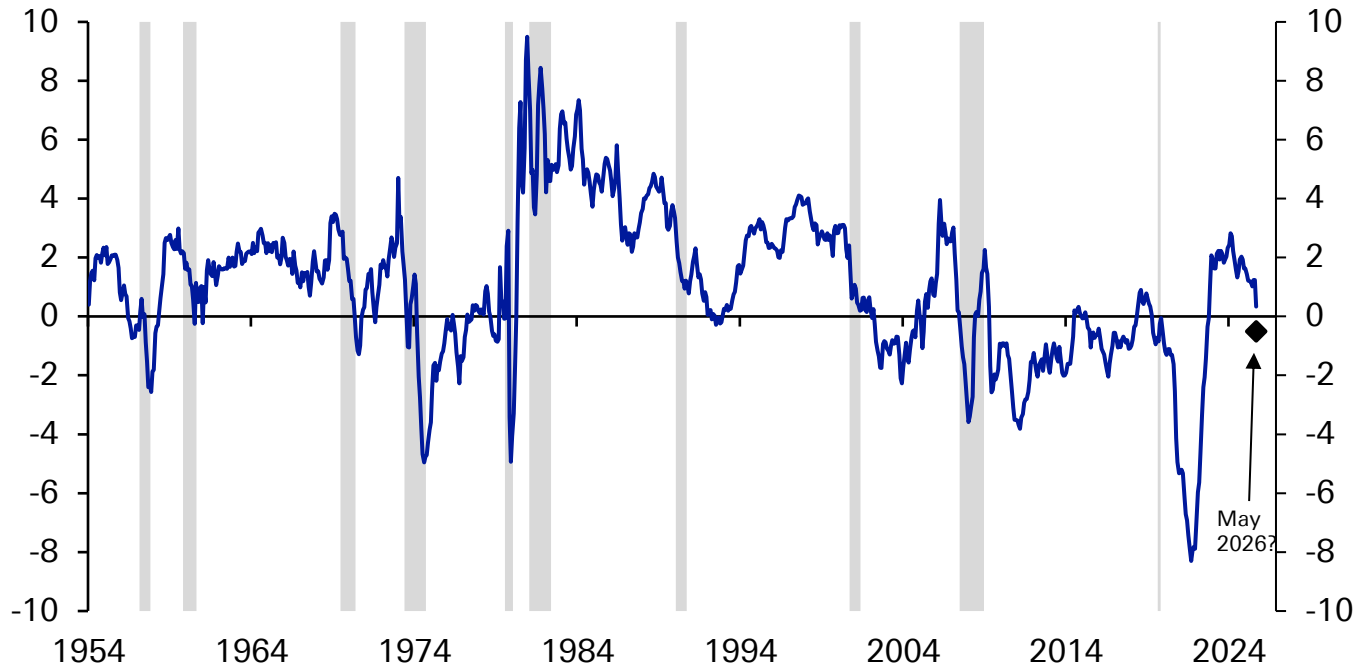


Source: Bloomberg Finance LP, Deutsche Bank; Last update May 4



The real federal funds rate (simply fed funds minus year-on-year CPI) is set to head back into negative territory because of the energy shock... under DB's US econ forecasts, it would be back to -0.5% in May...

Real federal funds rate (fed funds minus year-on-year CPI)



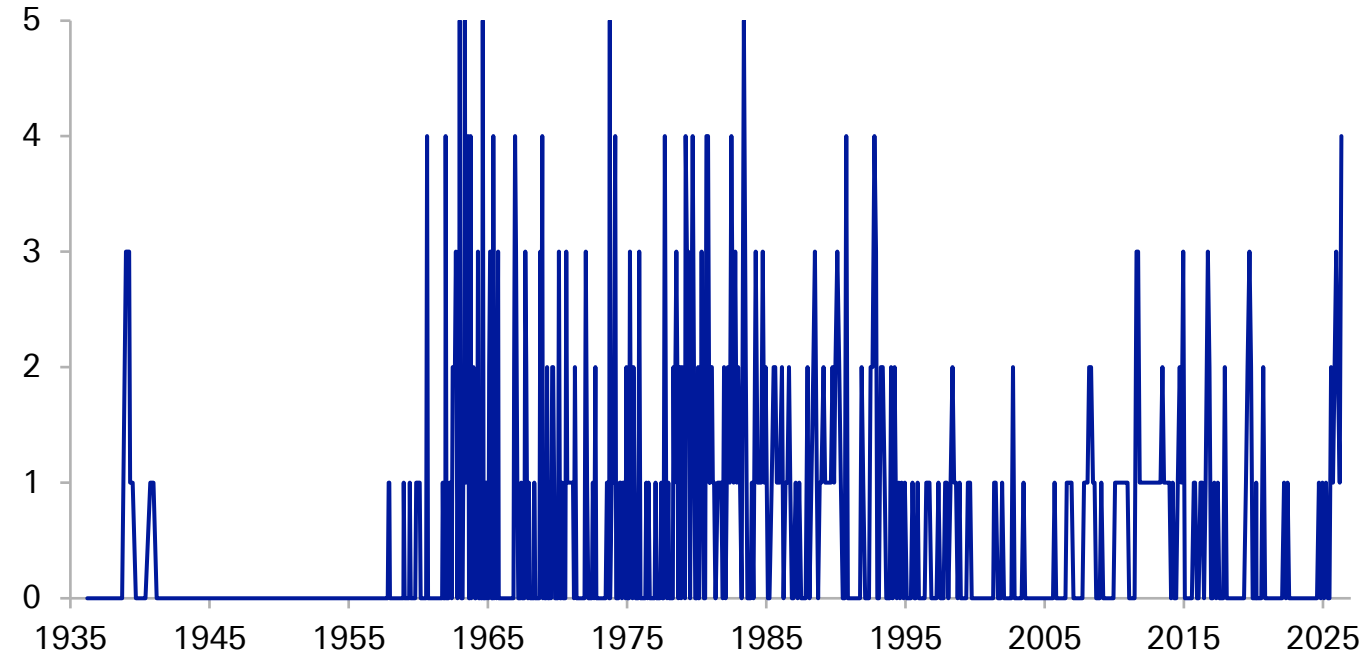
Source: Haver Analytics, Deutsche Bank



The energy shock has exacerbated recent FOMC disagreements. The April meeting saw 4 members dissent from the decision in some way, the most dissenters on a vote since 1992...

US FOMC Dissents Votes Against Action

of persons



Source: Bloomberg Finance LP, Deutsche Bank

As summer holidays approach, the risk is high stress for the aviation sector and travellers.....



Jet Fuel Singapore FOB Cargoes (USD/barrel)

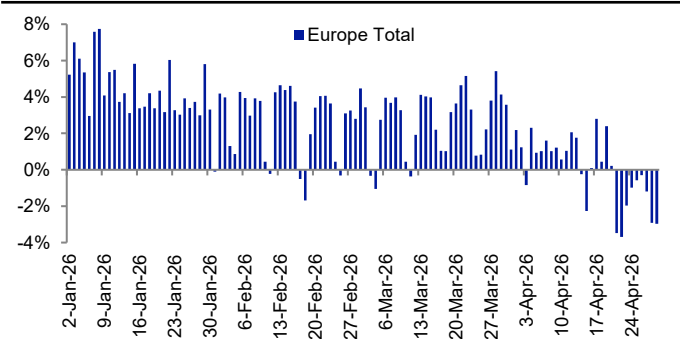


Source : Bloomberg Finance LP, Deutsche Bank.

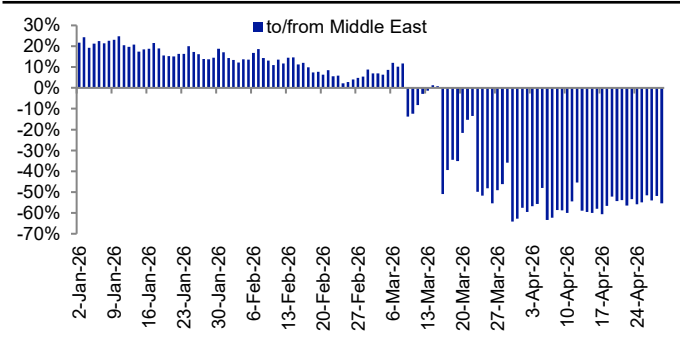


The number of flights to and from Europe are trending down. Initially because of a cancellation of most Middle East flights but increasingly other flights are being cancelled due to being unprofitable at current fuel prices...

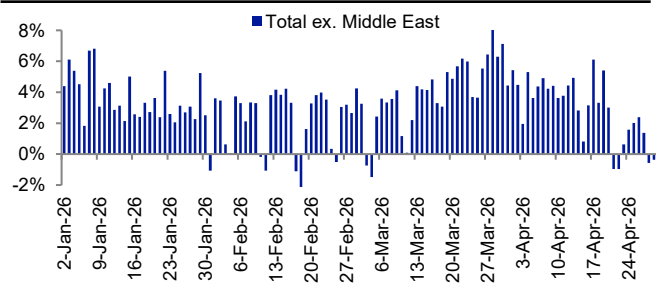
Daily flights to/from Europe (%y/y)



Daily flights to/from Middle East (%y/y)



Daily flights to/from Europe ex Middle East (%y/y)



Source: Cirium, Deutsche Bank; Note: The Diio Mi sourced data in this report has been extracted from a Cirium product. Cirium has not seen or reviewed any conclusions, recommendations or other views that may appear in this document. Cirium makes no warranties, express or implied, as to the accuracy, adequacy, timeliness, or completeness of its data or its fitness for any particular purpose. Cirium disclaims any and all liability relating to or arising out of use of its data and other content or to the fullest extent permissible by law

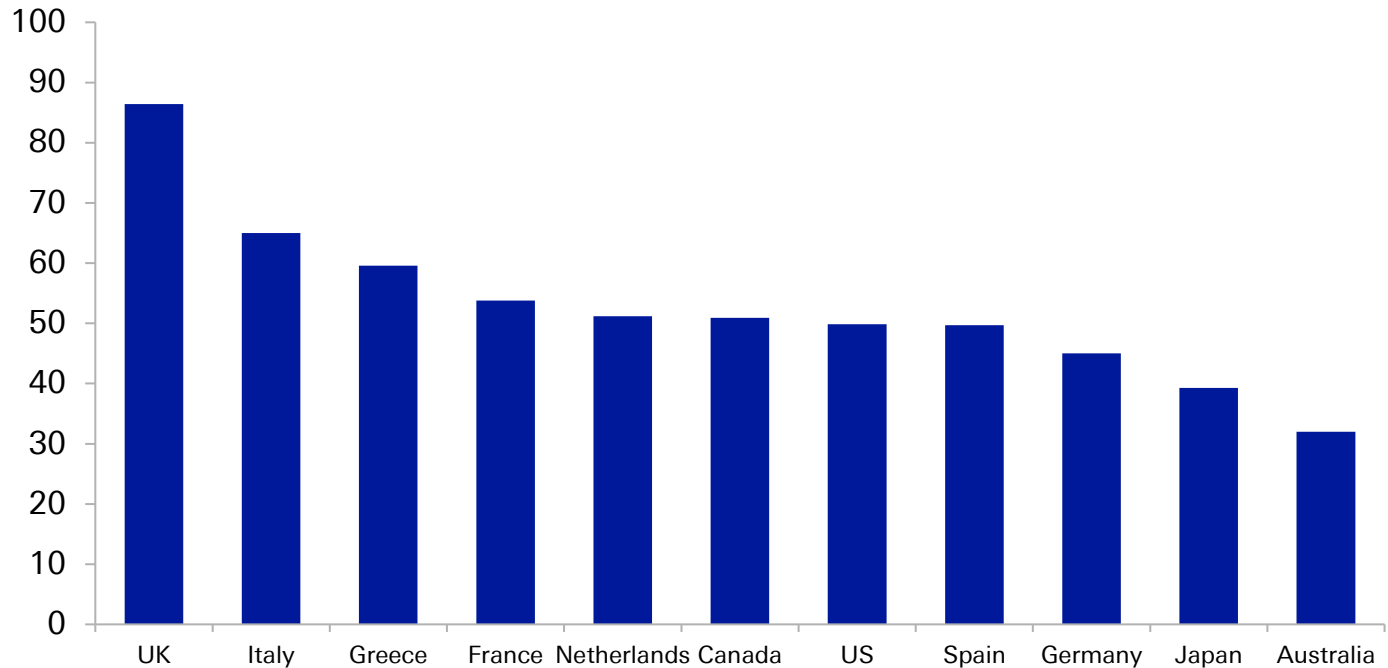


Rates are under pressure...



Yields have risen substantially since the conflict began a couple of months ago... which will constrain governments from providing fiscal support packages... The UK is underperforming on additional political concerns that any move to replace the PM could bring additional spending plans....

Change in 10y yield since Feb 27th (bps)

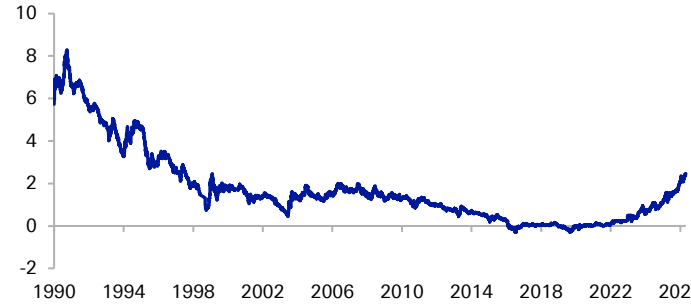


Source: Bloomberg Finance LP, Deutsche Bank, Last update May 5



Long-term government bond yields are up to multi-year highs... Japan's 10yr yield is above 2.5% for the first time since 1997, UK 30yr gilt yields are the highest since 1998, and Germany's 30yr yield is the highest since 2011 during the Euro crisis... Even 30yr USTs are with 10bps of 18yr highs

Japan 10yr bond yield (%)



UK 30yr gilt yield (%)



Germany 30yr bond yield (%)



US 30yr Treasury yield (%)



Source: Bloomberg Finance LP, Deutsche Bank

In fact, on a total return basis, US Treasuries still haven't got back to their peak before the 2021-22 inflation spike...



iBoxx US Treasuries (Total Return Index)

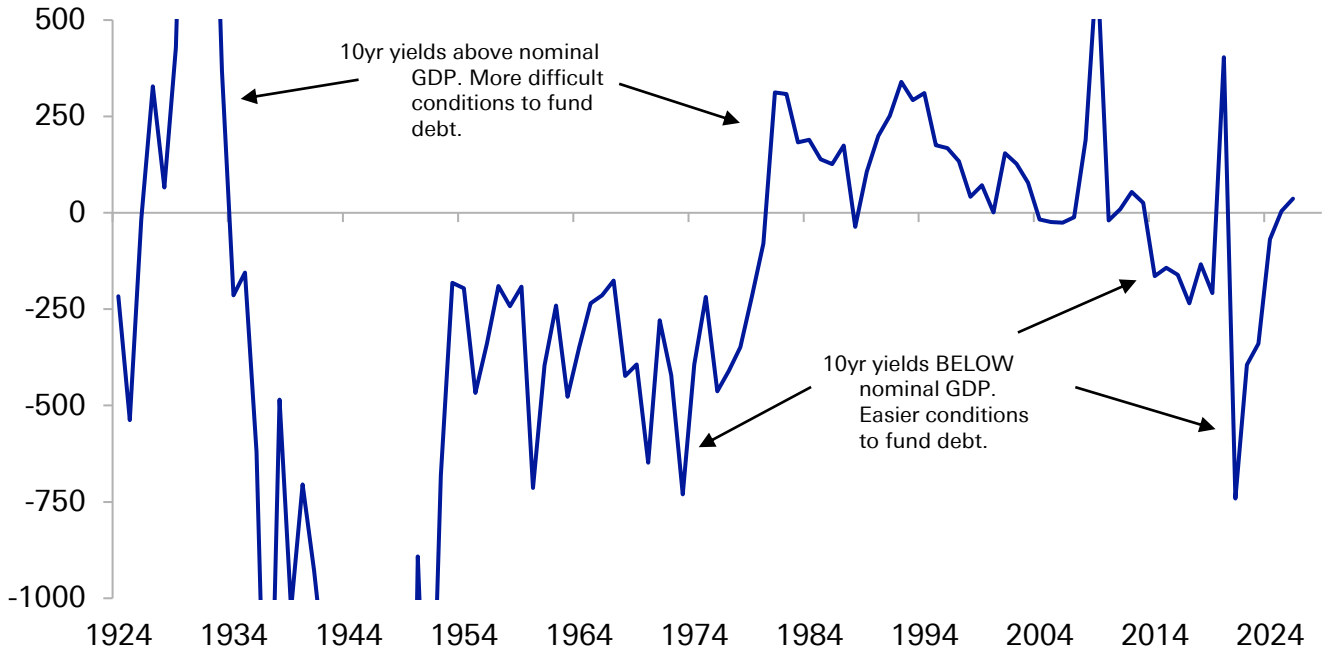


Source: Bloomberg Finance LP, Deutsche Bank



This backdrop of higher yields comes as DM Funding costs (using 10yr yields as a proxy) are now getting close to or even above Nominal GDP growth. In the post-GFC period of high debt and financial repression, yields were consistently below nominal GDP.... But that's now over, making it more difficult for governments to cushion economic shocks...

Average difference between Nominal GDP growth and 10yr yields for US, UK, Germany, France, Italy and Japan

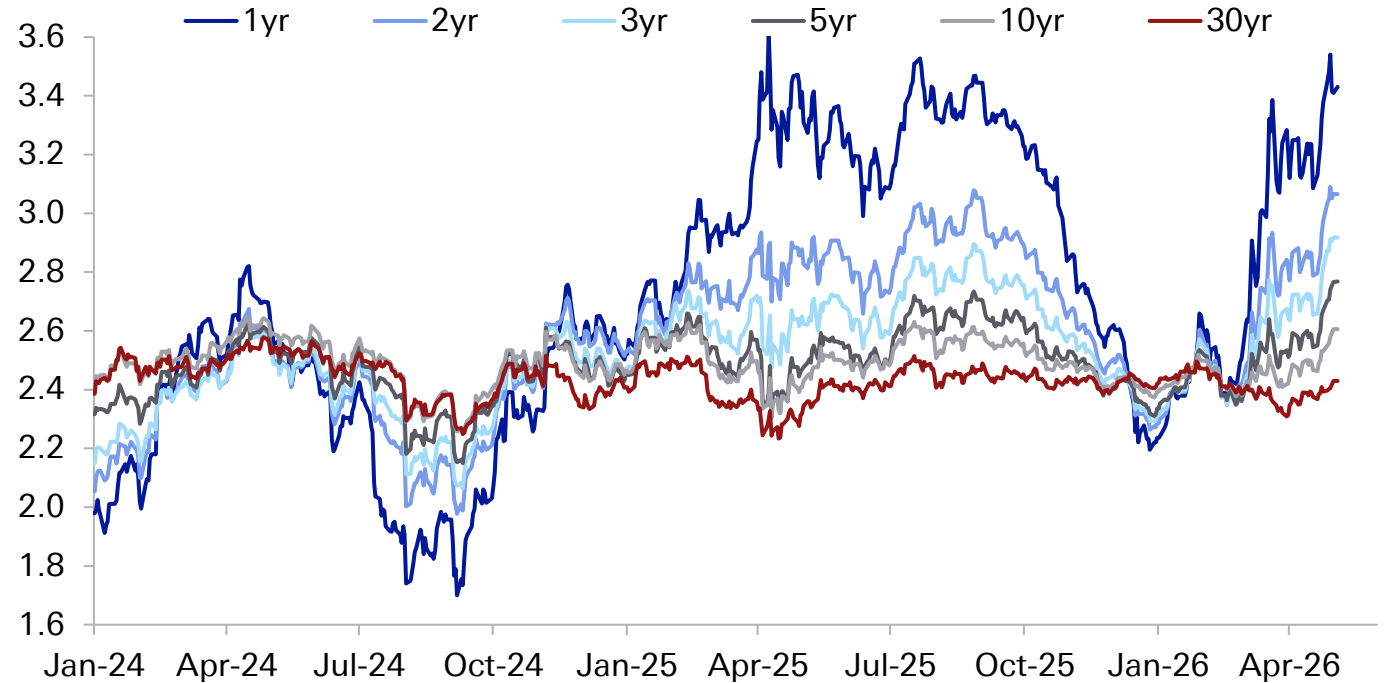


Source : Finaeon, Deutsche Bank



One of the problems for rates is that we've now had two US inflation expectations shocks in the last 12 months. Tariffs and now energy... this follows a period where inflation calm had been restored after the shock of 2021-22.

US Inflation Swaps (%)

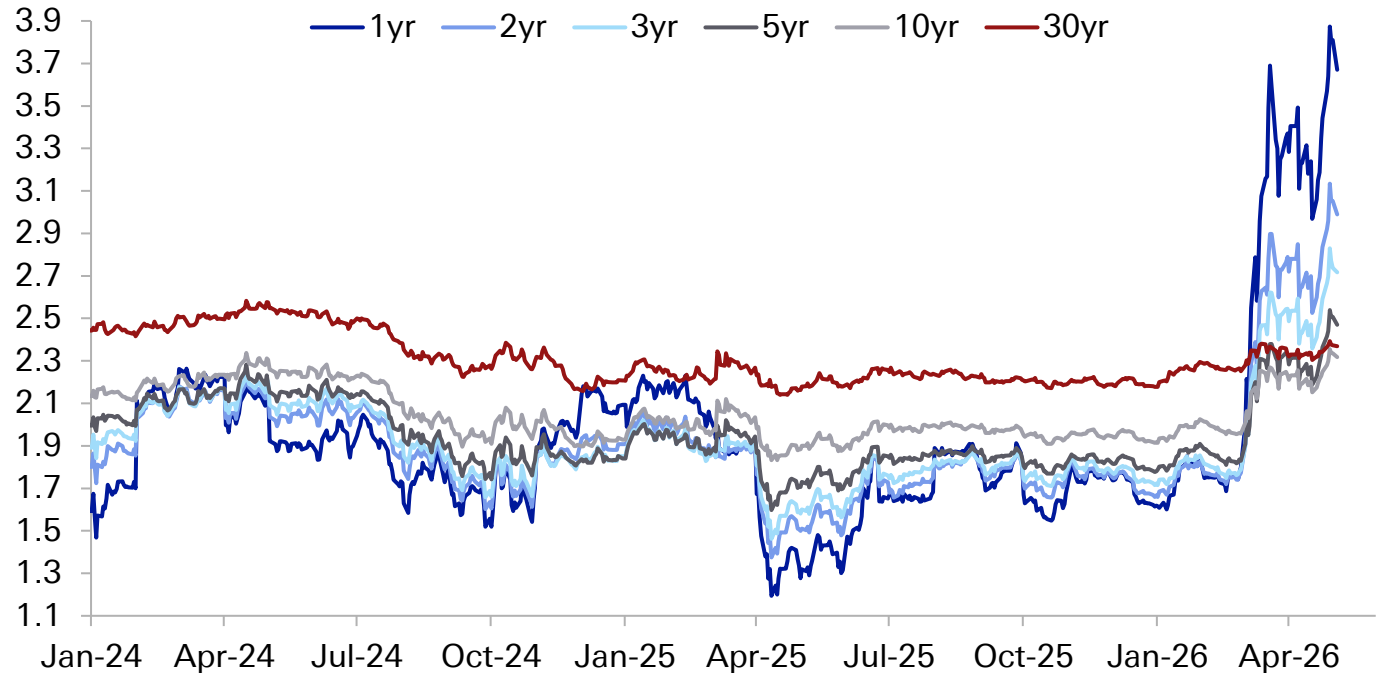


Source: Bloomberg Finance LP, Deutsche Bank



Europe had got inflation expectations back to target levels and didn't see a shock due to US tariffs (in fact the opposite). However, that has changed since the war in Iran, and markets are now pricing in multiple ECB rate hikes this year...

EUR Inflation swaps

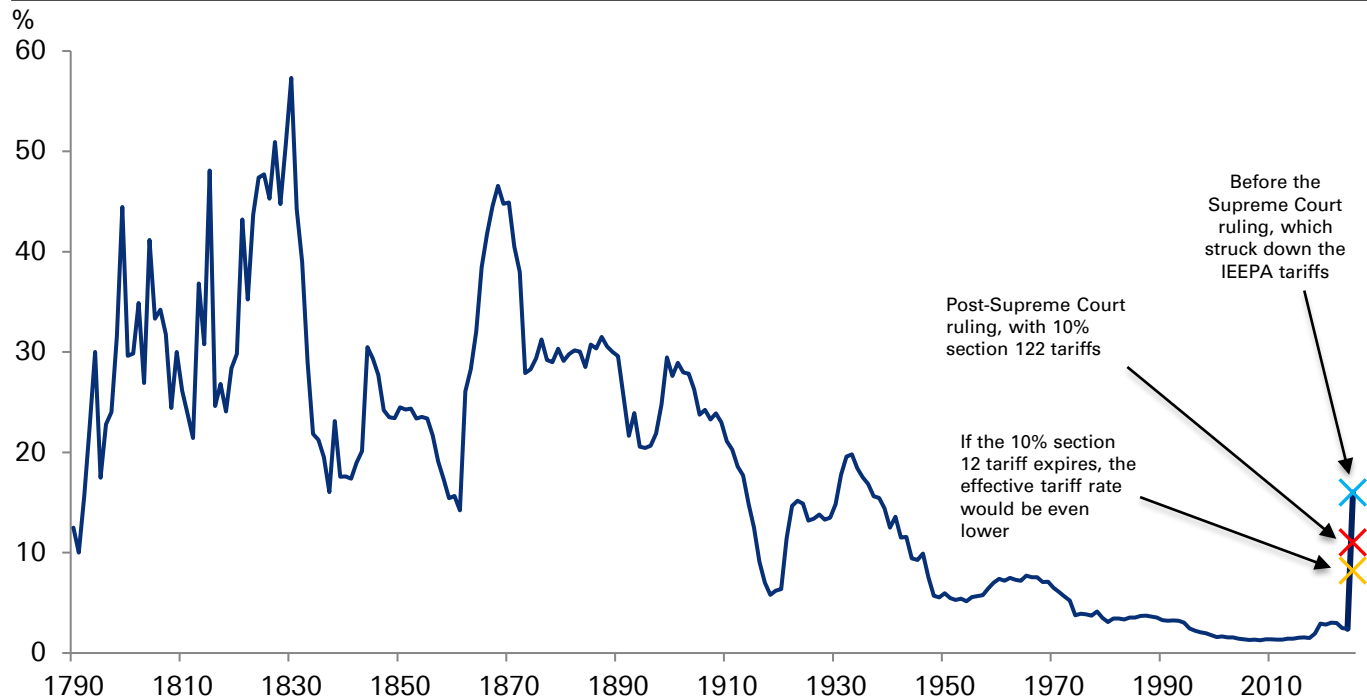


Source: Bloomberg Finance LP, Deutsche Bank



But whilst many inflation pressures are on the upside right now, the tariff impact has been coming down versus last year and worst-case fears... the 10% global tariff that replaced the IEEPA tariff has brought down the effective tariff rate, and if that expires in late-July (Congress would have to extend), then the tariff rate would fall even further...

US Average Effective Tariff Rate



Source: budgetlab.yale.edu, Deutsche Bank

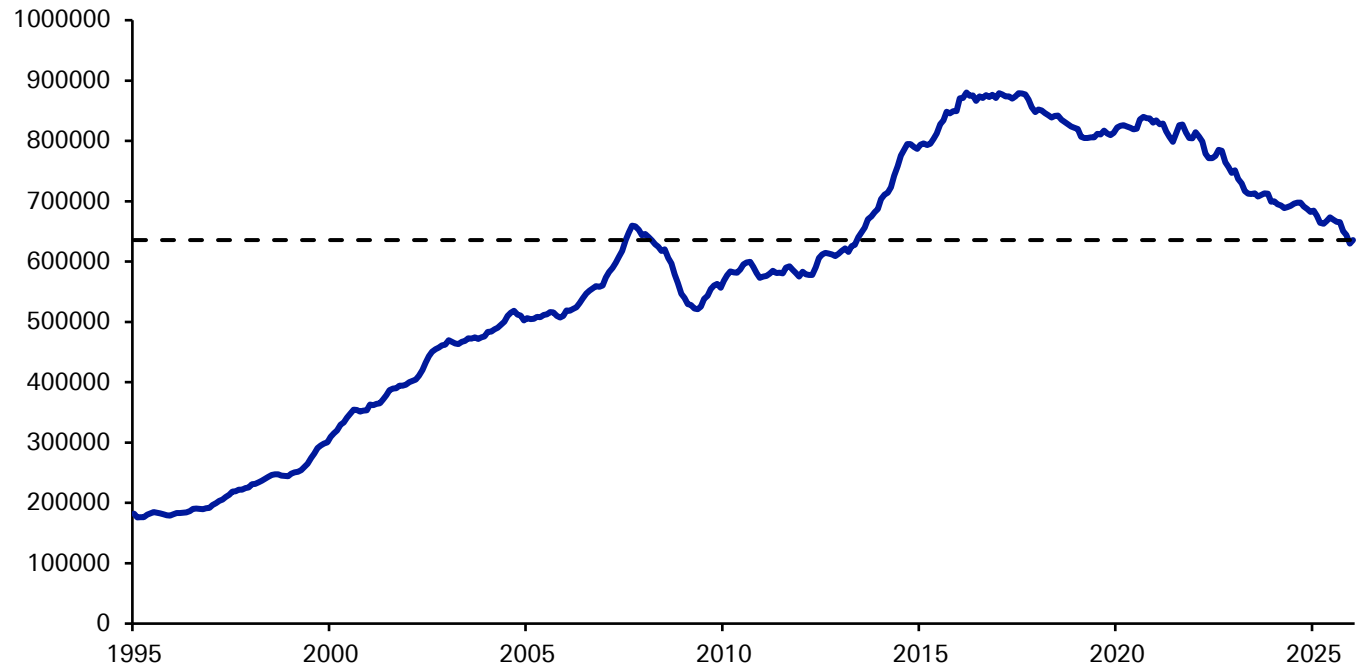


Rising yields aren't great for housing...



In inner London, where some of the most expensive housing in the UK is, prices have been coming down for the last decade in real terms... they're now beneath the pre-GFC peak nearly 20 years ago...

Inner London House prices in real terms (Jan 2026 prices, UK pound sterling)

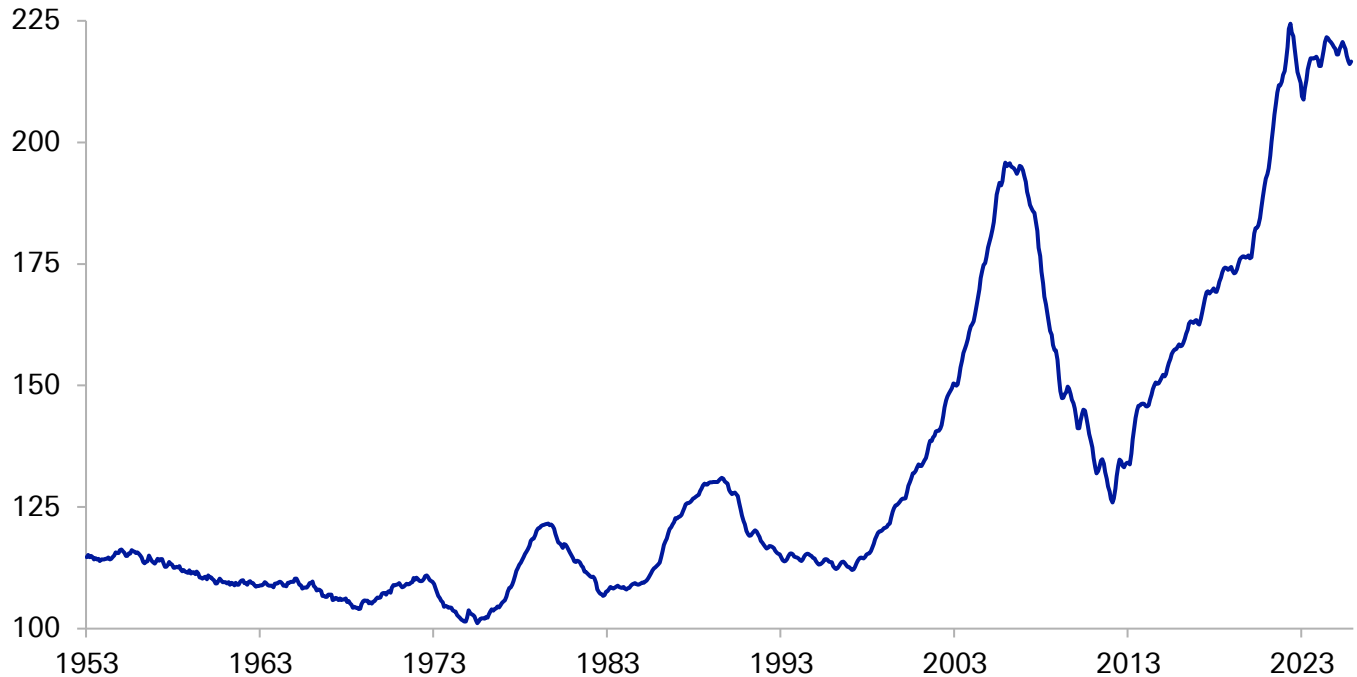


Source: Haver Analytics, Deutsche Bank

In real terms, US house prices have been stagnant for 4 years... although they're still hovering around record highs...



Real Home Price Index (NSA, 1890=100)



Source: Robert Shiller, Haver Analytics, Deutsche Bank



Will the conflict impact the dollar status

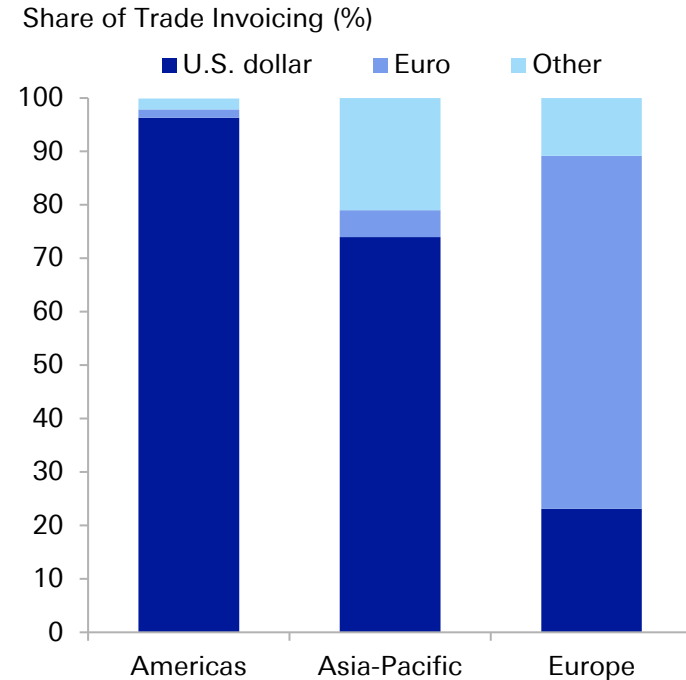
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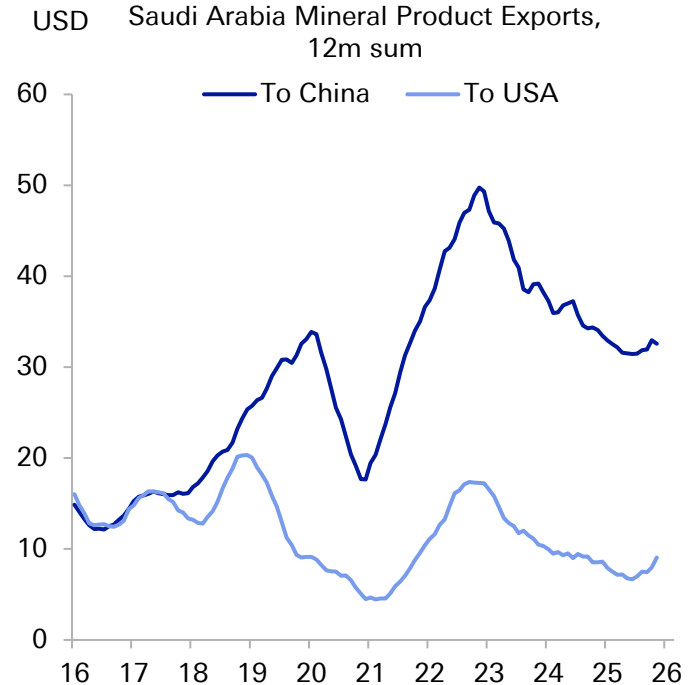
The petrodollar era saw oil priced in dollars with the US providing a security umbrella to the Middle East... the risk for the dollar is that era may be slowly fraying at the edges.... Mallika Sachdeva has written about it [here](#).

The dominance of the dollar in cross-border trade has a lot to do with the petrodollar

There were already signs of instability to petrodollar foundations: Saudi Arabia sells four times as much oil to China now as to the US



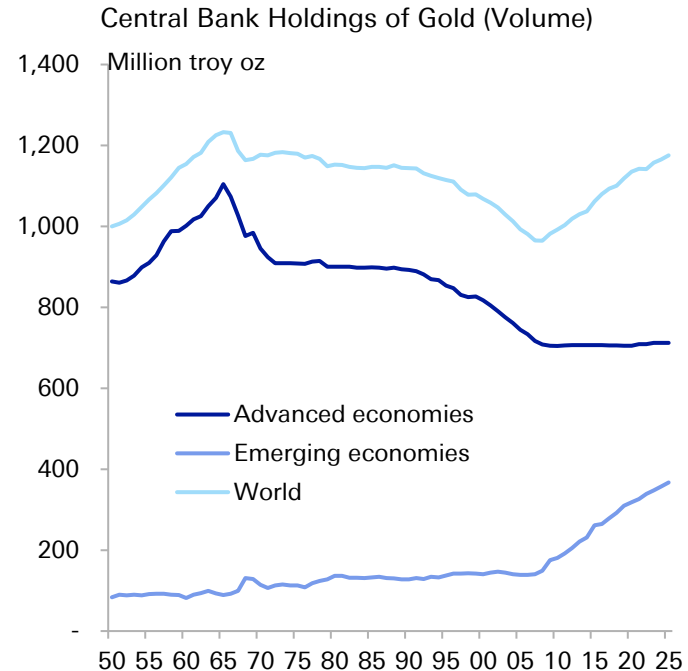
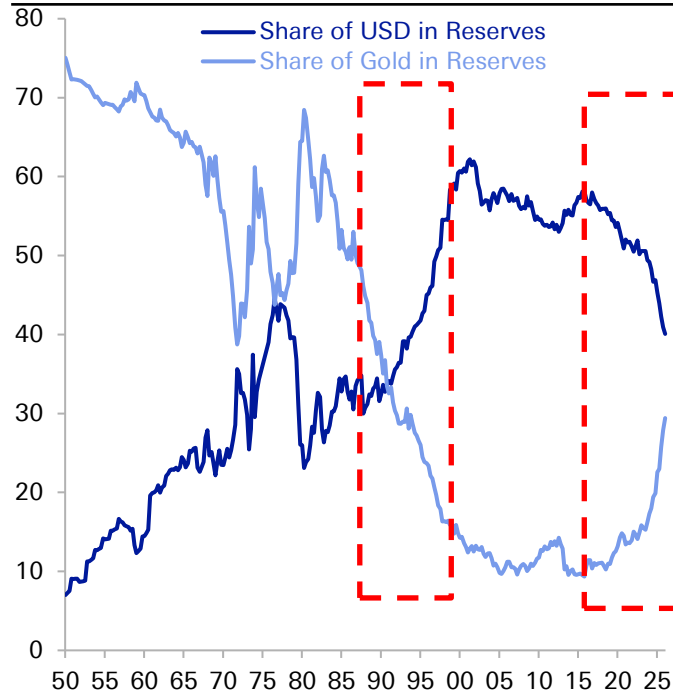
Source: Federal Reserve, CEIC, Deutsche Bank





Linked to the petrodollar theme, there is also de-dollarisation in reserves. Mostly from EM countries. Mallika Sachdeva has written about it [here](#).

The "end of history" in the 1990s saw a sharp rise in the USD and decline in gold's share of reserves. We think history has returned, and it is emerging markets that are leading the charge for gold.



Source: Bloomberg Finance LP, Eichengreen, Chitu and Mehl (2014), IMF International Liquidity Statistics, Deutsche Bank

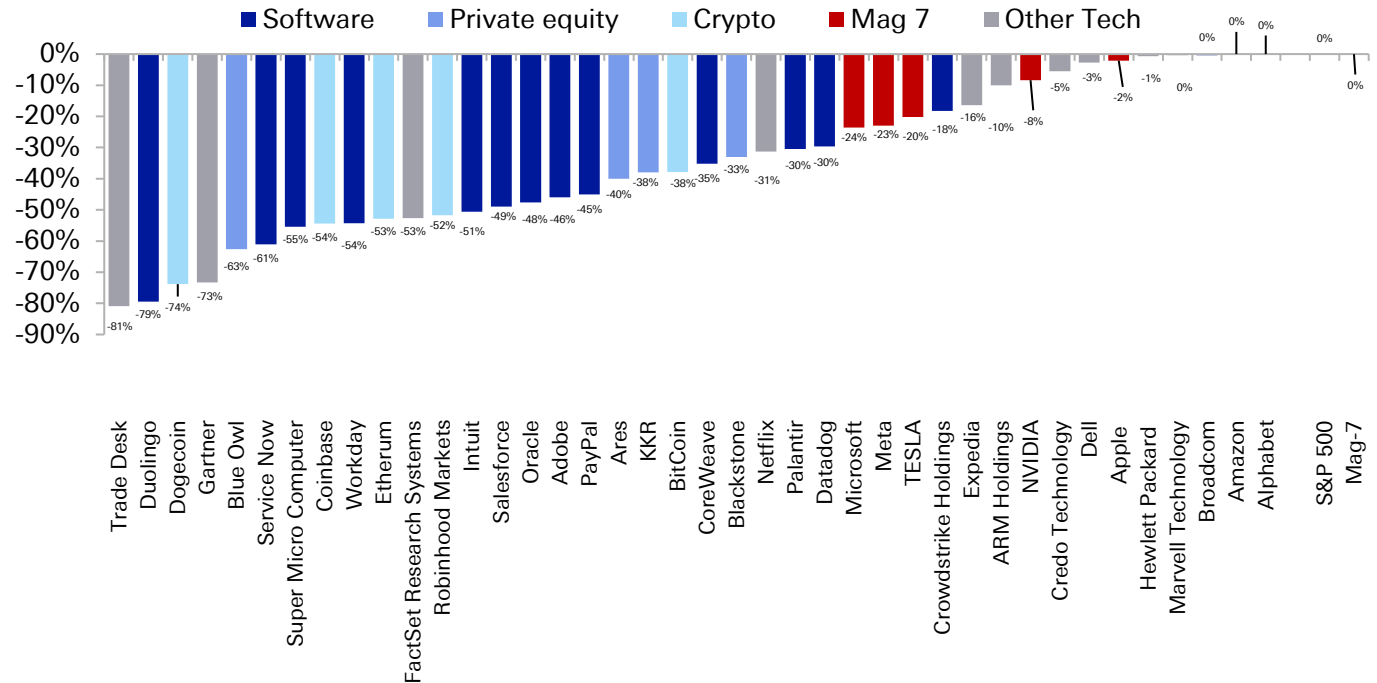


And finally... when the war is over we have
to remember disruption....



The 2026 drawdown in many tech/AI stocks and related or disrupted assets (e.g. crypto and software) has been pretty severe. Private equity has also been hit due to fears on private credit. So remember these themes if and when the war is over...

How Far Are a Selection of AI & Tech Stocks From Their highs? Drawdowns From 52-Week Highs for Selected US Tech and Related Stocks



Source: Bloomberg Finance LP, Deutsche Bank, Last update: May 1



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Will Claude's AI finance agents change your life?

May 6, 2026

Finance professionals may be quaking in their gilets today after Anthropic's latest release of Claude agents for financial services. But their alarm may be premature.

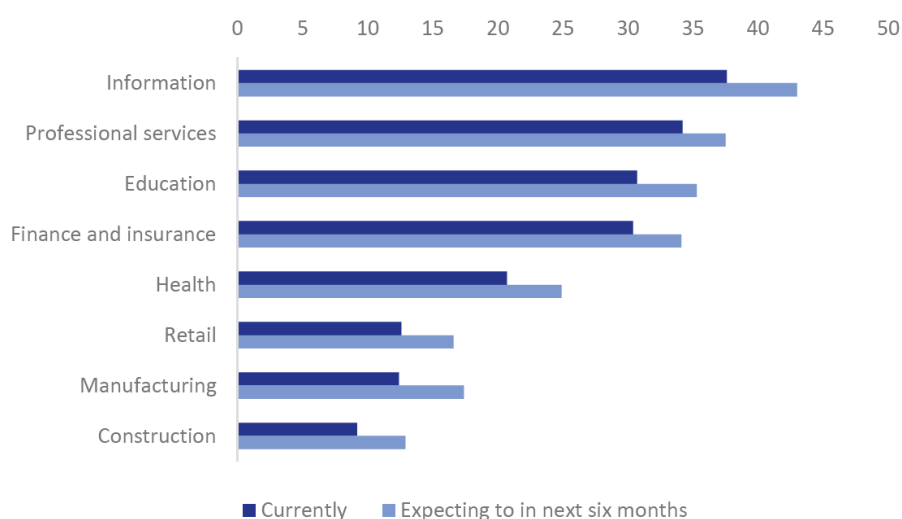
Anthropic [released](#) 10 "ready-to-run" agent templates yesterday to handle time-consuming tasks from research and client coverage to finance and operations with a webinar that included CEO Dario Amodei and JPMorgan CEO Jamie Dimon.

The templates bundle together skills, connectors (third-party market and research data) and sub-agents to help bankers build pitchbooks and models, screen KYC files, close month-end accounts and more.

While the internet was full of (AI-generated) memes of young bankers in tears this morning, the reality is that they are likely to continue to be working late nights and weekends to serve their clients for some time to come.

On the one hand, finance is the perfect industry for AI. It has abundant data; high-value, repeatable decisions often performed with manual processes; and operates in a digital, low-friction environment. US Census Bureau data shows that 30 percent of US banks and insurers were using AI in February and about 34 percent are expecting to use it in the next six months. That puts it not far behind information technology, the front-runner at 38 percent and 43 percent.

Figure 1: US companies using or expecting to use AI (% , Feb 2026)



Source: US Census Bureau, Deutsche Bank Research

Authors

[Adrian Cox](#)
Research Analyst



But on the other hand, adoption is easier said than done, particularly among the largest organisations operating in a highly regulated industry where 99 percent accurate is not accurate enough.

Barriers to adoption include:

1. **Technology and data:** the barriers to any new tech adoption include complex integration with legacy systems; the need for tailored, secure, high-quality – and accessible – data; and issues like data privacy and cybersecurity.
2. **Governance:** barriers specific to AI include the need for heightened guardrails around hallucination, bias, intellectual property and aligning stakeholders. There is also regulatory uncertainty as policymakers take differing and rapidly evolving approaches. AI activities must not only be understood but also explainable.
3. **People:** organisations need to rely on a limited pool of technology experts experienced in implementing a fast-evolving technology. They also must deal with a wide range of capabilities – and support or resistance – from managers and employees. At the very least, that requires training and cultural change.
4. **Customers and clients:** it is far from a given that customers and clients will be universally supportive of having their financial affairs handled by AI or dealing directly with AI agents, let alone giving enough data to the systems to make them valuable.

Diffusion is "much slower than the technology is capable of"

Amodei acknowledged the so-called capability overhang in the webinar, saying: "Our ability to commercialise this technology... is held back not by the power of the models and the economic value you can generate in principle but by the diffusion in the workplace... Diffusion is faster than we've seen in any technology previously in the world but it's still much slower than the technology itself is capable of."

Anthropic and its rival OpenAI are seeking to increase adoption by creating teams to help enterprises adopt AI – a crucial step in monetising the technology and paying for the rapid increase in demand for compute – not least to service the all-but-free service available to consumers.

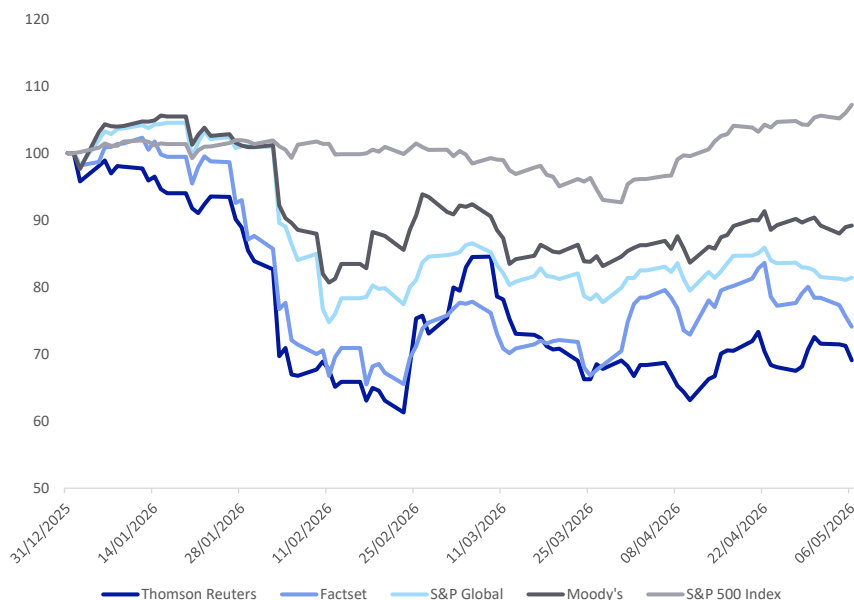
On Monday, Anthropic announced a joint venture focusing on deploying enterprise services, backed by companies including Blackstone, Hellman & Friedman and Goldman Sachs. On the same day, OpenAI was reported to be raising funds for a similar venture called The Development Company.

The ventures will reportedly raise money from the alternative asset managers in return for sales access to their investors' portfolio companies. It echoes the "forward-deployed engineer" model pioneered by Palantir, where engineers are embedded directly within client teams.

The news has had knock-on effects in data industry stocks, including Thomson Reuters and Factset. They had already been dented by the "SaaSocalypse" earlier this year, when the launch of Anthropic's Cowork led investors to question whether AI agents would disintermediate providers of traditional subscription software – or give them an opportunity by deploying AI themselves or being the golden source for leading AI providers.



Figure 2: Data providers and S&P 500 Index share price YTD, indexed to 100



Source : Bloomberg Finance LP, Deutsche Bank Research

Two life-changing scenarios?

Despite the challenges of adoption, there is no doubt that life for junior bankers in particular will change over time.

Anthropic described two scenarios.

In one, a natural resources group banker responds to a client who is considering taking their company private by forwarding a message to Claude using voice mode and Cowork Dispatch messenger. The tool then runs a pitch builder, including comps (comparable company) analysis, drawing in information from third-parties such as Standard & Poor's and Factset; creates an adjustable spreadsheet and an interactive slide deck; and prepares a draft to mail to the client.

In another scenario, a fund accountant uses a valuation review agent to pull tailored quarterly financial information before handing over to a statement auditor agent and then to a client.

The finance agents are available as plugins for users of Claude Cowork – which executes multi-step office tasks on a user's behalf – and Claude Code – an AI-powered coding assistant – running alongside Microsoft Excel, Powerpoint, Word and (soon) Outlook.

They are also available as a so-called cookbook for Claude Managed Agents, running autonomously on the Claude Platform for more in-depth work. They run best on Claude's latest Opus 4.7 model, Anthropic said. OpenAI and Google offer their own AI finance tools to perform some of the same tasks.

Three levels of adoption

Longer term, there are likely to be three levels of adoption of AI in financial services, with the first and parts of the second well-advanced, but the third still nascent:



1. **Individual productivity augmented:** tools like Cowork and Microsoft 365 Copilot.
2. **Processes automated:** streamlining operations and increasing efficiency, notably in back-office functions. Tasks like triaging incoming emails, and processes like onboarding, credit scoring and claims processing, may be among those that can be enhanced.
3. **Systems transformed:** building on AI platforms to create new workflows, rather than automating parts of existing workflows, and creating new revenue streams and business models. This may include autonomous agents running areas like trading and operations end-to-end; 24/7 market monitoring, client service and trade execution; and agents using multimodal data to update in real time.

Some areas are likely to remain elusive for longer still. Handing the keys to AI to trade the markets, the Holy Grail of capabilities, continues to be risky. Bloomberg reported today on a series of trading contests including one on [Alpha Arena](#), run by Nof1, where a portfolio run by eight major frontier systems lost about a third of its capital in two weeks.

Obstacles to extracting AI value

Getting the most out of the remarkable capabilities of AI will take time and depends on factors including:

1. **Adoption:** as mentioned above, only about a third of US finance companies are using AI in their business functions.
2. **Intensity:** most AI users still only dabble, with only 12 percent using it daily, according to St Louis Fed survey data.
3. **Suitability:** most jobs are a bundle of tasks, with some studies suggesting only one fifth of all US tasks can be performed by AI.
4. **Economic threshold:** even where tasks can be performed by AI at all, it may still be cheaper for humans to do them. Two years ago, admittedly a lifetime in AI terms, Nobel Prize-winning economist Daron Acemoglu's [Simple Macroeconomics of AI](#) paper cited data suggesting only a quarter of AI-labile tasks would be able to be automated.
5. **Overall cost:** while the capabilities of AI are increasing by the day, so is the cost of using more and more (albeit cheaper) tokens to execute more and more complex operations. At some point, companies will need to rein in token use and consolidate rather than letting every high-rolling employee develop an army of their own agents.

(These still low levels of deployment suggest that there could be significant latent demand justifying US hyperscaler capital expenditure on data centres reaching \$725bn this year. Shares in AMD, the No. 2 AI chipmaker, some way behind Nvidia, surged as much as 21 percent today after it boosted its sales forecast for the second quarter and said the market for generalist Central Processing Unit (GPU) chips will grow at greater than 35 percent a year to reach over \$120bn by 2030.)

What about us?

Humans still have a lot to offer. Beyond physical dexterity, most finance professionals will have an edge in qualities that clients are likely to want to pay for in the future:



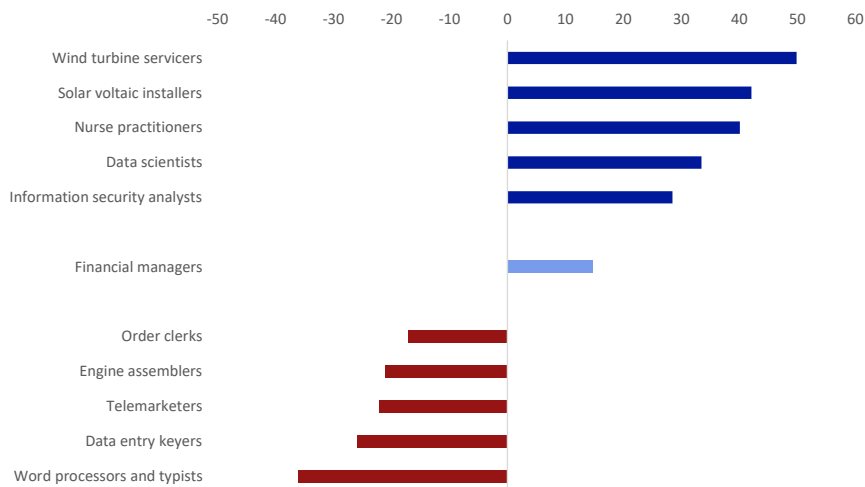
1. **Judgement** in defining problems and assessing outcomes, particularly unusual and risky edge cases;
2. **Genuine creativity**, rather than safe, centre-of-the-bell-curve amalgamations of what has gone before;
3. **Emotional intelligence**, including rapport, common sense and ethics.

An open question is how humans can develop these qualities if AI takes away much of the early-career grunt work where they are forged.

AI agents can boost those qualities while outperforming on aspects they do better and faster: analysing vast amounts of data; generating content; and executing tedious multi-step processes, with lower operational risk, an auditable workflow and humans in the loop only where necessary.

Consequently, the US Bureau of Labor Statistics predicts that financial managers will see a 15 percent increase in numbers over the coming decade.

Figure 3: Fastest growing and (selected) fastest declining occupations in US, 2024 to projected 2034 (% change)



Source : US Bureau of Labor Statistics, Deutsche Bank Research

That puts them among the winners in a list led by wind turbine servicers and solar voltaic installers, themselves benefitting from the need for energy to power the mushrooming of data centres. After nurse practitioners, another two occupations are also likely to be direct beneficiaries of AI: data scientists and information security analysts.

The larger of the fastest declining occupations will include word processors and typists, down by more than one third, the Bureau predicts, as well as data entry keyers, both performing tasks squarely in the sights of AI agents.



Recent relevant research

[Deutsche Bank Research Institute: Technology](#)

[AI, Iran and the Gulf 101](#) (April 13, 2026)

[AI is not convinced AI is a major disinflationary force](#) (April 9, 2026)

[Tracking the effects of AI on the US labor market](#) (March 10, 2026)

[What AI says about AI eating itself and the world...](#) (Feb 16, 2026)

[AI Q&A: Why have market vibes suddenly shifted on AI?](#) (Feb 16, 2026)

[Three AI themes for 2026: the honeymoon is over](#) (Jan 20, 2026)



Appendix 1

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The significance of the UAE

May 6, 2026

- The significance of the UAE extends beyond its oil production share within OPEC. The UAE is second only to Saudi Arabia with regard to production flexibility, both for reducing output (discipline) and increasing it (spare capacity). Notably, the UAE also played a larger role in production cuts during the GFC than Qatar, another former "OPEC 'Core'" member, which exited in 2019. Therefore, with the UAE's absence from May, OPEC and the OPEC+ coalition will have a diminished ability to manage market surpluses and deficits.
- The economic and fiscal motivations for the UAE's decision appear obvious, quite apart from the well-known political issues. The UAE stands out among much of its OPEC brethren due to its lower fiscal breakeven oil price. In fact, amongst oil-exporting countries surveyed by the IMF, the UAE (USD 45/bbl) is matched in its low fiscal breakeven only by Qatar (USD 43/bbl) in 2026, while Iran (USD 165/bbl) is at the higher end of the spectrum.
- In light of the UAE's pledge to "continue to act responsibly, bringing additional production to market in a gradual and measured manner", we anticipate that the UAE will most likely increase its production towards sustainable capacity over a period of a year following the end of commercial shipping restrictions.

UAE exit preceded by policy dissents

The UAE exit from OPEC follows several dissents from group production decisions, notably in November 2020 and July 2021 when these disagreements became public knowledge, and its possible departure from OPEC was discussed in the media as a "trial balloon".¹ The exit also occurred despite some OPEC recognition and accommodation of the UAE's increased production capacity to 4.85 mmb/d, announced in May 2024.² In 2024, the UAE successfully lobbied for a higher production quota to be phased in over the course of 2025, resulting in the largest percentage increase among all OPEC-9 countries subject to a quota (Figure 1).

Authors

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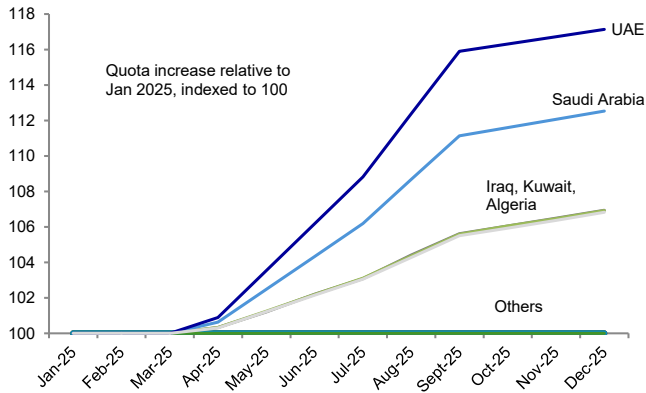
¹ UAE gets higher OPEC+ crude production quota", S&P Global, 4 Jun 2024, (<https://www.spglobal.com/energy/en/news-research/latest-news/crude-oil/060424-uae-gets-higher-opec-crude-production-quota>).

² UAE's ADNOC ups crude production capacity to 4.85 million b/d", S&P Global, 2 May 2024, (<https://www.spglobal.com/energy/en/news-research/latest-news/crude-oil/050224-uaes-adnoc-ups-crude-production-capacity-to-485-million-bd>).

4 May 2026
Hsueh On Oil



Figure 1: UAE saw the largest proportional quota increase in 2025 (Jan'25=100)

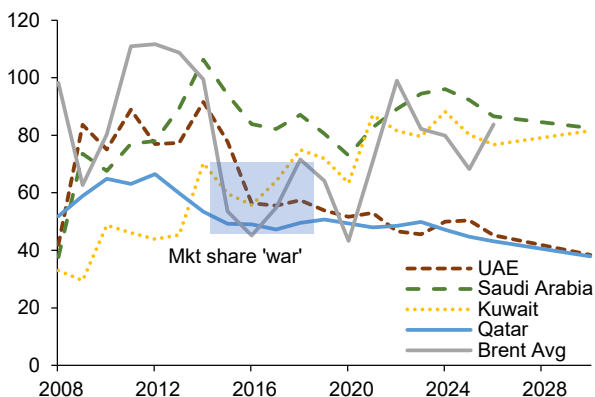


Source : OPEC, Deutsche Bank

UAE and Qatar share low fiscal breakeven oil prices

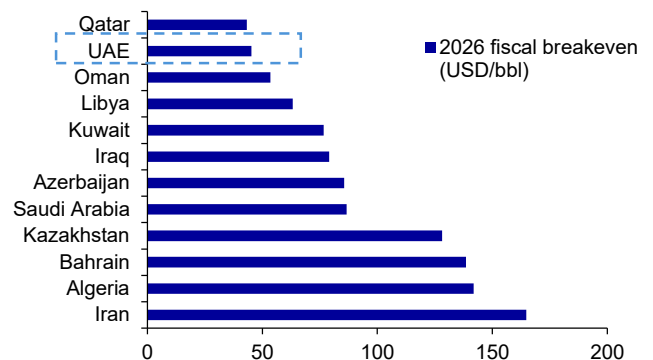
The economic and fiscal context for the UAE's decision appears obvious, quite apart from the well-known political issues.³ The UAE stands out among many of its OPEC counterparts by its lower fiscal breakeven oil price. Since the market share "war" starting from the end of 2014 through 2019, the UAE's fiscal breakeven has diverged from those of Kuwait and Saudi Arabia (Figure 2). In fact, amongst the oil-exporting countries surveyed by the IMF, the UAE (USD 45/bbl) is matched in its low fiscal breakeven only by Qatar (USD 43/bbl) in 2026, with Iran (USD 165/bbl) at the opposite end of the spectrum (Figure 3).

Figure 2: UAE diverged from Saudi Arabia and Kuwait in fiscal breakevens (USD/bbl)



Source : IMF, Deutsche Bank

Figure 3: UAE and Qatar (2019 exit from OPEC) have lowest fiscal breakevens (USD/bbl)



Source : IMF, Deutsche Bank

High contribution to supply discipline

The UAE's contribution to OPEC supply discipline extends beyond its share of oil production falling under the OPEC umbrella. Along with OPEC founding members

3 "The Saudi Arabia-UAE Dispute Is About More Than Just Yemen", Foreign Policy Magazine, 8 Jan 2026, <https://foreignpolicy.com/2026/01/08/saudi-arabia-uae-dispute-yemen-sudan/>.

4 May 2026
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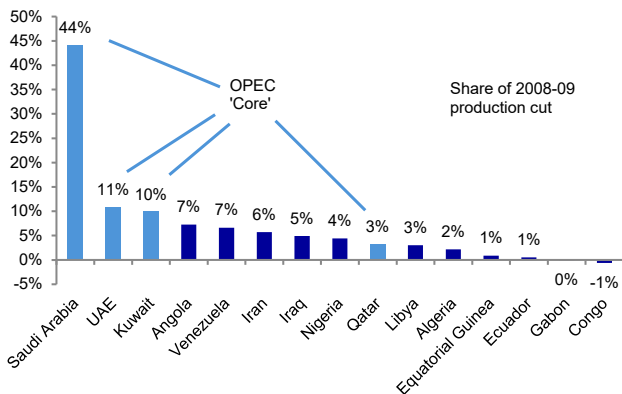


Saudi Arabia and Kuwait, the UAE (joined in 1967) and Qatar (joined 1961, exited 2019) were considered to be OPEC 'Core' countries in the sense that they are "immediate neighbors of Saudi Arabia and have usually followed similar policies."⁴

In early 2008, the UAE's share of OPEC production was only 8%, yet it contributed 11% of the production cut (Figure 4). The UAE's "sacrifice" as a percentage of its own production was comparable to Saudi Arabia's (Figure 5), while non-Core countries generally made lesser sacrifices. By 2026, the UAE's share of OPEC production had risen to 12% just before the US-Iran war. While the UAE has typically produced above its quota, it has not been a remarkable standout in this regard (Figure 6), with Saudi Arabia and Kuwait also being similar or larger over-producers in recent years (2024-2025).

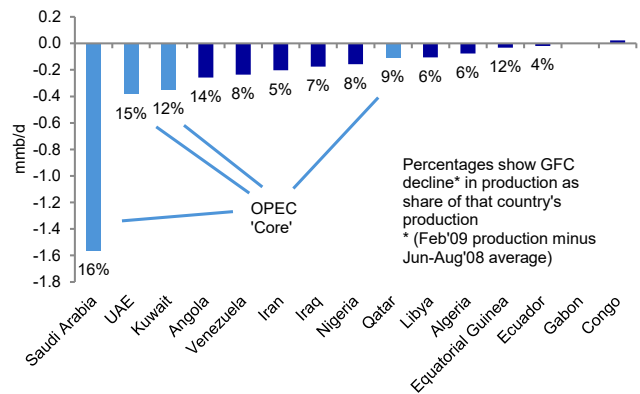
This implies that to achieve the same impact in addressing demand shortfalls, other countries would need to sacrifice slightly larger shares of production and revenue. In 2020, the lower compliance demonstrated by OPEC+ coalition partners under the Declaration of Cooperation (shown as 'non-OPEC' in Figure 7), suggests there is more room for improvement here than within OPEC proper. The nine OPEC+ coalition partners produced 13.4 mmb/d in January 2026, compared to 24.6 mmb/d from the OPEC-9 countries subject to a quota, and 30.1 mmb/d from all twelve OPEC countries including Libya, Venezuela and Iran.

Figure 4: UAE share of 2008-09 OPEC cut was second-largest



Source : IEA data from Monthly Oil Data Service © OECD/IEA 2026, www.iea.org/statistics, Licence: www.iea.org/t&c; as modified by Deutsche Bank

Figure 5: UAE 'sacrificed' nearly as high a share of its production as Saudi Arabia in 2008-09 (mmb/d)



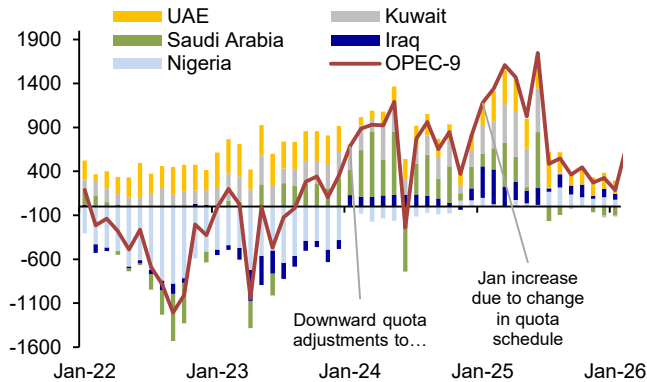
Source : IEA data from Monthly Oil Data Service © OECD/IEA 2026, www.iea.org/statistics, Licence: www.iea.org/t&c; as modified by Deutsche Bank

4 Gately, Dermot. "Lessons from the 1986 Oil Price Collapse", Brookings Papers on Economic Activity, June 1986, https://www.brookings.edu/wp-content/uploads/1986/06/1986b_bpea_gately_adelman_griffin.pdf.

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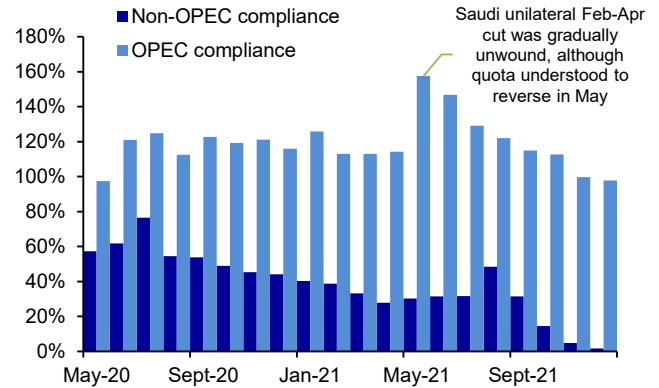


Figure 6: Notable OPEC-9 countries with production above or below quota (kb/d)



Source : Bloomberg Finance LP, IEA data from Monthly Oil Data Service © OECD/IEA 2026, www.iea.org/statistics, Licence: www.iea.org/t&c; as modified by Deutsche Bank, Kpler

Figure 7: OPEC+ coalition partners' compliance was lower than OPEC-9 countries in 2020-21



Source : IEA data from Monthly Oil Data Service © OECD/IEA 2026, www.iea.org/statistics, Licence: www.iea.org/t&c; as modified by Deutsche Bank

Second-highest spare capacity in OPEC-9

Regarding the capacity to raise production, the UAE is also relatively important, though not nearly as crucial as Saudi Arabia. Its average spare production capacity since 2020 has been 0.8 mmb/d, compared to 2.5 mmb/d for Saudi Arabia. If ADNOC's May 2024 estimate of sustainable capacity is used, the UAE's average spare capacity would rise to 1 mmb/d, as the IEA's estimate of sustainable capacity remained at 4.3 mmb/d in 2025. Consequently, after the UAE's departure, OPEC and the OPEC+ coalition will have less flexibility in mitigating market deficits.

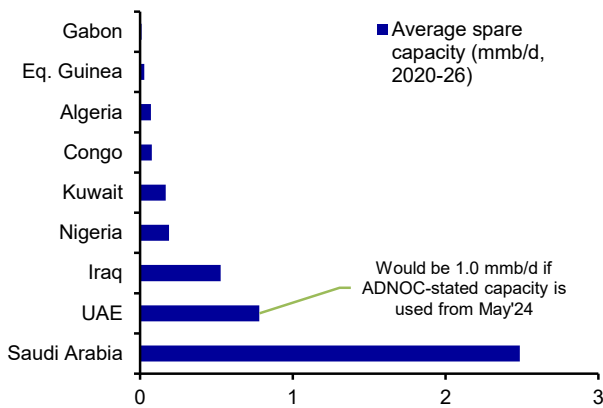
This point can also be illustrated by comparing the UAE to other countries that have recently departed from OPEC (Figure 9), showing that the UAE has a greater capacity to create oversupply. However, as the Minister of Energy and Infrastructure rightly points out, "the timing now is perfect" because "the impact is going to be minimal on all of the producers", and from the consumer's point of view, "the world will need much, much more production of energy."⁵

5 "Opec exit 'purely a policy move', UAE energy minister says", The National News, 29 Apr 2026, <https://www.thenationalnews.com/business/energy/2026/04/28/opec-exit-purely-a-policy-move-uae-energy-minister-says/>.

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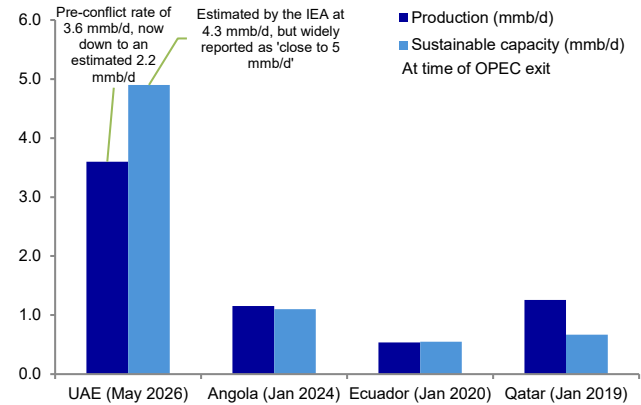


Figure 8: UAE possessed second-largest spare capacity out of OPEC-9 countries subject to a quota



Source : IEA data from Monthly Oil Data Service © OECD/IEA 2026, www.iea.org/statistics, Licence: www.iea.org/t&c; as modified by Deutsche Bank, Kpler

Figure 9: UAE has more potential to raise production than other countries at time of exit



Source : Bloomberg Finance LP, IEA data from Monthly Oil Data Service © OECD/IEA 2026, www.iea.org/statistics, Licence: www.iea.org/t&c; as modified by Deutsche Bank

UAE will continue to act responsibly

Beyond these important facts of the UAE's supply flexibility, it is worth noting from the Ministry of Energy and Infrastructure's public statement that:

"Following its exit, the UAE will continue to act responsibly, bringing additional production to market in a gradual and measured manner, aligned with demand and market conditions."⁶

Additionally,

"This decision does not alter the UAE's commitment to global market stability or its approach based on cooperation with producers and consumers. Rather, it enhances the UAE's ability to respond to evolving market needs."

From these statements, we infer that following a Strait of Hormuz re-opening, there may not necessarily be a very rapid increase in UAE production, even considering the extreme supply deficits currently estimated. We could also infer that to demonstrate its commitment to market responsibility, the Ministry of Energy may be especially careful with its production profile after a Strait of Hormuz reopening, since its behaviour will likely be under close scrutiny by the market. Accordingly, we believe it most likely that the UAE will raise its production towards sustainable capacity over a period of a year.

6 "UAE announces decision to exit OPEC & OPEC+", Emirates News Agency, 28 Apr 2026, (<https://www.wam.ae/en/article/bzxzuh7-uae-announces-decision-exit-opec-opec>).



Appendix 1

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May 5, 2026

Q&A with Matthew Barnard

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As equity research adapts to rapid shifts in technology, markets and client behaviour, Matt Barnard, Head of US Equity Research, is helping steer how Deutsche Bank evolves its research offering. In this edition of our Q & A series, Matt shares his perspective on the new ways clients are consuming research, his outlook for US markets, and what it's like leading research teams through a period of significant change.

Q: You've spoken about the equity research landscape undergoing a fundamental shift. What has changed most in how clients discover, engage with and use research today?

AI is fundamentally changing how clients consume sell-side research. As information becomes more easily accessible through AI tools, the value of differentiated insights from trusted sources has moved higher up the value chain. Investors are increasingly using AI to discover new ideas and challenge consensus views more quickly and efficiently. This allows them to filter out noise and spend more time engaging in deeper, more meaningful conversations with our Deutsche Bank research analysts.

Q: Against a complex backdrop for US geopolitics and policy, how do you currently assess the outlook for US markets – and which themes matter most for clients as we move through 2026?

Geopolitical developments in the US and globally continue to be a significant driver of volatility across risk assets. Deutsche Bank research's macro and strategy teams have consistently taken bold, often counter-consensus views, which are especially valuable in this environment. While conflicts in the Middle East will likely continue to create ripple effects across markets, the underlying fundamentals for US equities remain relatively strong. The rapid acceleration of AI adoption – and the associated capital investment – remains a key thematic focus for clients, both in the US and globally.

Q: You now lead teams through a period of rapid change. What do you find most rewarding about working in research management at this point in the cycle?

At its core, the role of an equity research analyst is to form forward-looking views with imperfect information. The strongest analysts thrive in periods of uncertainty and rapid change, identifying emerging trends and uncovering new investment opportunities early. From a management perspective, the



Q&A with Matthew Barnard

most rewarding part of my role is seeing some of the brightest minds in research stay ahead of these shifts and deliver true alpha for our clients.



Q&A with Matthew Barnard



Q&A with Matthew Barnard

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This Month in Geopolitics: May 2026

Key events, themes and trends to watch in the month ahead

May 2026

Dr. Helen Belopolsky | Miha Hribernik

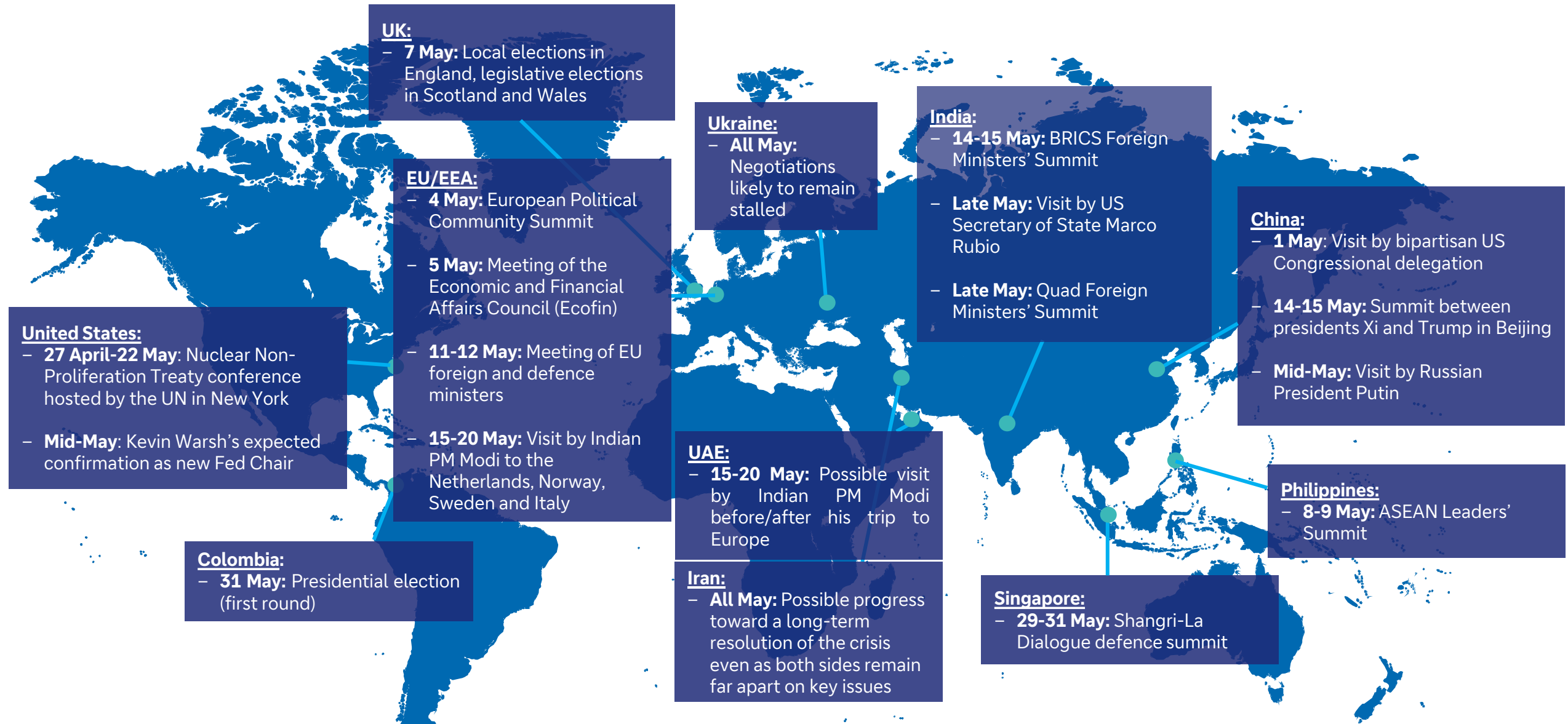
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IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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At a Glance: Geopolitical events to watch in May 2026

Our monthly Look-Ahead covers a *non-exhaustive* list of key events, issues and themes to watch in the month ahead. The below map intends to serve as an executive summary, providing a list of the issues at a glance. These are covered in more detail in subsequent slides.



Source: Deutsche Bank

In focus for May: The war in Iran and the Xi-Trump summit in Beijing



The War in Iran: An End in Sight?

Key points of contention remain between the two sides, including Iran's nuclear programme, the closure of the Strait of Hormuz and sanctions relief. The US rejection of Iran's latest peace proposal on 1 May, and Iran's warning of possible military action on 4 May, underscored that both sides remain far apart on key issues. But despite the precarious ceasefire, the probability of a longer-term resolution is assessed to be strong. The US is facing diminishing returns and increasing domestic and international pressure as the economic consequences of the crisis grow, which will likely push the administration towards a settlement.

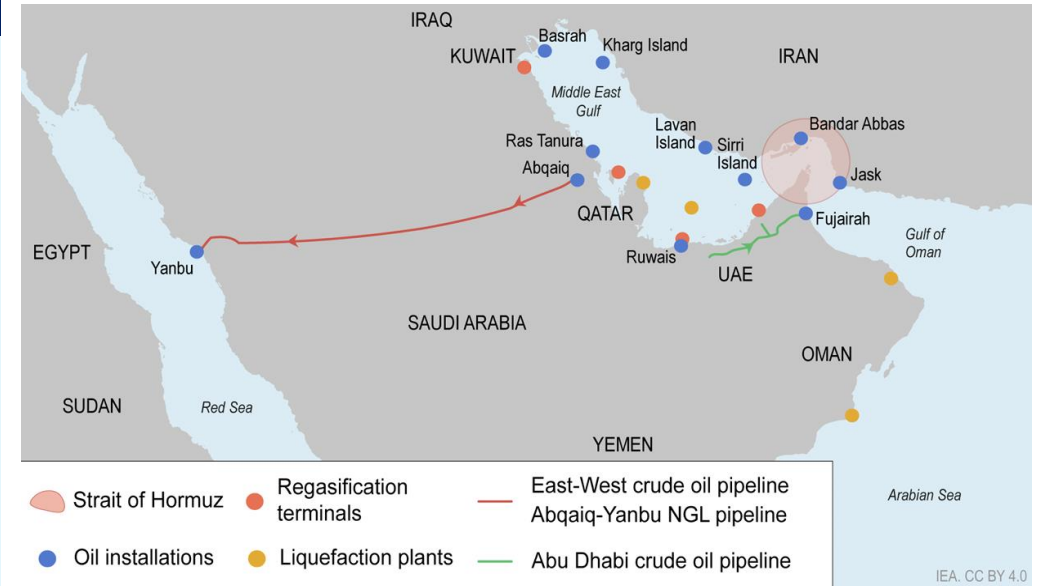
This conflict is a transformative event for the Middle East, shattering the previous geopolitical equilibrium and creating a new, unpredictable reality – as illustrated by the UAE's decision to leave OPEC on 1 May. Iran is expected to emerge with enhanced influence, while Gulf states will likely re-evaluate their security posture and further diversify their international relationships beyond the United States. Post-conflict normalization and economic recovery will be a lengthy process, likely taking months. Rebuilding critical infrastructure like Qatar's Ras Laffan gas field and restoring confidence for shipping through the Strait of Hormuz – which remains far below pre-war levels (see chart) will take considerable time, with knock-on effects like inflation likely lasting well into the second half of 2026.

Xi-Trump Summit: Toward a Tactical Deal

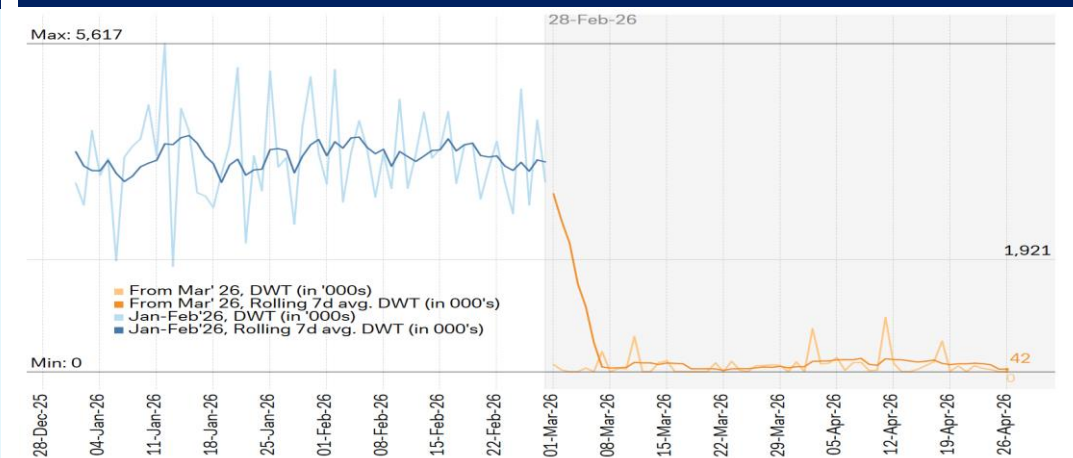
Barring a significant escalation in the Iran war, the summit between presidents Xi and Trump on 14-15 May will likely go ahead as planned – and deliver a package of tactical agreements. The meeting will likely coalesce around limited, but potentially more achievable, deliverables. These could include Chinese commitments to purchase up to 500 Boeing jets, US soybeans and energy, as well as Chinese requests for the US to ease export controls, including on chips and aircraft engine parts. Both sides could also agree to set up a 'Board of Trade' to increase bilateral trade in non-sensitive areas. Progress on a proposed 'Board of Investment' could be more limited owing to mutual national security concerns linked to investment, although the summit could still deliver a small number of investment deals.

Despite recent [signs of friction](#) over Iran, both sides appear to be prioritizing bilateral stability. The threshold for another summit deferral is likely high. But a breakdown of the US-Iran ceasefire and a return to fighting would be a notable downside risk, increasing the probability of another summit delay or even a cancellation.

Oil and gas infrastructure in the Gulf



Tanker activity exiting the Strait of Hormuz (in DWT'000s)



Other key events, issues and themes to watch in May 2026

Listed in chronological order



Ukraine war (all May)

Negotiations for a ceasefire in Ukraine will likely remain stalled during May. While there is ongoing diplomatic activity, no new rounds of ceasefire talks are publicly scheduled for May or June. Ukrainian President Zelenskyy has stated that the US wants Russia and Ukraine to reach a peace agreement by June. However, the specifics of how this would manifest in terms of scheduled meetings are unclear.

While a ceasefire over the coming months is far from guaranteed, even credible steps toward a deal would have implications for markets. These could include a possible easing of some sanctions on Russia, the potential for increased Ukrainian agriculture exports, and for enabling reconstruction – with implications for investments in minerals, gas infrastructure, technology, city reconstruction, data centres and artificial intelligence.

Nuclear Non-Proliferation Treaty conference (27 April – 22 May)

The UN-sponsored conference in New York is taking place against the backdrop of rising geopolitical tensions and the prospect of a new nuclear arms race. The expiry of the New START Treaty on 5 February marked the end of the last binding nuclear arms control agreement between the US and Russia. Both countries retain the world’s largest nuclear warhead inventories (see chart). While these have shrunk slightly in recent years, the unravelling of the New START deal could reignite nuclear competition.

Meanwhile, China is pursuing a significant expansion of its nuclear stockpile. There is little evidence to suggest that either the New York conference or other recent initiatives – including a proposal by President Trump for a new arms control treaty between China, Russia and the US – are likely to translate into binding agreements in the near term.

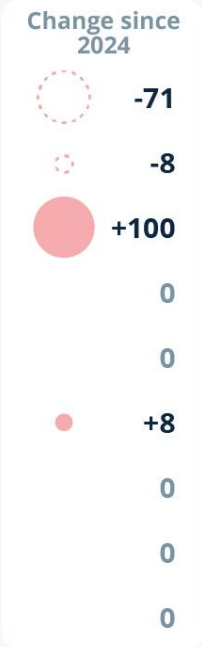
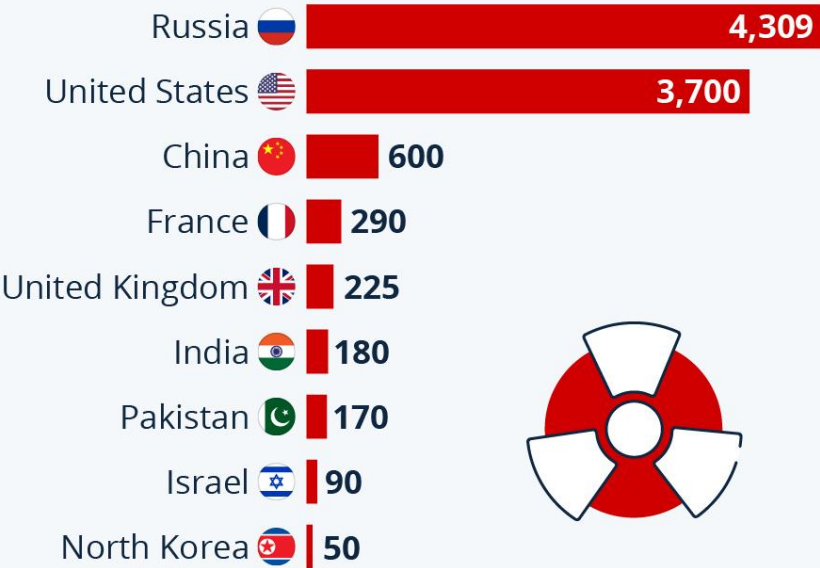
Bipartisan US Congressional delegation visiting China (1 May onward)

The reported visit reflects continued engagement and relative stability in China-US ties ahead of the Xi-Trump summit. The five-person delegation led by Republican Senator Steve Daines reportedly visited Beijing and Shanghai. The delegation is purportedly interested in better understanding China’s technological innovation and infrastructure, including its high-speed rail network.

Estimated nuclear warhead inventories, 2024-25

The Countries Armed With Nuclear Weapons

Estimated nuclear warhead inventories (as of Jan. 2025)*



* Deployed warheads as well as warheads in central storage. Excludes retired warheads
Source: SIPRI



Source: Statista



Other key events, issues and themes to watch in **May 2026**

Listed in chronological order



European Political Community Summit (4 May)

The summit will likely cover issues including energy security, defense, technology cooperation, and the conflicts in Iran and Ukraine. The European Political Community was established in the wake of the Russian invasion of Ukraine in 2022, with the goal of bolstering Europe-wide geopolitical cooperation. The group currently has 47 members, including non-EU members such as Türkiye, Armenia and Azerbaijan. The 4 May summit will take place in the Armenian capital Yerevan.

EU Economic and Financial Affairs Council meeting (5 May)

The meeting of the EU's financial and economic ministers will cover topics including the economic impacts of the Russian invasion of Ukraine, European economic recovery and tax fraud. The council will also hold a debate on the market integration and supervision package – a series of measures intended to strengthen the integration and efficiency of the EU's financial system.

Elections in the UK (7 May)

The elections could see a realignment in voter preferences toward the Reform and Green parties. In **England**, local elections will see more than 5,000 council seats up for grabs, as well as six directly elected mayoral posts. Recent polls suggest a surge in support for the Reform Party – which is among other policies advocating for tighter immigration curbs and lower taxation – and the Green Party. Labour and the Conservatives are facing significant seat losses.

Polls ahead of **Scotland's** parliamentary election on the same day indicate a lead for the ruling pro-independence Scottish National Party, with Labour and Reform virtually tied for second place. The Conservative Party – which came in second with almost a fifth of all votes in 2021 – could see its support halved.

For the legislative election in **Wales**, polls suggest support for the ruling Labour party could almost halve, with Reform and pro-independence Plaid Cymru nearly tied for first place, each with around 25%-29% of the projected vote.



Source: [Guillaume Périgois on Unsplash](#)

Other key events, issues and themes to watch in **May 2026**

Listed in chronological order



ASEAN Leaders' Summit in the Philippines (8-9 May)

The Iran crisis and its economic impacts on Southeast Asia – which include inflationary pressure, rising fiscal burdens due to energy subsidies, and jet fuel shortages – are expected to dominate discussions. Talks will also likely focus on the war's implications for energy, food prices, and for the region's large expatriate community in the Middle East, as well as critical minerals, and the South China Sea Code of Conduct.

EU foreign and defence ministers' meetings (11-12 May)

The session is expected to include discussions on the EU's response to the war in the Middle East and the management of any knock-on implications of the conflict for the EU's support for Ukraine. Ministers will also discuss inputs into the development of the upcoming European Security Strategy.

Kevin Warsh's expected confirmation as new Fed Chair (mid-May)

The US Senate is expected to confirm the nomination of Kevin Warsh as the new Chairman of the Board of Governors of the US Federal Reserve. The vote will likely be scheduled during the first half of May. Current US Fed Chair Jay Powell's term as Chair expires on 15 May, by which point Trump administration officials and Senate Republicans are likely to push for Warsh's confirmation to be complete. If confirmed, Warsh's first FOMC meeting will take place on 16-17 June.

Russian President Putin to visit China (mid-May)

Putin's visit to China will likely seek to cement the China-Russia partnership (including more energy supplies for Beijing) and coordinate US policies after the Xi-Trump summit. Media reports indicate Putin could visit during the week of 18 May. The timing could suggest an exchange of views with China on the outcomes of the Xi-Trump summit, and an effort to coordinate policies toward Washington. Putin and Xi could also seek to deepen diplomatic and security ties, and coordinate on the Iran conflict, given Tehran's status as both countries' close partner.

For Beijing, securing increased Russian energy imports may be another objective. Russia's Foreign Minister Lavrov stated on 15 April that Moscow is willing to boost exports to China. The supply of Russian crude already rose by a [reported](#) 31% y/y during 2026-Q1.



Source: [Rafik Wahba on Unsplash](#)

Other key events, issues and themes to watch in **May 2026**

Listed in chronological order



Indian Prime Minister Narendra Modi expected to visit Europe and the UAE (15-20 May)

The expected visit to Italy, Norway, Sweden, and the Netherlands will likely seek to further strengthen trade and investment ties. This will include attendance at the India-Nordic Summit on 15 May. Talks will likely cover issues including energy, technological cooperation and defence, with the goal of deepening strategic cooperation as India pursues economic and trade diversification amid the ongoing global energy shock. Reports suggest Modi's itinerary could include a stop in the United Arab Emirates – India's key Gulf partner – for talks on topics including regional security and energy cooperation.

India to host BRICS (14-15 May) and Quad (late May) foreign ministers' meetings

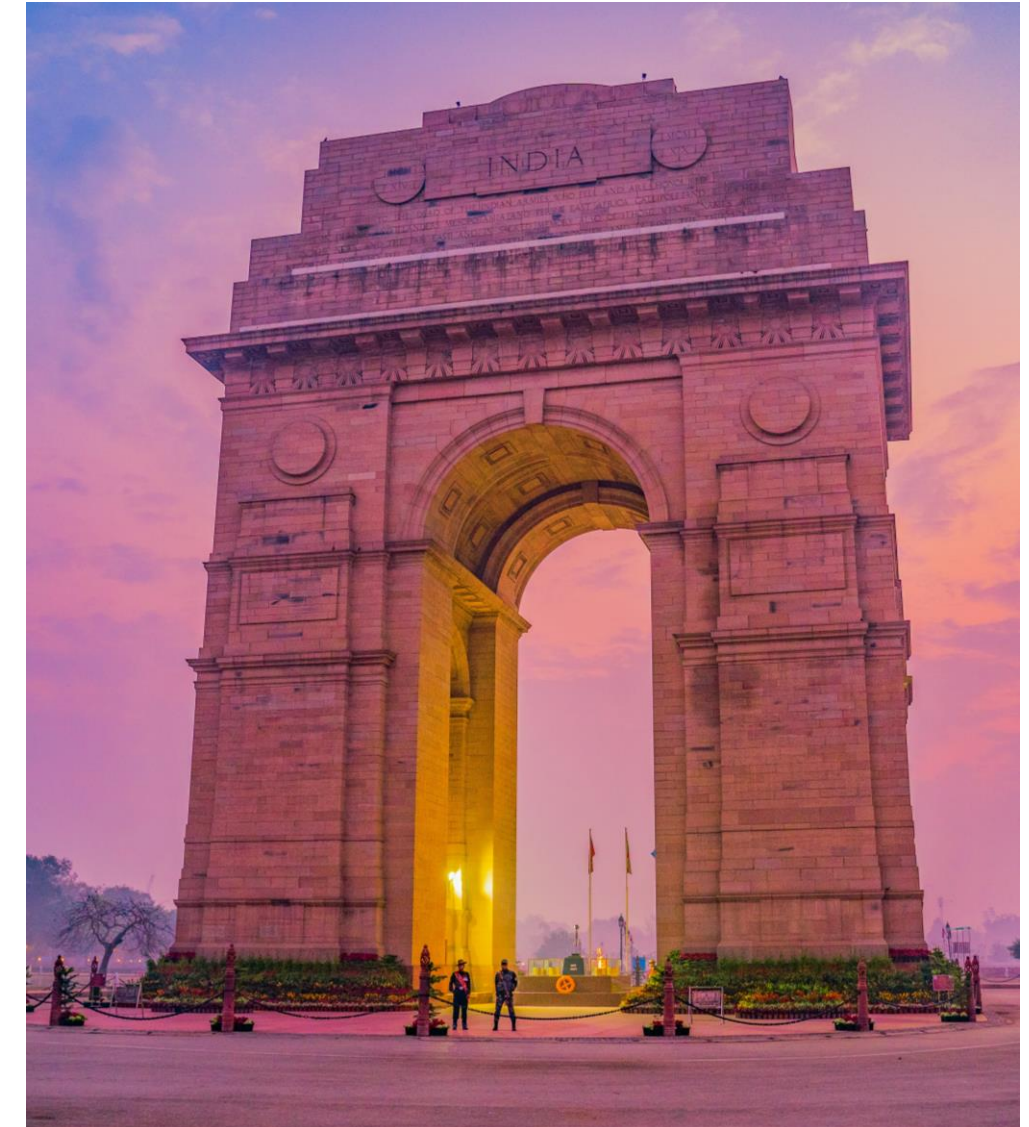
The hosting of two geopolitically competing meetings reflects new Delhi's multi-aligned policy approach. Marco Rubio's visit to India will reportedly be accompanied by the first meeting of the **Quad** group of countries (India, US, Japan and Australia) since July 2025. The Quad intends to bolster cooperation in areas including maritime security, infrastructure resilience, and emerging technologies.

New Delhi will also host a meeting of the **BRICS** foreign ministers, with talks expected to lay the groundwork for the BRICS Leaders Summit in India in September. While the meeting could address the Iran war, BRICS is unlikely to adopt a coordinated response to the crisis given members' diverging views on the issue. But as we have argued, the absence of a BRICS consensus on a geopolitical issue is not unusual. It should not be interpreted as either a setback for the organisation or as a potential point of fracture (see [Thematic Research: Geopolitical Crossroads: India's Strategic Balancing Act](#)).

US Secretary of State Marco Rubio to visit India (late May)

Rubio's India visit is expected to include talks on trade, critical minerals and energy. The visit will follow in the wake of the latest round of India-US trade talks in Washington in late April. Both sides are reportedly close to finalising the trade framework agreement from early February.

Rubio's visit and the ongoing trade talks reaffirm our view that India-US ties remain largely on track despite recent trade friction. This is due to a convergence of interests, including around long-term economic cooperation and shoring up India's position as an investment and manufacturing alternative to China ([Thematic Research: The US-India Bargain: Tariffs and Geopolitical Alignment](#)).



Source: [Swapnil Bapat](#) on [Unsplash](#)

Other key events, issues and themes to watch in **May 2026**

Listed in chronological order



Shangri-La Dialogue defence summit in Singapore (29-31 May)

Singapore will host the annual Shangri-La Dialogue defence summit from 29-31 May. The event will bring together military and government officials from across the Asia Pacific and outside the region. The event can provide insights into how countries such as China and the US intend to approach major geopolitical challenges, including second-order security impacts from the Iran war.

Presidential election in Colombia; first round (31 May)

Colombia will hold the first round of its presidential elections on 31 May. A second round, if no candidate secures more than 50% of the vote, is expected and scheduled for 21 June.

Left-wing candidate Iván Cepeda is currently leading in most polls, with a tight race expected between two centre-right/right candidates, Paloma Valencia and Abelardo de la Espriella, who are competing for a position in the second-round runoff.



Source: [Random Institute](#) on [Unsplash](#)

Appendix 1



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Deutsche Bank
Research

Q2 Corporate Bank Quarterly

A guide to Macro, Markets, Geopolitics & Tech for CFOs
and Corporates

April 2026

Marion Laboure | Camilla Siazon | Luke Templeman,
Adrian Cox | Helen Belopolsky | Miha Hribernik | Jim Reid

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

Deutsche Bank
Research Institute

Introduction: Macro, Markets, Geopolitics & Tech

A guide for CFOs & Corporates



We're pleased to share our April 2026 Guide for CFOs and Corporates, providing essential insights into macroeconomics, markets, geopolitics and technology. This report offers critical analysis and perspectives to help corporate clients navigate today's complex economic environment.

Key highlights:

- **Macroeconomic outlook:** The Iran War is driving an energy price shock that threatens a global inflation shock, despite growth remaining resilient outside the euro area.
- **Market analysis:** Higher energy costs and rates are pulling on corporate earnings growth. Brent crude averages \$84.80/bbl in our baseline. Dollar weakness narrative resumes.
- **Geopolitics:** A long-term solution between the US and Iran is likely, but post-conflict normalization will take months. Key Q2 events to watch: Xi-Trump summit (14-15 May) and Ukraine ceasefire talks.
- **Artificial Intelligence:** AI disruption concerns have waned somewhat, however, the software sector remains under intense scrutiny and Q1 outlook announcements will be critical.
- **The future of monetary policy:** The energy shock has materially shifted central bank trajectories. Fed on hold through 2026; ECB hiking twice. The US continues to lead stablecoin adoption.

Upcoming key macro events

04 \ April	05 \ May	06 \ June
27-28 JN Bank of Japan decision		10-11 EZ ECB decision
28-29 US Federal Reserve meeting		14-16 G7 G7 summit in France
29-30 EZ ECB decision		15-16 JN Bank of Japan decision
30 UK Bank of England decision		16-17* US Federal Reserve meeting
		18 UK Bank of England decision
		18-19 EU European Council meeting
07 \ July	08 \ August	09 \ September
01 IR Ireland holds EU Council presidency		- FR France Senate election
22-23 EZ ECB decision		09-10 EZ ECB decision
28-29 US Federal Reserve meeting		15-16* US Federal Reserve meeting
30 UK Bank of England decision		17 UK Bank of England decision
30-31 JN Bank of Japan decision		17-18 JN Bank of Japan decision
		30 US Fiscal-year ends- multiple agencies' funding expires
10 \ October	11 \ November	12 \ December
04-25 BR Brazilian general election	03 US US midterm elections (House & Senate)	- CH China Central Economic Work Conference
27-28 US Federal reserve meeting	05 UK Bank of England decision	08-09* US Federal reserve meeting
28-29 EZ ECB decision	09-20 COP31 COP31 in Türkiye	14-15 G20 G20 summit in US
29-30 JN Bank of Japan decision		16-17 EZ ECB decision
		17 UK Bank of England decision
		17-18 JN Bank of Japan decision

Source: [The House View: Infographic - 2026 Global Macro Events Calendar – 06-Jan-2026](#)



GDP Growth (%)				
	2024	2025	2026F	2027F
Global	3.4	3.4	3.3	3.3
US	2.8	2.1	2.4	2.2
Eurozone	0.9	1.5	0.5	1.1
Germany	-0.5	0.2	0.5	1.3
France	1.1	0.9	1.0	1.1
Italy	0.6	0.7	0.8	1.0
Spain	3.5	2.8	2.2	2.1
Japan	-0.2	1.2	0.8	1.0
UK	1.1	1.4	0.9	1.2
China	5.0	5.0	4.9	4.5
India	7.2	7.5	6.7	7.2
Russia	4.9	1.0	0.8	1.3
Asia (ex Japan)	5.4	5.5	5.2	5.1
EEMEA	3.4	3.0	2.9	3.3
LatAm	2.5	1.9	2.0	1.8
EM	4.6	4.5	4.3	4.4

- *CPI (%) forecasts are period averages. **Germany National CPI taken.
EEMEA: Czech Rep., Israel, Egypt, Hungary, Poland, Russia, Saudi Arabia, South Africa, Turkey, UAE
- LATAM (ex Venezuela): Argentina, Brazil, Chile, Colombia, Mexico, Peru
- ASIA (ex Japan): China, HK, India, Indonesia, Korea, Malaysia, Philippines, Singapore, Sri Lanka, Taiwan, Thailand, Vietnam
- Current prices as of April 28, 2026.
- *China rates are 7d OMO rates.
- **Gold, Brent and WTI forecasts are for Q2 / Q3 / Q4 2026.

CPI Inflation, YoY* (%)				
	2025	2026F	2027F	2028F
US	2.7	3.3	2.5	2.5
Eurozone	2.1	2.8	2.2	2.2
Germany**	2.2	2.9	2.4	2.9
Japan	3.2	1.8	2.2	2.2
UK	3.4	3.1	2.6	2.0
China	0.1	1.5	2.0	2.0
India	2.2	4.6	4.2	
Russia	8.7	5.7	5.4	

Central Bank policy rate (%)				
	Current	Jun 26F	Sep 26F	Dec 26F
US	3.625	3.625	3.625	3.625
Eurozone	2.00	2.25	2.50	2.50
UK	3.75	3.75	3.75	3.75
Japan	0.75	0.75	1.00	1.00
China*	1.40	1.40	1.40	1.40
India	5.25	5.25	5.25	5.25

Key market metrics				
	Current	Jun 26F	Sep 26F	Dec 26F
US 10Y Yield (%)	4.34	4.30	4.45	4.45
EUR 10Y Yield (%)	3.03	2.90	3.00	3.10
EUR/USD	1.17	1.20	1.22	1.25
USD/JPY	159	155	152	150
USD/CNY	6.83	6.80	6.80	6.70
S&P 500	7174	7450	7675	8000
**Oil WTI (USD/bbl)	97.51	90	82	73
**Oil Brent (USD/bbl)	109.56	96	87	78
**Gold	4669	5,100	5,500	5,800



01. Macroeconomics



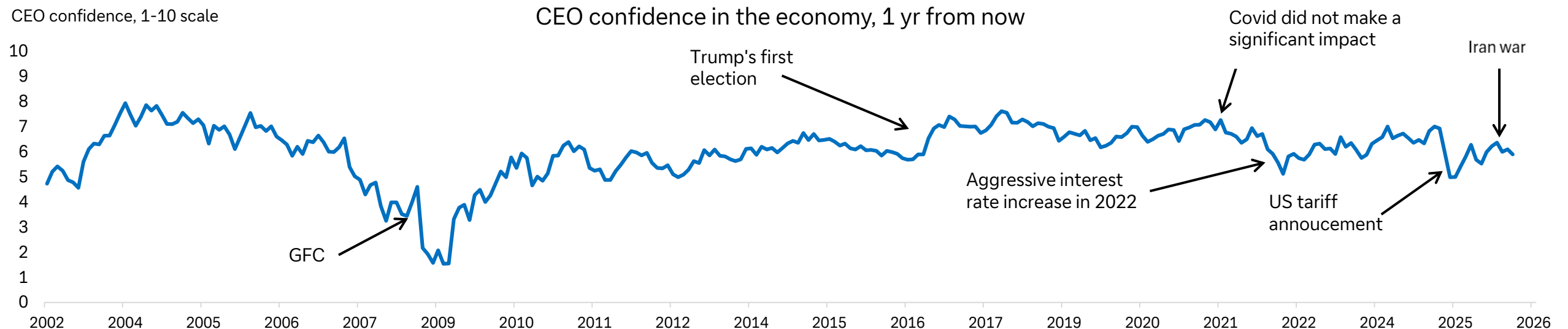
2026 Global growth

Energy shock weighs on growth, especially in the Euro Area



View	Rationale	View	Rationale
 Growth at 3.3% in 2026 and 2027.	The energy price shock from the Iran war is reverberating through the global economy, although the impact varies across regions.	 Growth increased to 4.9% in 2026, 4.5% in 2027.	Strong performance driven by a +15% y/y export surge in Q1, rebound in domestic investment, and some stabilization in the property sector.
 Growth revised to 2.4% in 2026 (2.3% q4/q4) and 2.2% in 2027 (2.2% q4/q4).	The downgrade reflects the drag from higher oil prices and softer consumer spending, counterbalanced by continued tailwinds from supportive fiscal policy, and strong AI investments.	EM	In Asia, fuel price controls have limited immediate inflation rise, but full effects will emerge as domestic prices adjust over time to higher imported input prices.
 Euro area growth decreased to 0.5% in 2026 and 1.1% in 2027. German GDP downgraded to 0.5% in 2026 and 1.3% in 2027.	The energy shock is impacting the region at a time when growth was already softening, lowering the bar to stagnation or even recession.		CEEMEA has largest upward inflation revisions, but more limited drag on activity and sentiment compared to Asia. LatAM impact less pronounced, electoral cycles remain in focus.

The drop in corporate sentiment from the Iran war is not as steep as one may think



Read: [Global Economics: Monthly Flyover \(April 2026\) - Deutsche Bank Research](#)

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3 key risks for the global economy

The growing economic effects of the war in the Middle East are raising further questions about the government response, not just in the US but also in Europe.

1. Iran conflict

- Oil-driven energy crisis risks global stagflation, leading to higher prices and slower growth for corporates.
- Despite volatile environment, the probability of a long-term solution is strong.

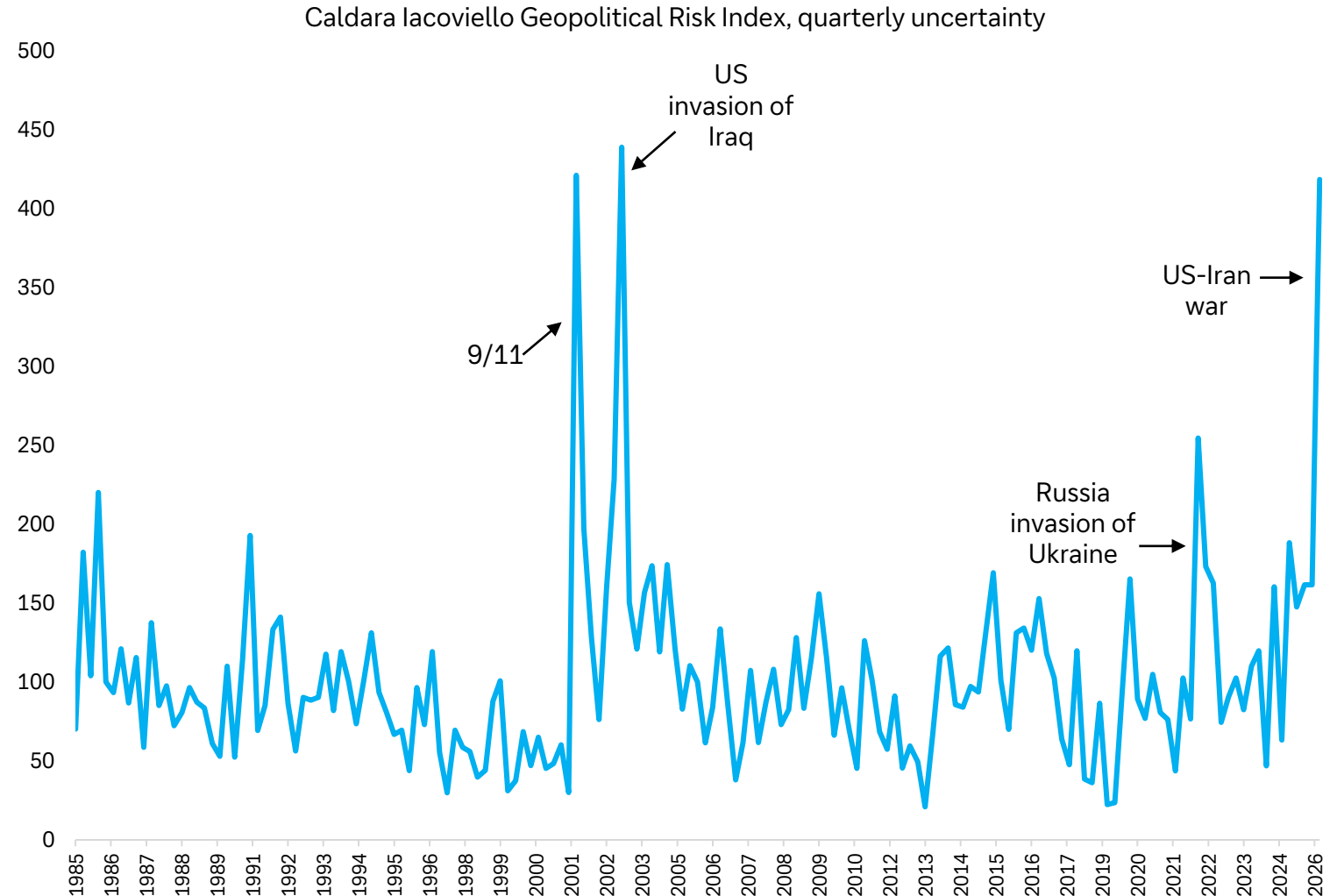
2. Private credit

- Liquidity pressures are building but default rates remain low while the most material exposures are with larger managers with more buffer.
- There are expectations of a severe shock (BDCs' discount to NAV is about 20%) however, this fear has been relatively muted in broader public credit markets.

3. AI disruption

- Concerns that AI will disrupt the labour force are, so far, unfounded in aggregate. The most likely scenario remains that job losses will be limited to specific roles and most will use AI to amplify existing roles.
- The Nasdaq has recovered from another AI-related sell-off.

Geopolitical uncertainty is now at its highest level since the US invasion of Iraq



Source: Deutsche Bank Research, Caldara, Iacoviello Geopolitical Risk Index. Takes automated text-search results of the electronic archives of newspapers

Central banks

The energy shock has shifted central bank trajectories



Fed

- Fed to remain on hold near neutral through 2026.
- Core PCE (2.9% q4/q4) set to remain above target, coupled with labour market stabilization, makes chances of easing unlikely.



ECB

- 25bp hikes in Jun and Sep to 2.50% year-end, previously on hold.
- 2.5% at the upper end of neutral; ECB is sending signal on their commitment to price stability.



BoE

- On hold at 3.75% through 2026. Upside risks stem from energy prices trickling down into core goods and food.



BoJ

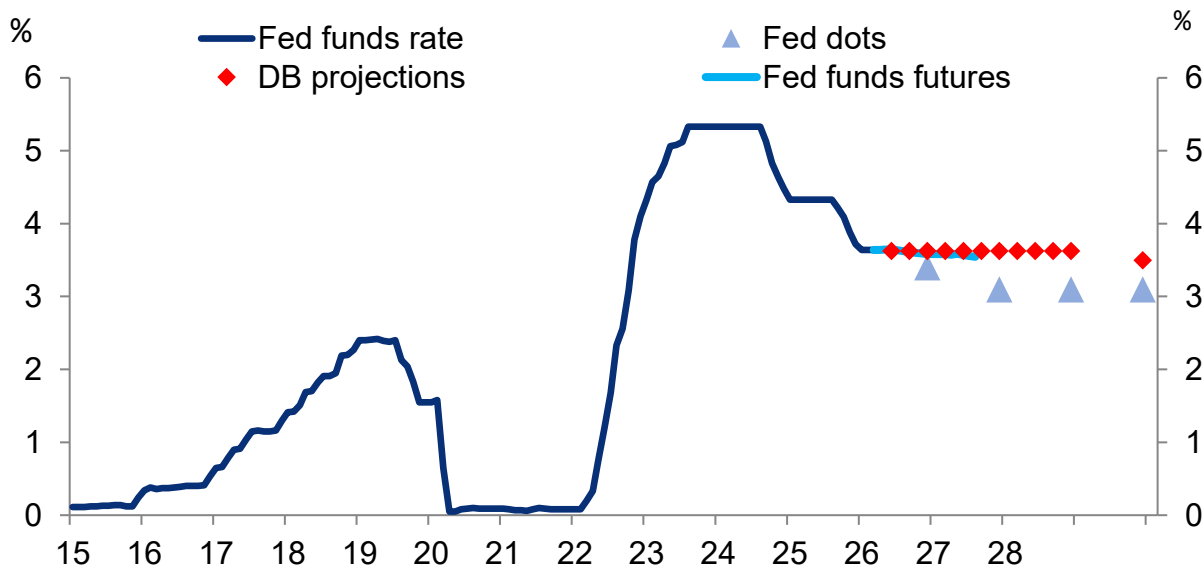
- 25bps hikes in Jul and Jan 27, 1.25% by Jan 2027.



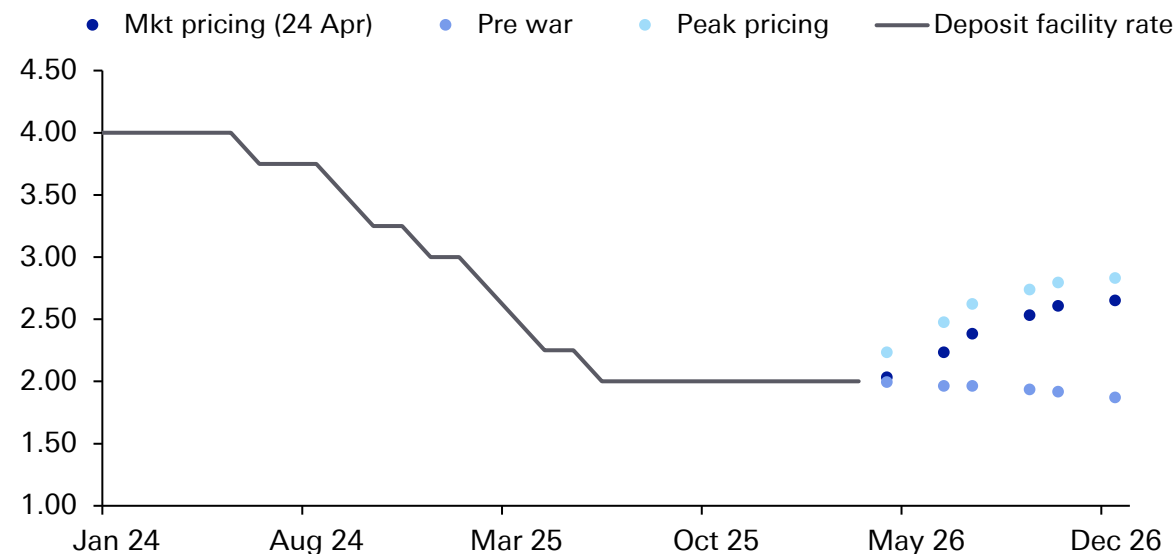
PBoC

- Hold at 1.40% through 2026.

FED: Central bank is done cutting interest rates from here



ECB: 2.50% YE, previously 2.0%



Read: [Focus Europe: ECB Preview: Tough talk, tough decisions - Deutsche Bank Research](#) and [Fed Notes: FOMC preview: Powell's last dance? - Deutsche Bank Research](#)



02. Geopolitics



Iran conflict

An end in sight?



While diplomacy is continuing, the situation remains extremely fragile. The two sides' postures continue to diverge on Iran's nuclear programme, the pace of sanctions relief, potential reparations, Iranian support for its proxies, and long-term control of the Strait of Hormuz. While the ceasefire has been extended, the crisis remains unresolved and we should expect continued escalations and bellicose rhetoric by all parties to the conflict.

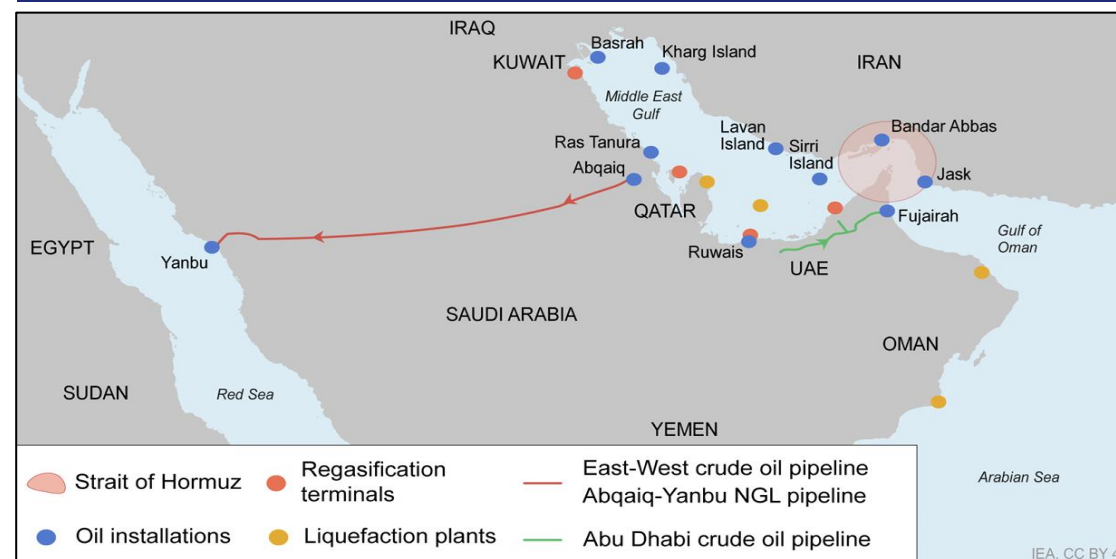
Despite the precarious nature of the ceasefire, and the added uncertainty due to the US blockade of the Strait of Hormuz, we assess that the probability of a longer-term solution is strong. The US is facing diminishing returns and growing costs. We have already witnessed increasing domestic and international pressure on Trump, which we expect will only increase as the economic fallout of crisis persists and expands.

However this ends, it will be a gamechanger for the Middle East. The long-standing Middle East geopolitical equilibrium has been shattered. Iran will likely walk away from this conflict with enhanced influence over the Strait of Hormuz, potential sanctions relief, and with greater political leverage having withstood the US and Israeli attacks. Meanwhile, the conflict illustrates that the Gulf is no longer buffered from conflict. Consequently, its relationship with the US will inevitably be impacted. We expect to see a diversification in Gulf relations in the aftermath of the acute crisis.

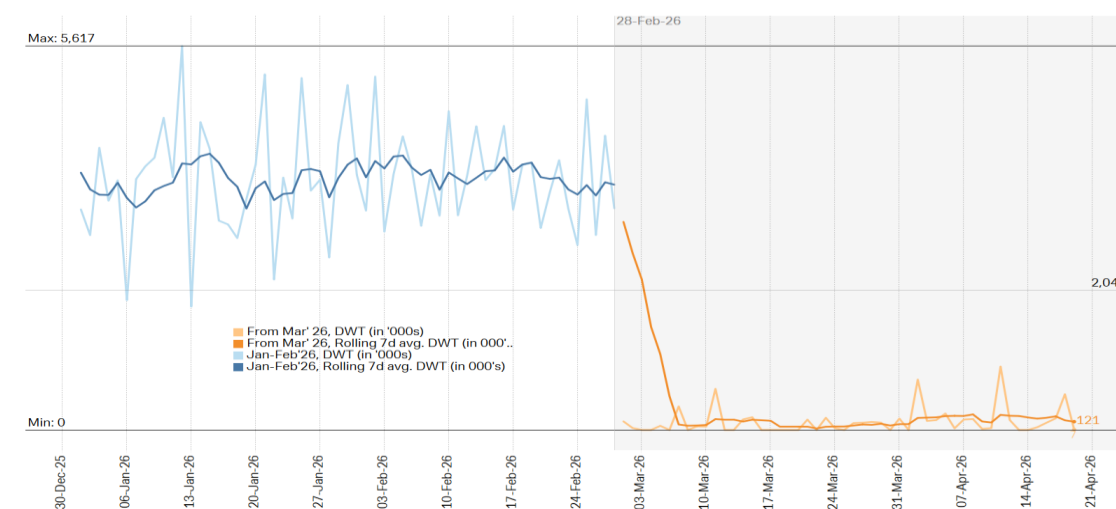
Post-conflict normalization could take months. Even if the war concludes in the near term, rebuilding critical infrastructure and restoring shipping and energy flows will take significant time. For example, repairs to Qatar's Ras Laffan natural gas field – the world's largest – could take months, inevitably delaying a rebound in global supply.

A rebound in shipping (see chart) will also be contingent on confidence – both by shippers and insurers – that the Strait of Hormuz is safe and clear of mines. This could take weeks beyond an end to the conflict. Significant knock-on effects, including for inflation, could continue well into 2026-H2 even with a near-term peace deal, including in the form of jet fuel and fertilizer shortages.

Oil and gas infrastructure in the Gulf



Tanker activity exiting the Strait of Hormuz (in DWT'000s)



Source: Deutsche Bank Research, International Energy Agency, dbDataInsights.

Key topics to monitor during 2026-Q2

- **Trump-Xi Summit:** Barring a significant escalation in the Iran war, the summit between presidents Xi and Trump on 14-15 May will likely go ahead as planned – and deliver a package of tactical economic agreements. Despite recent signs of friction over Iran, both sides appear to be prioritizing bilateral stability for 2026, and the threshold for another summit deferral is likely high. However, a breakdown of the US-Iran ceasefire and a return to fighting would be a notable downside risk, which would increase the probability of another summit delay, or even a cancellation.
- **Putin's visit to China:** It will likely aim to cement the China-Russia partnership (including more energy supplies for Beijing), and possibly to coordinate US policies after the Xi-Trump summit. Unconfirmed Russian media reports suggest Putin could visit during the week of 18 May. Putin and Xi could also seek to deepen diplomatic and security ties, and coordinate on the Iran conflict, given Tehran's status as both countries' close partner.
- **Ukraine ceasefire talks:** Negotiations for a ceasefire in Ukraine are largely stalled. While there is ongoing diplomatic activity, no new rounds of ceasefire talks are publicly scheduled for May or June. Ukrainian President Zelenskyy has stated that the US wants Russia and Ukraine to reach a peace agreement by June. However, the specifics of how this would manifest in terms of scheduled meetings are unclear.
- While a ceasefire over the coming months is far from guaranteed, even credible steps toward a deal would have implications for markets. These could include a possible easing of some sanctions on Russia, the potential for increased Ukrainian agriculture exports, and for enabling reconstruction – with implications for investments in minerals, gas infrastructure, technology, city reconstruction, data centres and artificial intelligence.

For more on geopolitical flashpoints to watch out for, read our recent publications here:

[Iran Ceasefire: An End in Sight?](#) ; [This Month in Geopolitics: April 2026](#) ; [Geopolitical Crossroads: India's Strategic Balancing Act](#) ; [High-Stakes Diplomacy: Navigating the Xi-Trump Summit in the Shadow of Iran](#)



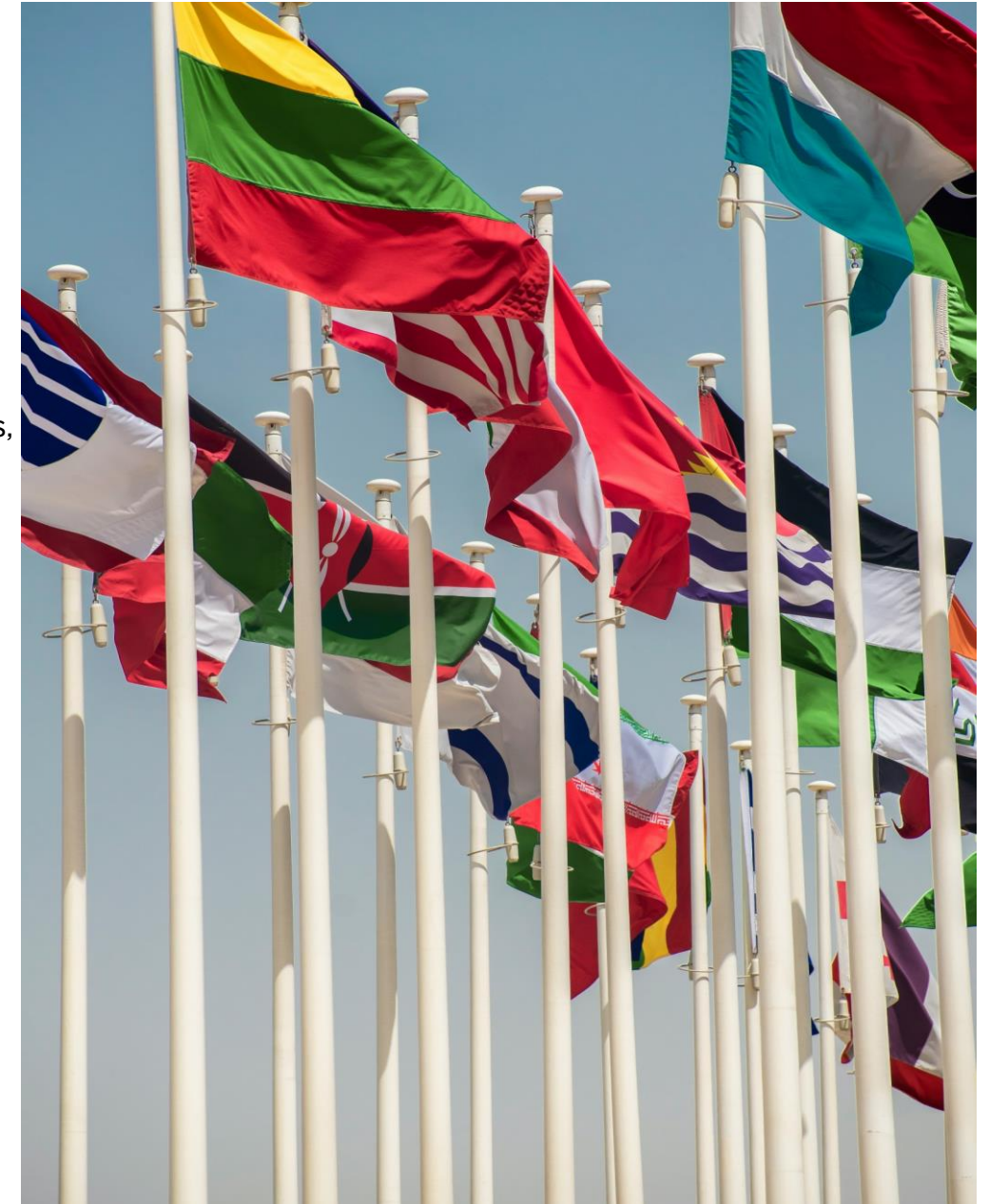
Source: [Weichao Deng](#) on [Unsplash](#)

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Selected geopolitical events in April - June



- **7 May:** **UK** local elections and Scottish and Welsh parliamentary elections.
- **8-9 May:** The **Philippines** will host the ASEAN Leaders' Summit.
- **11-12 May:** **EU** foreign and defence ministers meetings.
- **14-15 May:** Presidents Xi and Trump are expected to meet in **Beijing**.
- **Mid-May:** President Putin is expected to visit **Beijing**.
- **Mid-May:** **Indian** Prime Minister Narendra Modi is expected to visit the Netherlands, Norway and Italy.
- **22 May:** Nuclear Non-Proliferation Review Conference concludes in **New York**.
- **29-31 May:** **Singapore** will host the annual Shangri-La Dialogue defence summit.
- **31 May:** **Colombia** Presidential election first round
- **Late May:** US Secretary of State Marco Rubio will visit **India** for talks.
- **Late May/Early June:** The OECD Ministerial Council Meeting is expected to take place in **Paris**.
- **11 June:** FIFA World Cup begins in **US, Canada and Mexico**
- **15-17 June:** **France** will host the G7 Leaders' Summit in Évian.
- **19 June:** **European** Council meeting
- **1 July:** The first review of the **US-Mexico-Canada** (USMCA) trade agreement begins.



Source: [aboodi vesakaran](#) on [Unsplash](#)



03. Markets



Commodity markets

The key question: When do the stresses in financial assets directly affected by the war begin to catalyse confidence problems in broader markets?



Oil: Brent \$96/bbl (Q2'26), \$84.8/bbl (2026) average, assuming the Strait of Hormuz will re-open in May.

- Improving but uncertain backdrop, thanks to (i) supply diversion and remaining flows limiting the magnitude of oil supply disruption; and (ii) significant demand reduction rising to -2.3mmb/d y/y in April, according to the IEA.
- **Risks:** Longer suspension of the SoH (reopening beyond May) could accelerate the run-up in Brent well above \$100/bbl near term. This would translate into higher-for-longer inflation.
- Central banks' ability to look-through supply shocks is tested.

Gold: moderate YE target to \$5,000/oz (prevs. \$6,000/oz)

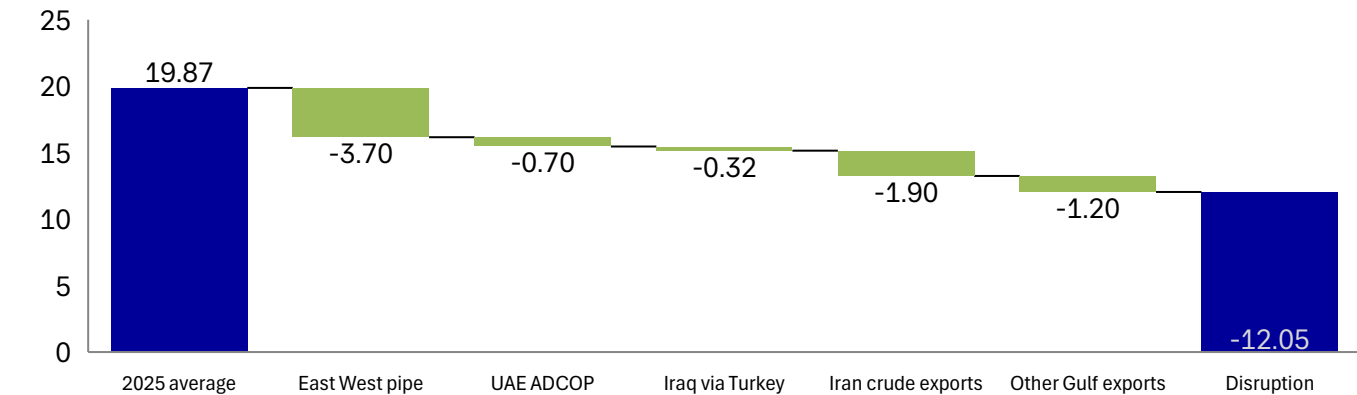
- Forecast based on belief that dollar weakness will resume.

Brent crude oil prices (\$/bbl) have risen but are clearly lower today than they were in 2022



Source: Deutsche Bank Research, Bloomberg Finance LP

Strait of Hormuz April supply disruption down to 12mmb/d



Source: Deutsche Bank Research, Bloomberg Finance LP, IEA, Kpler

Latest oil price forecasts

USD	2025	Q1 2026	Q2 2026	Q3 2026	Q4 2026	2026	2027
WTI (bbl)	64.9	72.7	90.0	82.0	73.0	79.4	71.0
% change		-0.5%	12.5%	9.3%	4.3%	6.6%	4.4%
Brent (bbl)	68.20	78.10	96.00	87.00	78.00	84.80	75.00
% change		0.1%	12.9%	8.7%	4.0%	6.6%	4.2%

Reports: [Hsueh On Oil: Problem solved? - Deutsche Bank Research](#) and [Precious Special Report: Gold investors to emerge from war bunker - Deutsche Bank Research](#)

Rates & Credit markets

Dual shock of higher rates and energy costs introduces some uncertainty whether strong corporate growth forecasts will continue into year-end



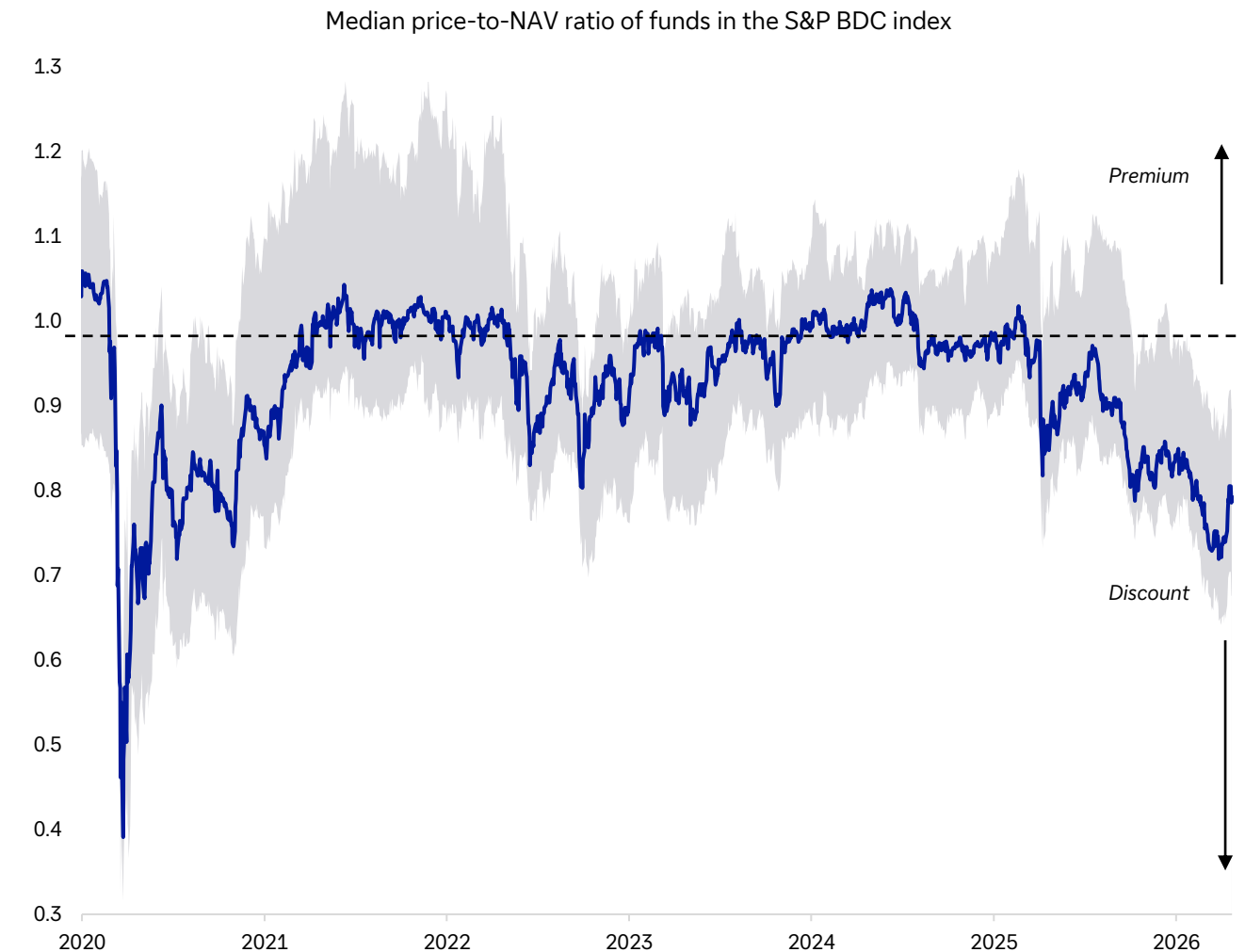
Private credit outflows have had some effect on broader leveraged loans and some larger US money centres/regional banks, tightening financial conditions somewhat.

- Private credits remain centred on AI disruption, particularly in the software sector, rather than the ongoing energy shock.
- We see credit spreads to widen throughout 2026, with €HY 420bps, \$HY at 410bps YE and €IG at 110bps, \$IG at 110bps YE.
- In Europe, weaker macro fundamentals plus energy shock driven restrictive policy rates also account for wider spreads. Political risks in Europe are growing.

We retain our conviction that the 10Y US Treasury Yield will continue to fall, impacting corporate funding costs.

- Difficult to justify further Fed rate cuts given (i) unemployment rate is broadly stable; (ii) inflation is sticky above 2% and; (iii) policy rates are unlikely to be restrictive.
- AI tailwinds (either bubble or labour market impact) potentially challenge our bearish view on 10Y UST (4.45% by YE).

US business development companies are a key proxy for the market's private credit concerns. The current discount to NAV of about 20% suggests investors expect severe stress events.



Reports: [IG & HY Strategy: The Narratives Are Turning Upside Down - Deutsche Bank Research](#) and [IG & HY Strategy: Private Credit: The gates are up, but the problem is within - Deutsche Bank Research](#) and [Global Fixed Income Weekly: Strategy Update - Deutsche Bank Research](#)

Source: Deutsche Bank Research, S&P Global. See: [Thematic Research: Private capital monitor: Fear and loathing - Deutsche Bank Research](#)

Foreign exchange markets

The dollar bullish narrative has eroded, staying dollar bearish

Despite some reprieve during war, big picture remains dollar bearish

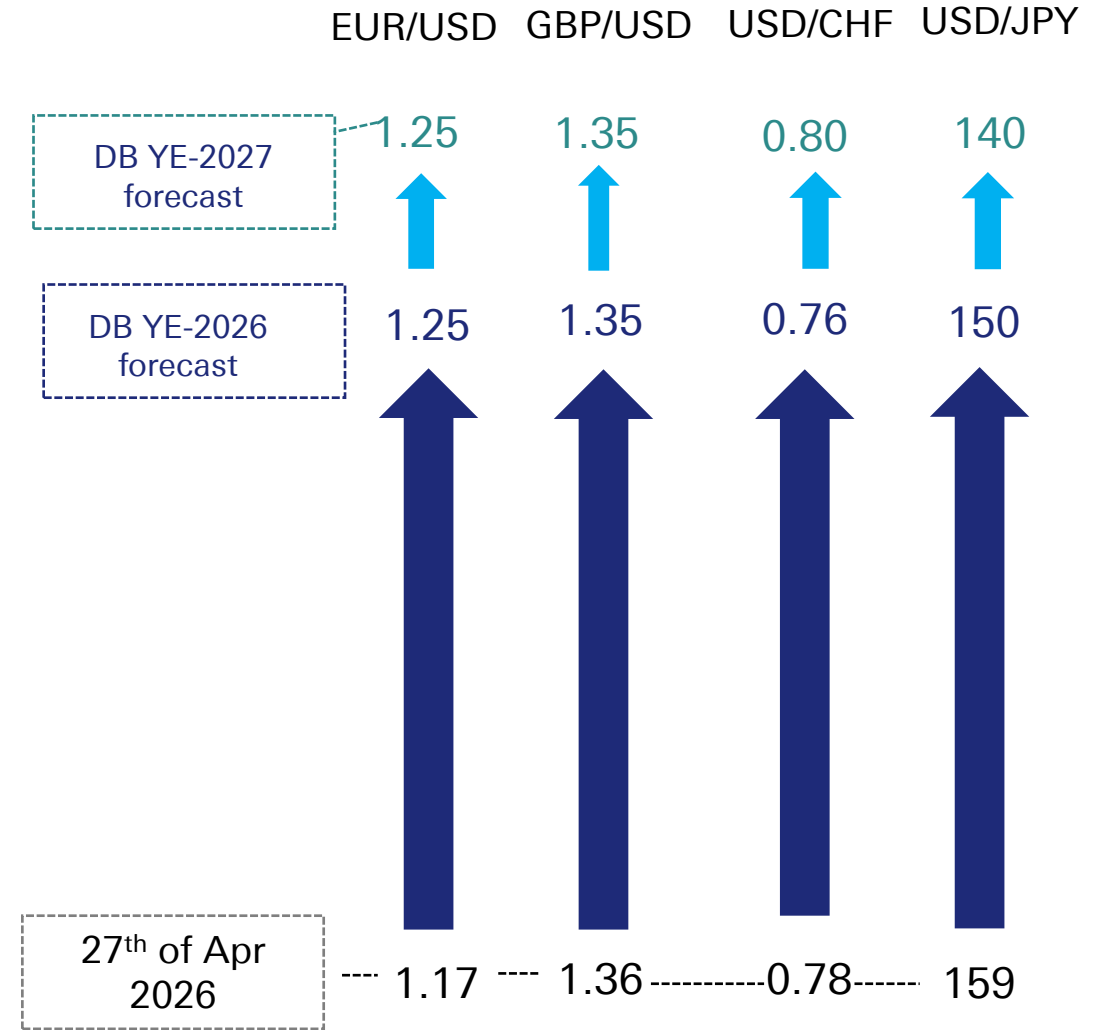
- Usually, we would expect the dollar (as a haven currency) to have rallied more than it did during the Iran war. That highlights some weakness.
- Strengthens the argument that the USD is still no longer as exceptional as a high-yielder and as a global growth outlier.
- The Iran war exacerbated dollar weakness due to Asia and the Middle East running down FX reserves and excess savings to finance higher energy costs. Countries were already moving away from the USD into smaller, unspecified reserve currencies (up from 3% to 6% of these assets in the last 3 years).

Currency outlook

- **EUR/USD:** Rise to \$1.25 by year-end. While the energy price spike complicates the Euro Area's growth picture, it has also brought forward the ECB rate hike cycle, leading to a stronger Euro.
- **USD/JPY:** Raised end-2026 to 150. JPY remains very cheap as the BOJ remains unlikely to tighten policy rates this year.
- **RMB:** Supported by de-escalation in US-China tensions and the government's continued aims to internationalise the yuan. RMB remains cheap but a stronger RMB would increase the appeal for global corporates to transition to using and saving in the yuan, especially if most Middle East energy continues to travel to Asia. Corporates continue to have scope to increase conversion of their large stock and flow of FX into RMB.

Read: [FX Blog: Time to sell USD - Deutsche Bank Research](#) and [The Big FX View: War and Peace - Deutsche Bank Research](#)

Currency movements and DB forecasts for YE-2026 and 2027



Source: Deutsche Bank Research, Bloomberg Finance LP.



04. Technology



Artificial intelligence

AI disruption concerns are rising in 2026, but there is no conclusive relationship between modest job declines and substitution with AI



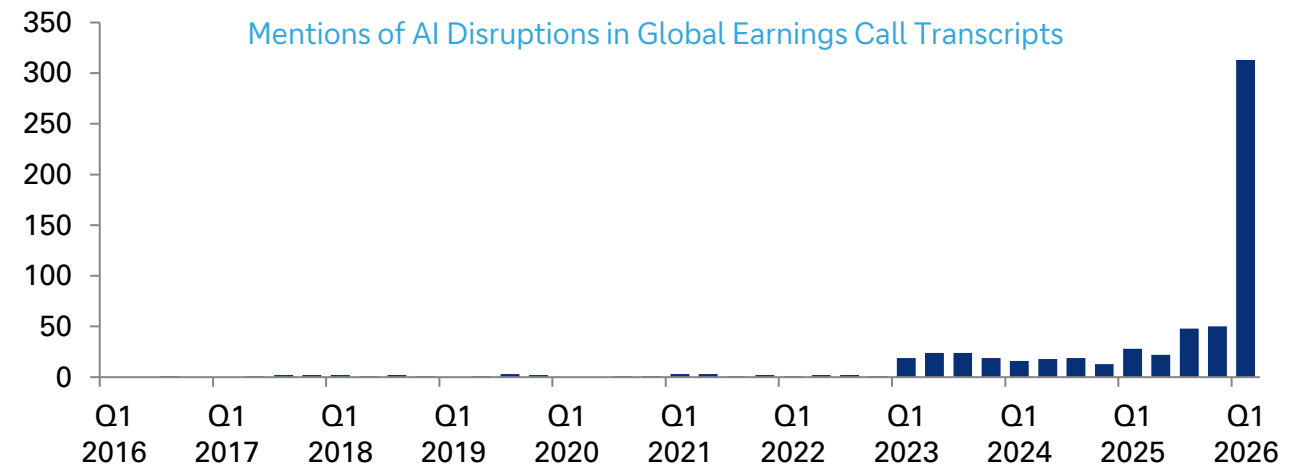
Job displacement from AI remains a concern, but the aggregate evidence so far is encouraging

- A Dec 2025 NBER study [suggests](#) that AI, by substituting primarily cognitive tasks, can shift relative demand away from high-skilled roles toward jobs with lower education, lower wages and more male workers.
- The evidence is still inconclusive, [as noted](#) by our US analysts. A survey by Indeed found no noticeable deterioration amongst roles of recent market concerns, such as legal, marketing, accounting and management.

The case for AI's positive benefits is growing

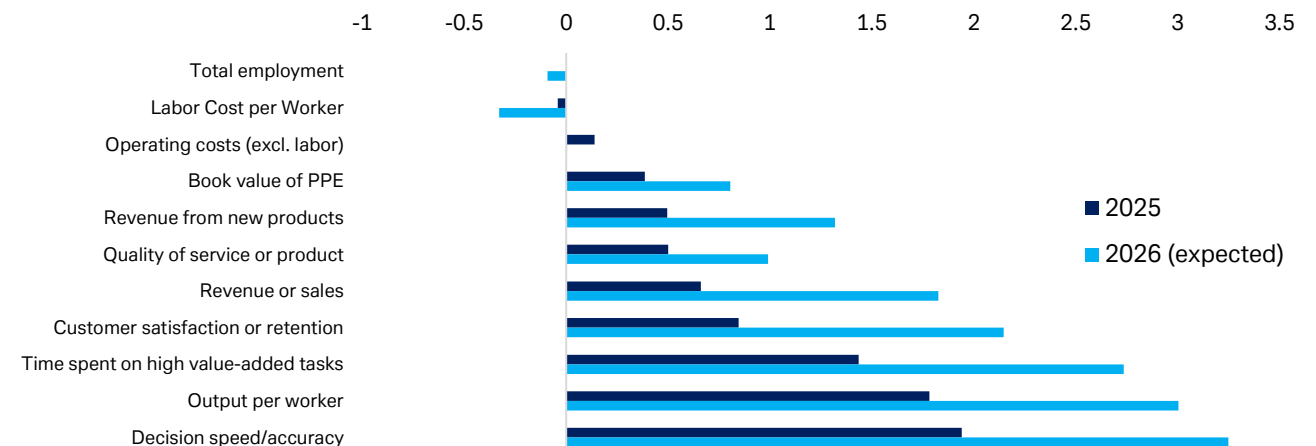
- According to an [Atlanta Fed survey](#), business executives are already seeing labour productivity gains from AI and anticipate it to double in 2026. Firms in high-skill services are seeing the greatest gains. Finance, for example, saw productivity growth of 0.8% last year, implied by revenues and employment, a figure which is set to exceed 2% in 2026.
- Firms in manufacturing and construction saw smaller but still positive implied gains of roughly 0.4 percent in 2025.
- The study found that productivity gains were associated with innovation and demand-orientated channels rather than cost reduction alone.

In Q1 26, mentions increased more than sixfold from Q4 2025 in earnings calls.



Source: Deutsche Bank Research, Bloomberg Finance LP.

Impact of AI usage on firm outcomes



Source: Deutsche Bank Research, The CFO Survey, Federal Reserve Bank of Richmond, Federal Reserve Bank of Atlanta. Question asks for likert scale options that are mapped to percentage outcomes (i.e. increased moderately, 5.1 to 10%. Above chart uses midpoint of ranges provided and +/- 5% above/below highest/lowest options to calculate numeric averages across respondents.

Artificial intelligence

Companies will continue to invest in AI to varying degrees, but lack of trust remains a key barrier



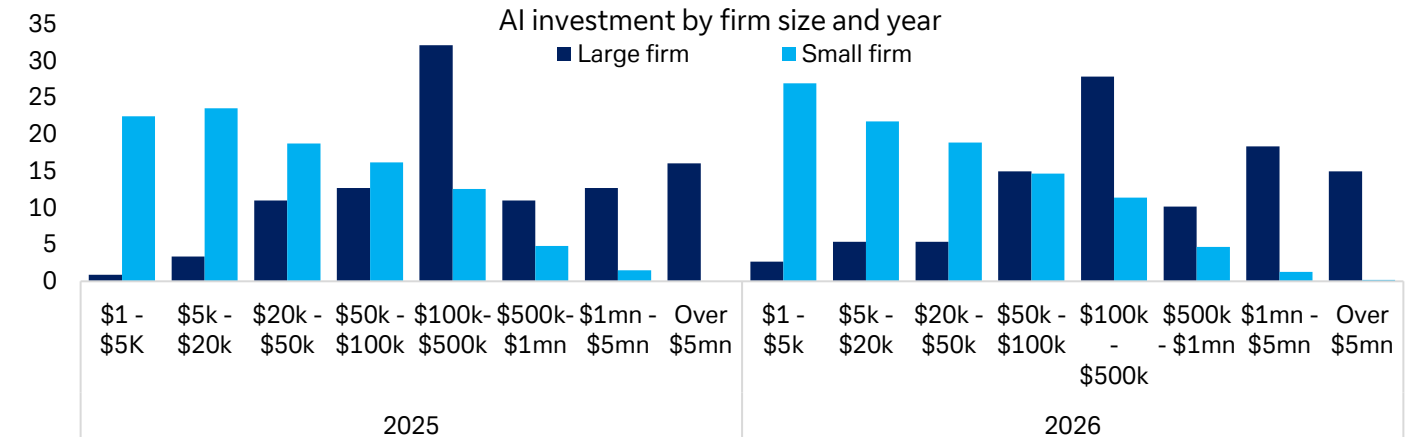
Company adoption continues to be underway, with more pickup in 2026

- Large firms led corporate AI adoption in 2025; smaller firms are only beginning to invest in AI now, according to a survey led by the Atlanta Fed and Richmond Fed.
- Around 30% of large firms expect to invest more than \$1mn in AI in 2026 (vs 1% of small firms). But nearly 60% of smaller firms plan to invest less than \$20,000 in AI in 2026 (vs 14% of larger firms).

Western economies lag in AI trust and adoption

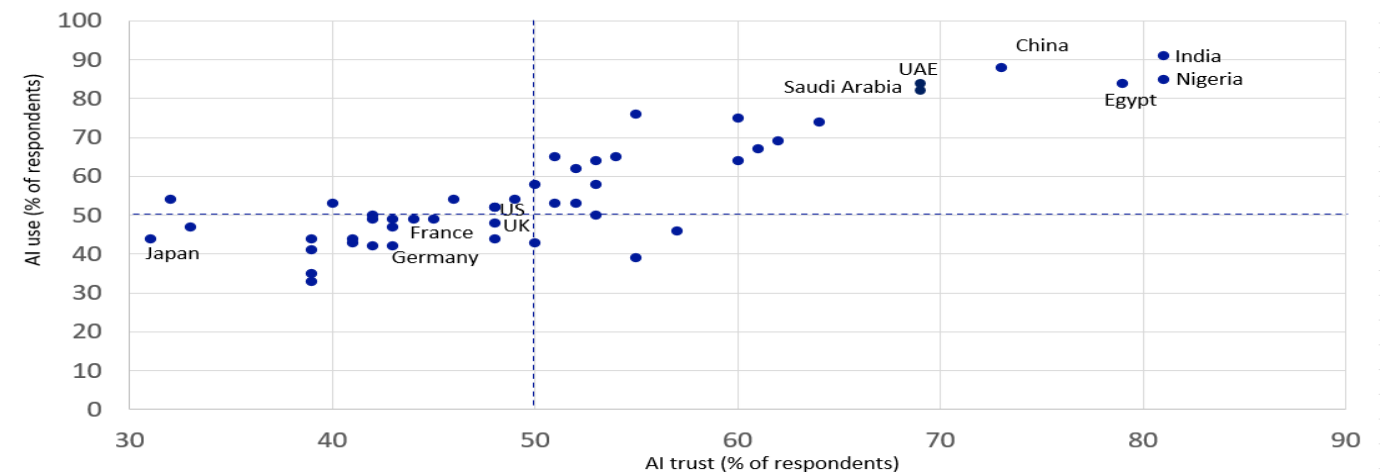
- Barriers to adoption continue to be lack of workforce training, belief AI is not yet sufficiently mature and privacy concerns (see [here](#))
- While the US leads in the most advanced models and chips, and accounts for half of the world's data centres, US consumers are more suspicious of AI than their peers in markets such as India, the UAE, China and Saudi Arabia. Western consumers also lag behind in adoption.
- 48% of US consumers say they trust AI for work purposes, and only 52% say they use it on a regular basis, according to a [survey](#).
- That compares with well over two-thirds trusting AI for work in the leading Asian and Gulf markets and around 85 to 90 percent using it.

80% of firms will invest in AI in 2026, with large firms to invest the most, according to a CFO survey led by the Atlanta Fed and Richmond Fed



Source: Deutsche Bank Research, The CFO Survey. Large firms have > 500 employees. What do you expect your company's total expenditure and financial investment in AI technology/solutions will be over the next 12 months? Federal Reserve Bank Richmond, Federal Reserve Bank Atlanta.

Western economies lag behind their peers in trust in AI and intentional use at work



Source: Deutsche Bank Research, University of Melbourne, KPMG.

Digital currencies

The digital euro is coming faster than most CFOs realise; Stablecoins are becoming a corporate treasury tool



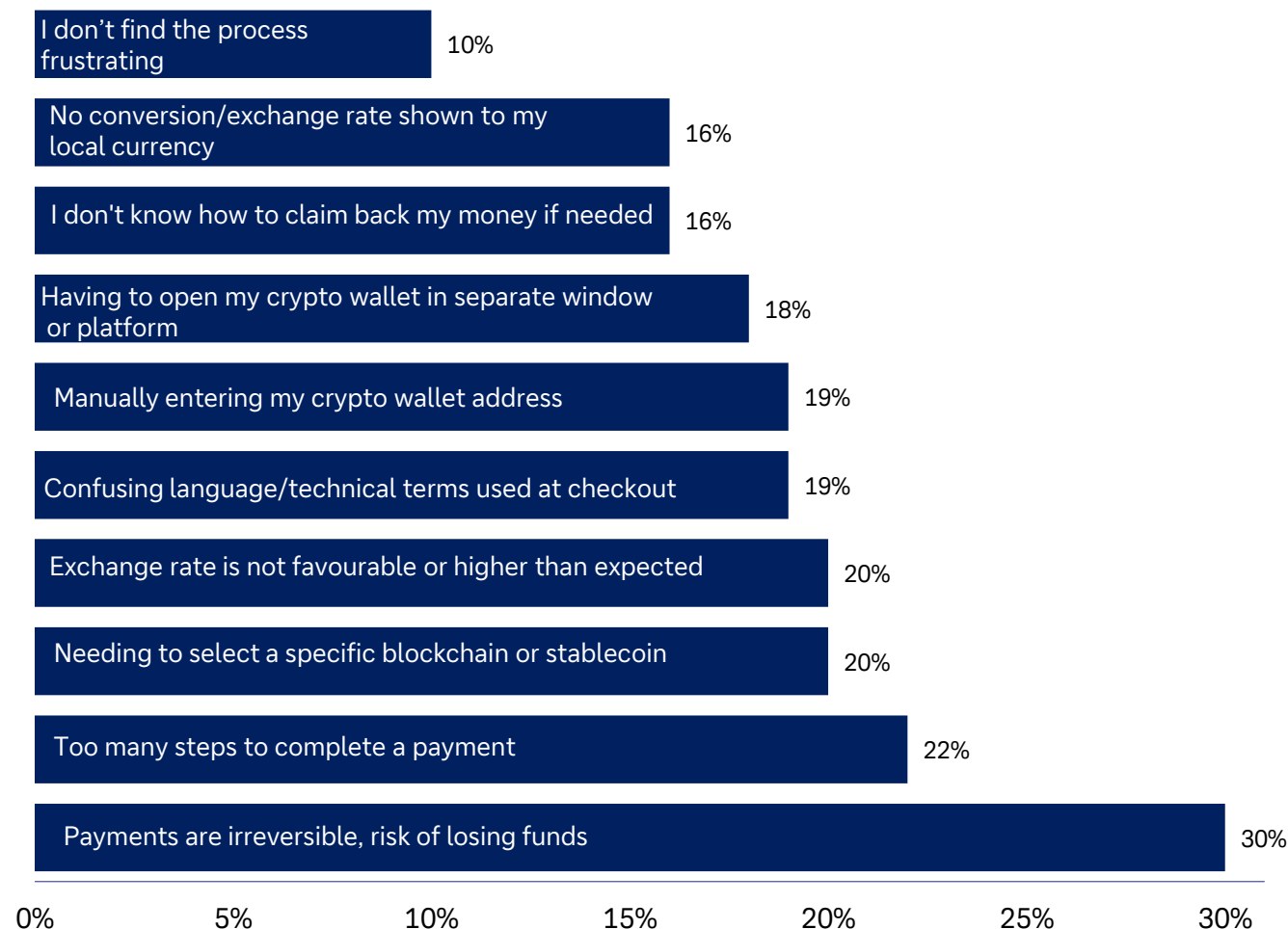
Stablecoins

- Over 80% of transitions occurred outside the US in 2025. Use cases expanding beyond crypto trading into treasury management, payments & settlements, banking & fintech, e-commerce & services.
- Top CFO risks: irreversible payments (30%), too many steps to complete a transaction (22%), fragmented wallet infrastructure.
- In Europe, political debate on permissibility of multi-jurisdictional stablecoins – agreement pending between ECB and European Commission.
- US CLARITY Act still pending. Key debate includes stablecoin yield and bank lending impact.
- March 2026: The OCC proposed stricter stablecoins rules, including a minimum capital floor of \$5mn and a ban on interest payments to holders. This favours established payment giants and raises barriers for new entrants. See [here](#).

CBDCS

- Wholesale: ECB's Project Pontes – a settlement solution for DLT-based large-value transaction – launches Q3 2026.
- Retail: European Parliament vote scheduled 23 June 2026. Pilot expected to begin in 2027; retail issuance targeted for 2029.

Challenges with stablecoin payments experience (BVNK Survey)



Source: Deutsche Bank Research, BVNK Survey (March 2026).



Appendix 1

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